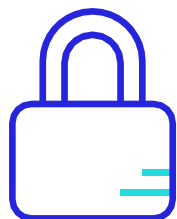


ACI Fraud Management

Proactive Risk Manager

ACI Fraud Management

Introduction



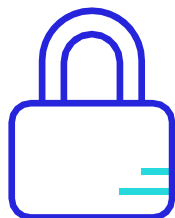
For over two decades, **ACI Worldwide** and **Interswitch** have maintained a strategic partnership, delivering cutting-edge, 21st-century payment solutions tailored to this market and beyond. This long-standing collaboration has played a crucial role in driving digital transformation, enhancing transaction security, and enabling seamless payment experiences.

As financial fraud continues to evolve, businesses face increasing threats from sophisticated cybercriminals who exploit gaps in traditional security systems. Fraudulent activities, including unauthorized transactions, identity theft, and payment system breaches, have created significant financial and reputational risks for organizations. The need for a **proactive, intelligent, and adaptive fraud prevention solution** has never been greater.

To address these challenges, ACI Worldwide and Interswitch introduce **Proactive Risk Manager (PRM)**—a state-of-the-art fraud detection and prevention technology. PRM leverages **real-time transaction monitoring, artificial intelligence (AI), and machine learning (ML)** to detect and prevent fraudulent activities before they impact businesses. With advanced analytics, customizable risk rules, and seamless integration with existing payment infrastructure, PRM provides a **robust, scalable, and intelligent fraud defence system**.

ACI Fraud Management

Key Benefits of PRM



- Real-Time Fraud Prevention:** Instant detection and blocking of suspicious transactions.
- AI & Machine Learning:** Continuous learning to improve fraud detection accuracy.
- Adaptive Risk Scoring:** Dynamically adjusts to emerging fraud patterns.
- Seamless Integration:** Works across multiple payment channels and systems.
- Regulatory Compliance:** Helps businesses meet fraud prevention and data security regulations

ACI Fraud Management

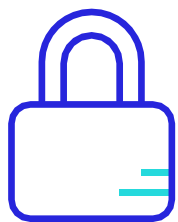
Competitive Advantage

Unlike conventional fraud management tools, PRM **combines real-time decisioning with AI-driven insights**, offering a superior fraud prevention mechanism tailored for financial institutions, fintechs, and enterprises. The solution not only **reduces fraud losses** but also ensures a **frictionless customer experience**, enabling businesses to operate securely without compromising transaction speed or convenience.

With ACI Worldwide and Interswitch's continued commitment to innovation, PRM empowers businesses to **stay ahead of fraud threats** while safeguarding their revenue and reputation in today's digital economy.

ACI Fraud Management

Solution Summary



ACI Fraud Management offers banks and intermediaries a comprehensive, real-time approach to fraud management that uses a superior combination of machine learning, fraud and payments data, and advanced analytics to prevent fraud and reduce the burden of compliance.

USING ACI MODEL GENERATOR, EVERLINK
IMPROVES ITS CNP FRAUD DETECTION RATE

11%

Best Enterprise Fraud Prevention Provider Europe

GLOBAL BANKING & FINANCING REVIEW, 2019



With ACI, we will continue to provide banks and merchants access to fast and simple payment options as well as the ability to leverage rich data and analytics, all without compromising security.

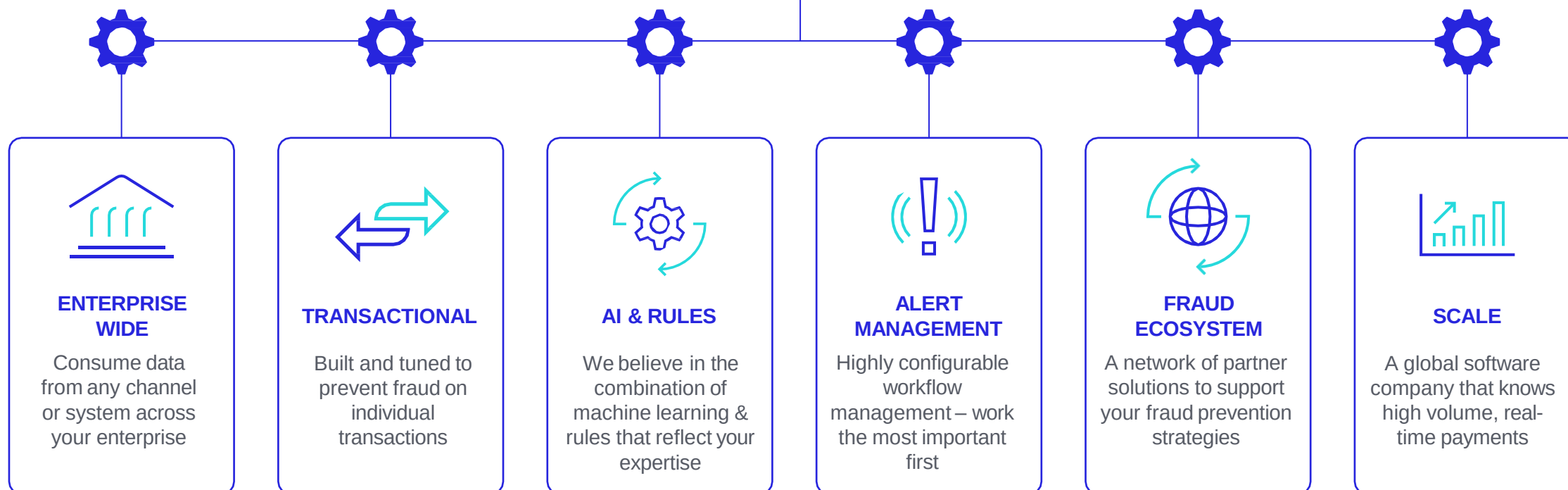
Christer Jonsson
Senior Vice President, Card Services
EVRY

What is Proactive Risk Manager?

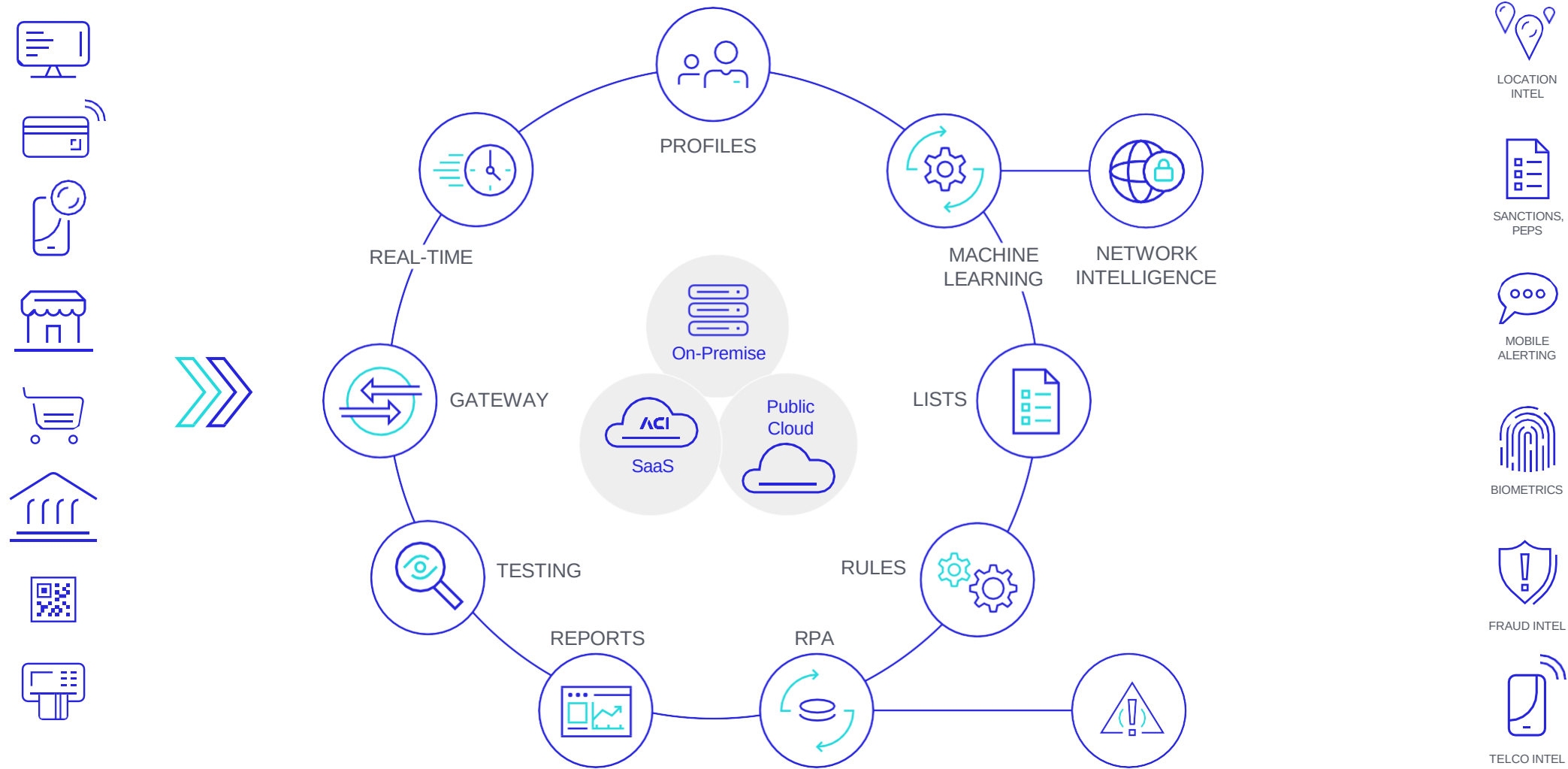
ACI Fraud Management

Proactive Risk Manager

An enterprise wide, transactional fraud prevention solution that scales, employing machine learning and rules with a highly-efficient alert management capability, and a supporting fraud prevention ecosystem



Proactive Risk Manager



Key Capabilities



Message Orchestration

- Simple integration through RESTful API Web Service
- Enrich data as required
- Configurable real-time responses



Analytics Engine

- Real-time rules created on any combination of data within the database – risk score, transaction history, profiles, demographics, 3rd party data
- Strategy simulation with Rule Tester



Machine Learning

- Calculate your own features and develop your own models, based on your data
- No cleansing or prep required
- Ready from Day 1



Security Management

- Use defaults roles or create custom access
- Only grant access required to perform the role, nothing more
- Each & every action is audited



Operational Control

- Prioritise workflow in real-time
- Work the most important alerts first
- Only grant users access to customer segments that are required

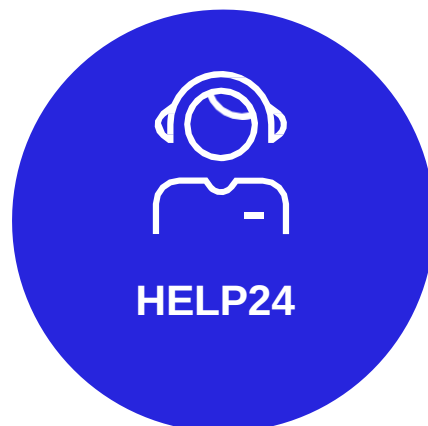


Customer Experience

- Behavioural profiling reduces false positives
- 2-way notifications appreciated by customers
- 360° / enterprise view of customer

Beyond Software

ACI has the people to support you



HELP24
First level support 24*7*365



User Groups / Community
Global & regional exchange meetings; connect & interact online with other PRM users



Fraud Consultancy / Training
ACI consultants partner with you to deliver your fraud strategy and goals, whilst optimising the customer experience

Fraud Primer

Stopping Fraud

The building blocks



VELOCITY RULES

More than x transactions within x hours

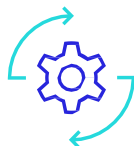
1:20



SOPHISTICATED RULES

Velocity *AND* Condition A *IF* Condition B *OR* Condition C

1:10



MACHINE LEARNING

Score *AND* Condition D *IF* Condition E / Condition F

1:4



BEHAVIOURAL PROFILING

Score *AND* Condition *AND* Unknown Device / IP Address

1:2



REFERENCE DATA

Score, Condition, Device ID *AND* Reference Match

1:1

Scores vs Rules

SCORES



- Score range 0-999
- Scores indicate the *level* of risk
- Can involve complex, manual analytics which requires expensive resources
- Once done, can be adaptive, learning from new fraud
- Optional, but enhances the strategy

RULES



- They trigger. Or they don't.
- More specific, lots of detail, can be complex
- Scores are used in rules, just another element
- Rules are always needed, they are the *construct* for alerts
- Don't need to be a data scientist
- Provide more agility

User Interface

User Interface

ACI Proactive Risk Manager

- Opening screen is a dashboard with configurable widgets
- Reviewer screen
 - Demographics in the top half of screen
 - Transaction grid in the bottom half of the screen
 - Action buttons on the left-hand side of the screen

The screenshot displays the ACI Proactive Risk Manager interface. The top navigation bar includes 'DASHBOARD', 'SEARCH', 'REVIEW', 'PENDING ALERTS', 'REPORTS', and 'FORWARDS & REMINDERS'. The user is logged in as 'prm'.

The dashboard on the left features a 'QUEUE STATUS / CARDALL, MERCHALL, UANALL, TERMINALALL, ...' table with columns: Queue, Queue Type, Portf..., U..., Active..., D..., and Amo... The table lists various queue types and their associated metrics.

The 'TRANSACTION VOLUME OVERVIEW / AGNEARREALTIMEWITHRTU...' table shows transaction data with columns: Analytic Gateway, Date, Hour, and Total Transactions.

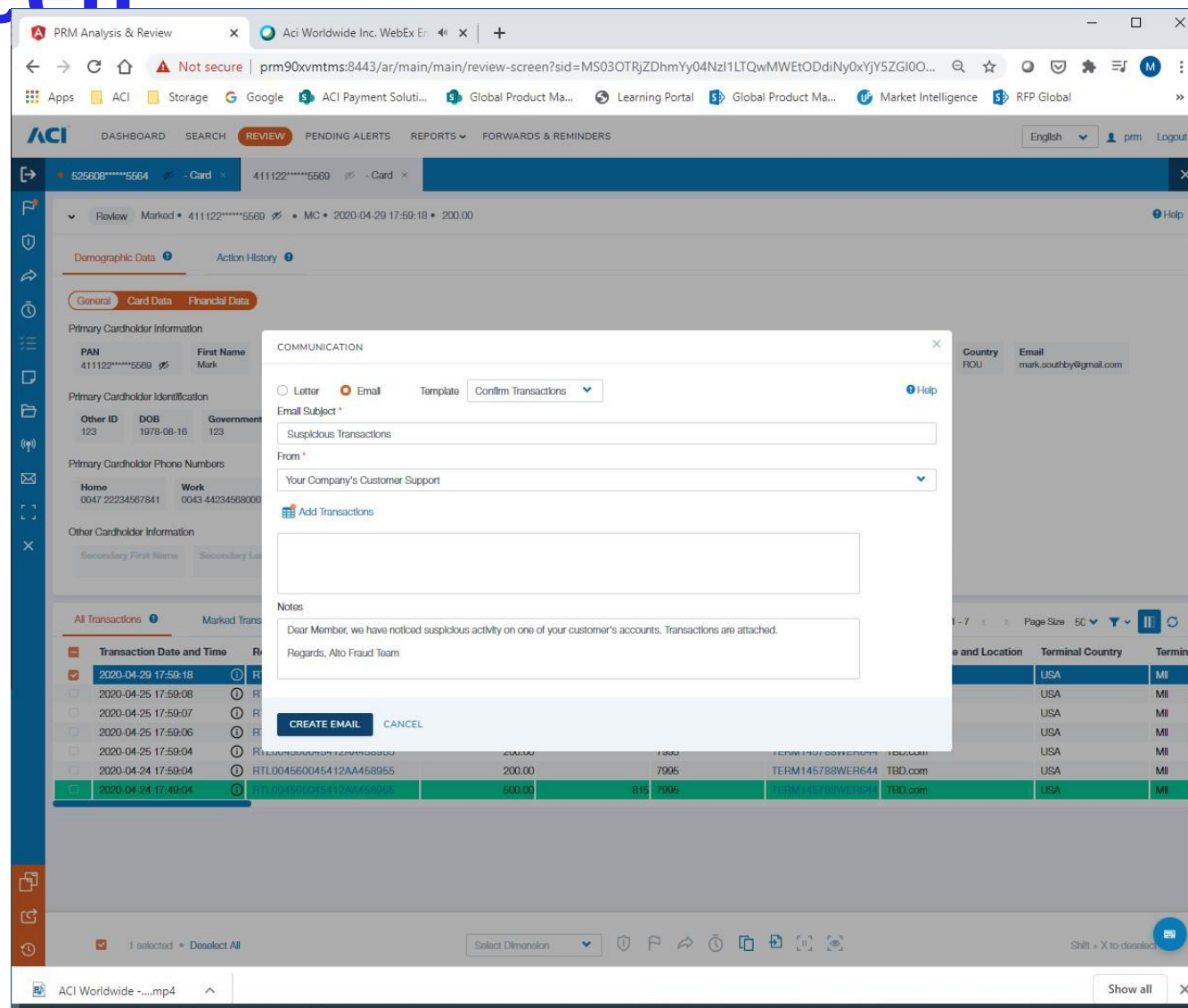
The main reviewer screen on the right shows a 'Review' tab for a specific cardholder. It includes a 'Demographic Data' section with tabs for 'General', 'Card Data', and 'Financial Data'. The 'General' tab displays primary cardholder information such as PAN, First Name, Last Name, Middle Initial, Suffix, Previous PRIM Name, Address 1, Address 2, City, and State/Province. It also shows postal code, country, email, and other identification details like Other ID, DOB, Government ID, and Financial Institution ID. Phone numbers (Home, Work, Mobile) and other cardholder information (Secondary First Name, Secondary Last Name, etc.) are also visible.

The bottom section of the reviewer screen displays a list of transactions with columns: Transaction Date and Time, Retailer ID, Amount 1, NRT Score, MCC/SIC, Terminal ID, and Terminal Name an... The transactions are listed in a table with checkboxes for selection.

Reviewer Screen

Action Buttons

- Mark Fraud / Suspicious
- Block
- Forward
- Remind
- Response Required
- Notes
- Documentation – investigations, SMRs
- Manual Notification
- Communication
- Travel Notification
- Override
- Watch



Machine Learning

w/ ACI Model Generator

ACI Model Generator

Design, develop, and deploy your own machine learning models



No additional data required, works on existing PRM database

Build models in-house - no data cleansing or calls to external systems

All PRM data is available:

- transactions
- profiles
- demographics
- lists
- marked fraud

Users define features, based on any data present in PRM

Written in standard SQL (as per rules)

Assess value of feature(s) and adjust

Group features to create model

Transforms scattered data and key risk indicators into a single number, 000-999

Model performance presented on UI:

- detection rate
- FPR
- savings

Multiple models if required e.g. different channels

Ready from Day 1

Adaptively recalculates after new fraud marked

Score is attached to transaction, just another data element

Rules can be simple:

- Decline > 800
- Alert > 600

Increased fraud detection rates

Reduction in false positives

Rules are less complex, the score does the “heavy lifting”

ACI Model Generator

Adaptive Machine Learning

- The output of the machine learning model is a risk score between 0 and 999
- The greater the score, the more likely the transaction is fraudulent
- The score can be utilised simply:
 - Alert when risk score greater than 600
 - Decline when risk score greater than 800
- Or in conjunction with other data:
 - Decline when transaction is greater than \$10,000 and risk score greater than 700
 - Alert when transaction is from previously unseen merchant, between 1 and 5 am, and risk score greater than 650

ACI Model Generator

CNP Model Example

- A feature is a mathematical expression that attempts to identify, in this case, fraud
- A machine learning model is built on a set of features
- Let's start with a typical CNP fraud indicator – MCC – for our first feature



MCC

- An example using specific MCCs:

```
CASE WHEN @SD_RETL_SIC_CDE IN ('6552', '6689', '2232') THEN 'H'  
      WHEN @SD_RETL_SIC_CDE IN ('9777', '3225', '9877', '1455') THEN 'M'  
      ELSE 'L'  
END |
```

ACI Model Generator

Weight of Evidence (WoE)

- To analyse the effectiveness of a feature, we split the data into buckets (or bins)
- Calculate the % of fraudulent and genuine transactions in each bucket
- Calculate the WoE of each bucket

$$\text{WoE} = \ln \left(\frac{\% \text{ of non-frauds}}{\% \text{ of frauds}} \right)$$

ACI Model Generator

Weight of Evidence (WoE)

- Back to our MCC feature, which had three buckets – 'H', 'M' and 'L'.

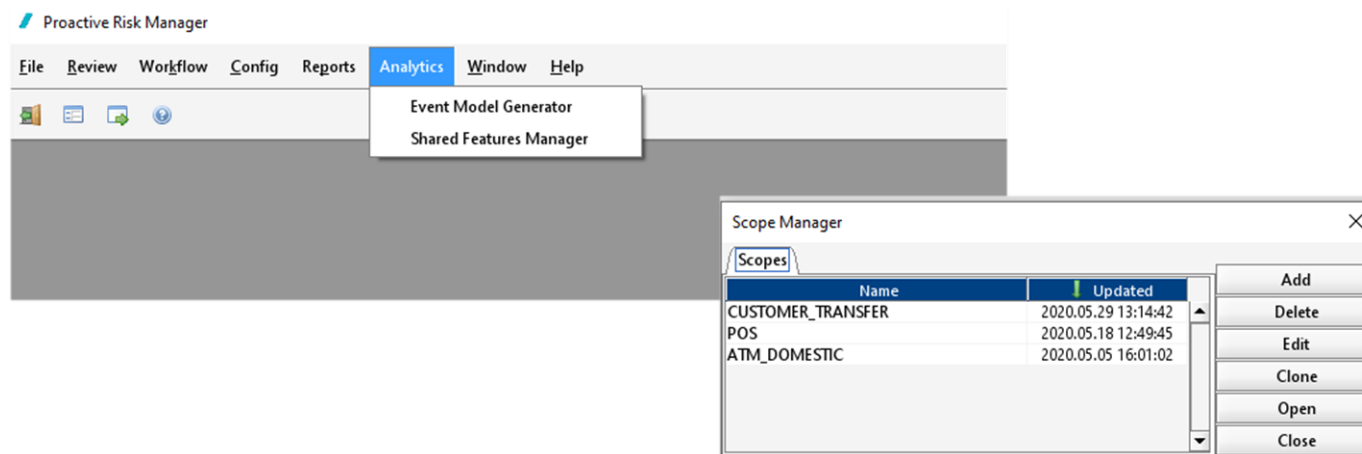
```
CASE WHEN @SD_RET_L_SIC_CDE IN ('6552', '6689', '2232') THEN 'H'
      WHEN @SD_RET_L_SIC_CDE IN ('9777', '3225', '9877', '1455') THEN 'M'
      ELSE 'L'
END |
```

- WoE is calculated for each, with the result being the indicator of how well each bucket:
 - Separates the fraudulent from genuine transactions
 - Compares to the whole set of buckets

ACI Model Generator

Weight of Evidence (WoE)

- Open ACI Model Generator
- Open the Scope Manager
- Select “POS”



ACI Model Generator

Weight of Evidence (WoE)

- Models
- Features
- The MCC feature

Adaptive Event Model Generator => READY!

Models

	Name (ID)	bias	weight	sr%	Rng	Inc	Cost	Benefit	updated	trained
☐	POS_MOD_1	23.278110	16.498440	100	-10	-2	\$30.00	\$370.00	2020.05.18 14:58:13	2020.05.18 15:26:33
☐	POS_MOD_2	38.360801	20.754391	100	-15	-5	\$30.00	\$370.00	2020.06.02 14:14:24	2020.06.02 14:15:02

Features

	Name (ID)	Type	RC#	NT-IV	TG-IV	IV	sr%	sk%	SKC	Eval	Rng	Inc	Offs	Expr	updated
☒	DaysSinceLastFraud	Local	9999	3.647788	-0.873926	4.521714	100	100	SD_PAN	3-STRONG	-15	-5	1	-3	2020.05.18 14:45:33
☒	MCC_Habituality	Local	9999	0.340729	-2.695616	3.036345	100	100	SD_PAN	3-STRONG	-15	-5	1	-3	2020.05.18 14:32:52
☒	DaysSinceLastFraud	Shared Community	397	0.852158	-0.438309	1.290467	100	100	SD_PAN	2-MEDIUM	-90	-1	1	-3	2020.06.12 14:30:45
☒	DaysSinceLastFraud	Shared Local	397	0.852158	-0.438309	1.290467	100	100	SD_PAN	2-MEDIUM	-20	-5	1	-3	2020.06.12 14:36:43
☒	VelocityAmount	Shared Community	815	0.261069	-0.665128	0.926198	100	100	SD_PAN	1-WEAK	-35	-3	1	-3	2020.06.12 14:23:35
☒	BenfordLaw	Shared Community	502	0.462770	-0.195789	0.658558	100	100	SD_PAN	1-WEAK	-90	-1	1	-3	2020.05.27 14:31:09
☒	BenFordLaw	Local	9999	0.163518	-0.473747	0.637266	100	100	SD_PAN	1-WEAK	-15	-5	1	-3	2020.05.18 14:11:05
☒	Retailer	Local	9999	0.470234	-0.050438	0.520673	100	100	SD_PAN	1-WEAK	-15	-5	1	-3	2020.05.18 14:31:01
☒	TerminalName	Local	9999	0.407143	-0.089490	0.496633	100	100	SD_PAN	1-WEAK	-15	-5	1	-3	2020.05.18 13:24:14
☒	TerminalZip	Shared Community	919	0.230569	-0.179875	0.410445	100	100	SD_PAN	1-WEAK	-35	-5	1	-3	2020.06.03 11:22:56
☒	VeloAmt	Local	9999	0.113472	-0.293238	0.406710	100	100	SD_PAN	1-WEAK	-15	-5	1	-3	2020.05.18 13:56:21
☒	VelocityAmount	Shared Local	815	0.108338	-0.262302	0.370640	100	100	SD_PAN	1-WEAK	-20	-5	1	-3	2020.06.12 14:39:11
☒	VelocityTxnCnt	Shared Community	574	0.079214	-0.117077	0.196291	100	100	SD_PAN	1-WEAK	-44	-4	1	-3	2020.06.02 11:47:56
☒	MCC_AmountGrp	Shared Community	81	0.091704	-0.090503	0.182206	100	100	SD_PAN	1-WEAK	-35	-3	1	-3	2020.06.12 14:25:30
☒	MCC_AmountGrp	Shared Local	81	0.067393	-0.076683	0.144076	100	100	SD_PAN	1-WEAK	-15	-5	1	-3	2020.06.16 17:58:29
☒	VeloCnt	Local	9999	0.055048	-0.078686	0.133734	100	100	SD_PAN	1-WEAK	-15	-5	1	-3	2020.05.18 13:27:12
☒	FraudulentTerminal	Local	9999	0.064066	-0.067264	0.131330	100	100	SD_PAN	1-WEAK	-15	-5	1	-3	2020.05.18 15:25:34
☒	BenfordLaw	Shared Local	502	0.068651	-0.047919	0.116569	100	100	SD_PAN	1-WEAK	-25	-4	1	-3	2020.05.29 08:02:56
☒	MCCAmtCtg	Local	9999	0.049361	-0.053911	0.103273	100	100	SD_PAN	1-WEAK	-15	-5	1	-3	2020.05.18 13:27:55
☒	MCC	Local	9999	0.040356	-0.043654	0.084010	100	100	SD_PAN	0-USELESS	-10	-2	1	-3	2020.05.19 09:38:13
☒	TxnAmt	Local	9999	0.038657	-0.041529	0.080186	100	100	SD_PAN	0-USELESS	-10	-2	1	-3	2020.05.19 09:39:49
☒	VelocityTxnCnt	Shared Local	574	0.023305	-0.033229	0.056535	100	100	SD_PAN	0-USELESS	-44	-4	1	-3	2020.06.02 11:50:43
☒	TestFeature	Shared Local	334	0.018823	-0.019841	0.038663	100	100	SD_PAN	0-USELESS	-45	-5	1	-3	2020.06.19 07:04:57
☒	CardHolderAge	Local	9999	0.015115	-0.011735	0.026850	100	100	SD_PAN	0-USELESS	-15	-5	1	-3	2020.05.18 13:00:03
☒	TerminalRiskLevel	Shared Community	579	0.009271	-0.009871	0.019142	100	100	SD_PAN	0-USELESS	-35	-3	1	-3	2020.06.12 14:29:14
☒	TerminalRiskLevel	Shared Local	579	0.007279	-0.007081	0.014360	100	100	SD_PAN	0-USELESS	-20	-5	1	-3	2020.06.18 08:43:21
☒	POSEntryMode	Local	9999	0.000646	-0.000620	0.001267	100	100	SD_PAN	0-USELESS	-15	-5	1	-3	2020.05.18 13:56:47
☒	TerminalZip	Shared Local	919	0.000000	0.000000	0.000000	100	100	SD_PAN	0-USELESS	-35	-5	1	-3	2020.06.12 14:33:18
☒	TestFeature	Shared Community	334	0.000000	0.000000	0.000000	100	100	SD_PAN	???	-90	-1	1	-3	2020.06.19 06:49:12
☒	TimeOfDay	Local	9999	0.000000	0.000000	0.000000	100	100	SD_PAN	0-USELESS	-15	-5	1	-3	2020.05.18 14:47:53

Model operations

Add

Delete

Compare models

Link features

Train/Normalize

Benchmark

Deploy NRT

Deploy RT

Feature operations

Change filters

Refresh

Add

Delete

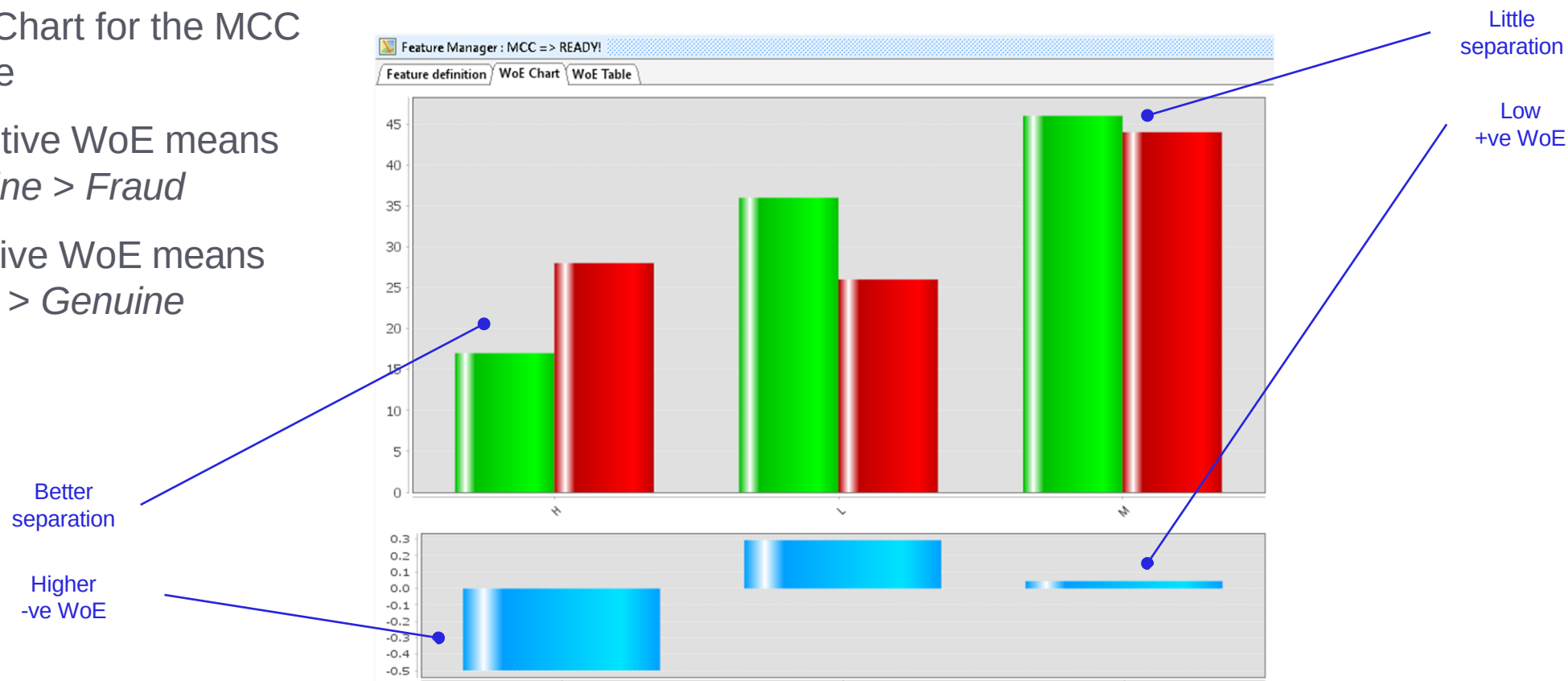
Purge datasets

System cleanup

ACI Model Generator

Weight of Evidence (WoE)

- WoE Chart for the MCC feature
- A positive WoE means *Genuine > Fraud*
- Negative WoE means *Fraud > Genuine*



ACI Model Generator

Simple CNP Model

- Add more CNP fraud indicators:



MCC



Country



Velocity

- Consider some additional data elements:



Time
of Day



Amount /
Velocity



Multiple
Countries

ACI Model Generator

Simple CNP Model

- We can then use PRM's behavioural profiles for deeper analysis:



MCC
Frequency



Retailer
Frequency

ACI Model Generator

Simple CNP Model

- The purpose of a feature is to:
 - Separate events into buckets so the *WoE* of those categories can be calculated
 - Calculate the *information value* (iv) of the whole set of categories
- The *iv* indicates how predictive of fraud the feature is:

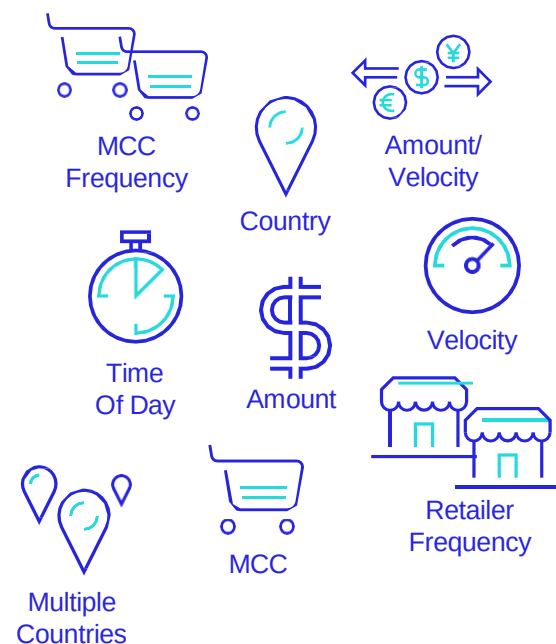
Information Value	Predictive Power
Less than 0.02	Useless
0.02 to 0.1	Weak
0.1 to 0.3	Medium
0.3 to 0.5	Strong
> 0.5	Excellent

- The lower the *iv* of a feature, the less that feature contributes to the score

ACI Model Generator

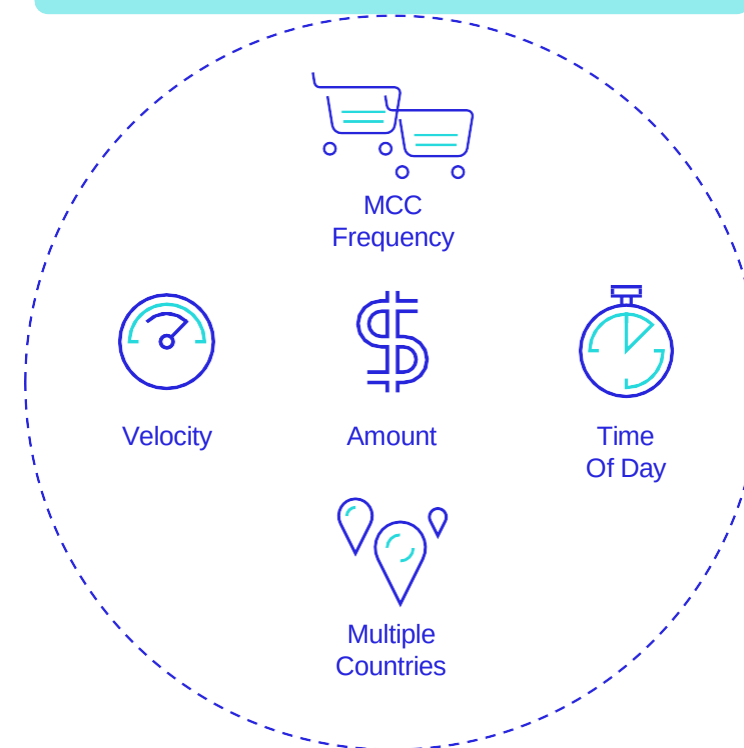
Simple CNP Model

Feature Candidates



- 1) Calculate WoE
- 2) Evaluate IV
- 3) Keep the predictive features
- 4) Create the model
- 5) Test and evaluate model

Predictive Model



ACI Model Generator

Evaluating the model

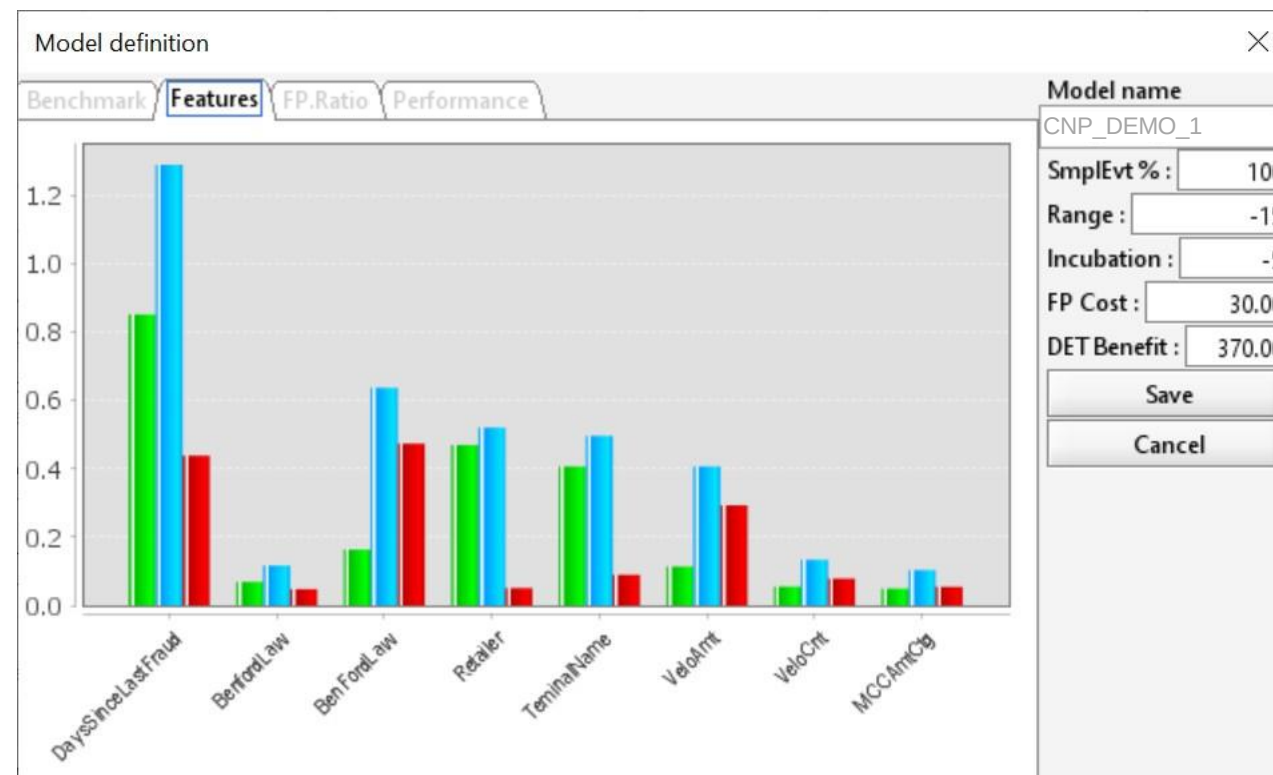
- Score range
- Total events
- Genuine transactions
- Fraud transactions
- Accumulated transactions
- Accumulated frauds
- % fraud detected at this score
- False positive ratio
- Value (using variables “FP Cost” and “Det Benefit”)

Model definition									Model name	
Benchmark	Features	FP.Ratio	Performance						CNP_DEMO_1	
Range	Thits	NTh	TGh	NTa	TGa	DET	FPR	Value	SmplEvt % :	100
950+	16	1	15	1	15	4.3%	0.1:1	5520	Range :	-15
900+	229	154	75	155	90	25.9%	1.7:1	28650	Incubation :	-5
850+	2,777	2,570	207	2,725	297	85.6%	9.2:1	28140	FP Cost :	30.00
800+	1,307	1,276	31	4,001	328	94.5%	12.2:1	1330	DET Benefit :	370.00
750+	184	184	0	4,185	328	94.5%	12.8:1	-4190	Save	
700+	5	5	0	4,190	328	94.5%	12.8:1	-4340	Cancel	
650+	0	0	0	4,190	328	94.5%	12.8:1	-4340		
600+	0	0	0	4,190	328	94.5%	12.8:1	-4340		
550+	0	0	0	4,190	328	94.5%	12.8:1	-4340		
500+	0	0	0	4,190	328	94.5%	12.8:1	-4340		
450+	0	0	0	4,190	328	94.5%	12.8:1	-4340		
400+	32	26	6	4,216	334	96.3%	12.6:1	-2900		
350+	176	167	9	4,383	343	98.8%	12.8:1	-4580		
300+	57	53	4	4,436	347	100.0%	12.8:1	-4690		
250+	685	685	0	5,121	347	100.0%	14.8:1	-25240		
200+	191	191	0	5,312	347	100.0%	15.3:1	-30970		
150+	83	83	0	5,395	347	100.0%	15.5:1	-33460		
100+	3,272	3,272	0	8,667	347	100.0%	25.0:1	-131620		
050+	1,011	1,011	0	9,678	347	100.0%	27.9:1	-161950		

ACI Model Generator

Evaluating the model

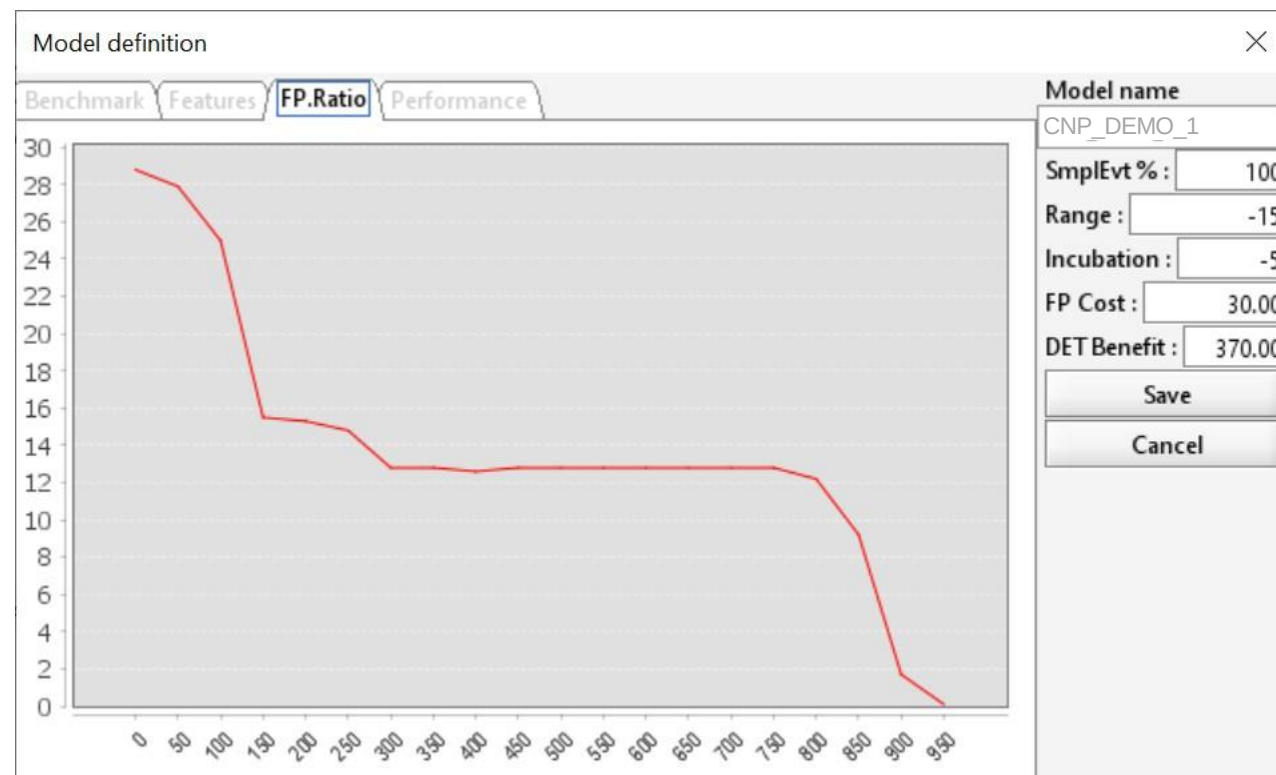
- Each feature displayed graphically for quick assessment
- The higher the column, the more it contributes to the model



ACI Model Generator

Evaluating the model

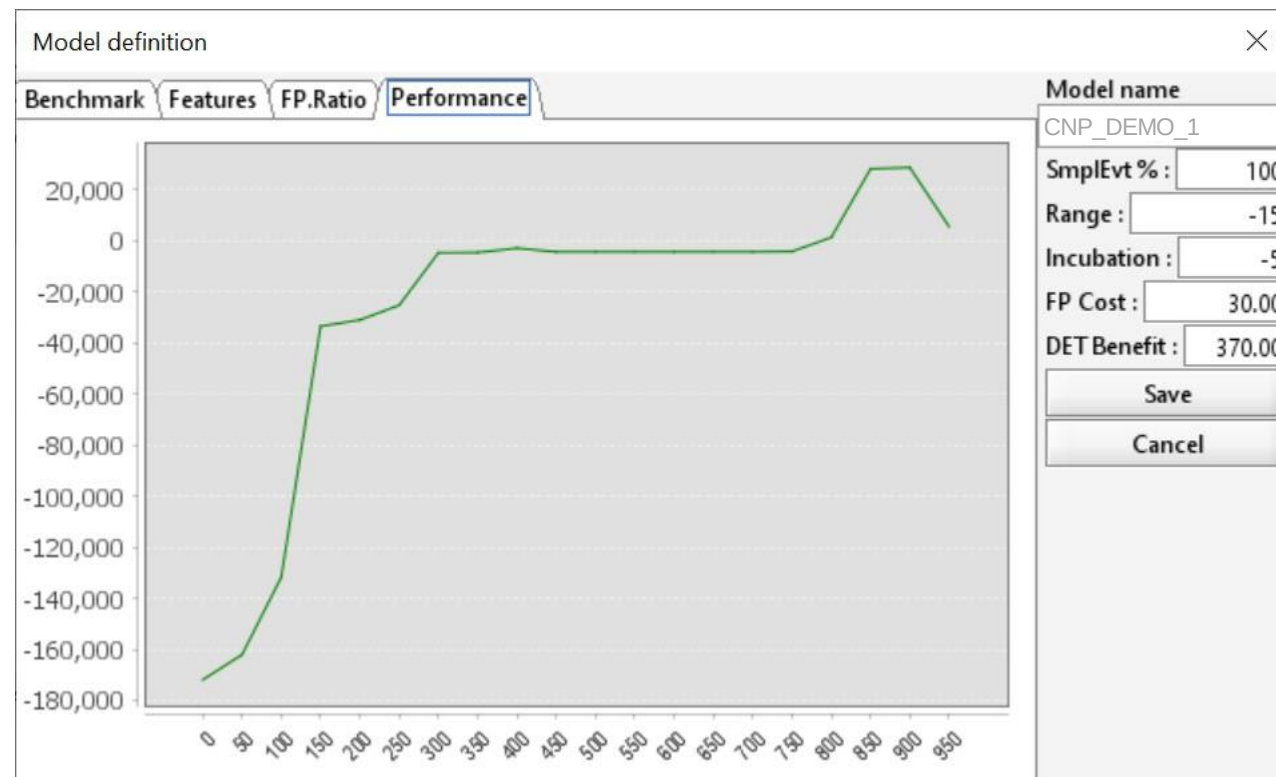
- “Y” = Number of alerts
- ‘X’ = Score



ACI Model Generator

Evaluating the model

- “Y” = \$
- “X” = Score



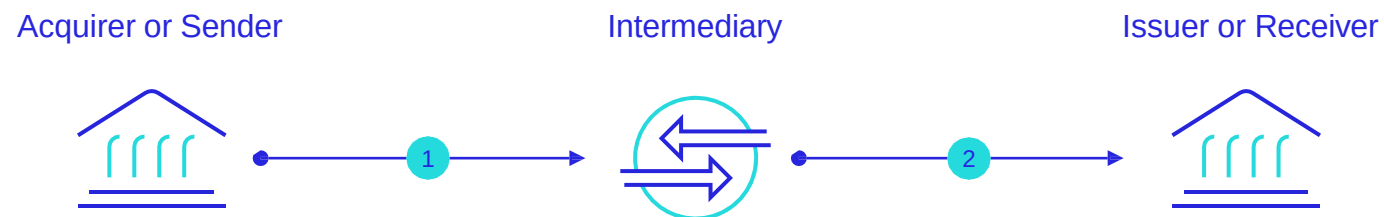
Intermediaries

Fraud Protection on Behalf of Members

Processing Today

Without fraud protection

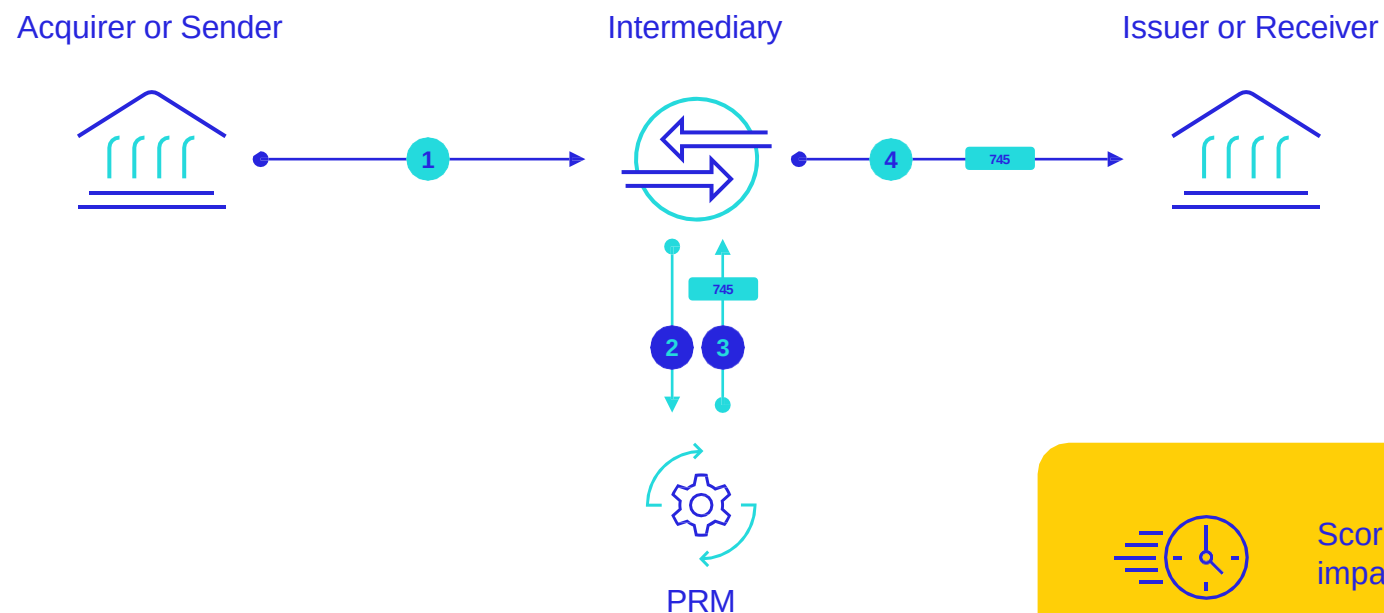
‘Standard’ processing has the Sender/Acquirer transaction sent to Intermediary
– if rudimentary checks are passed, the transaction is sent to the Receiver/Issuer



Add Risk Scoring

With PRM's ACI Model Generator

Transaction is sent to intermediary, but risk scored in real-time before being sent to Issuer/Receiver



Scoring performed in milliseconds, little impact to SLA



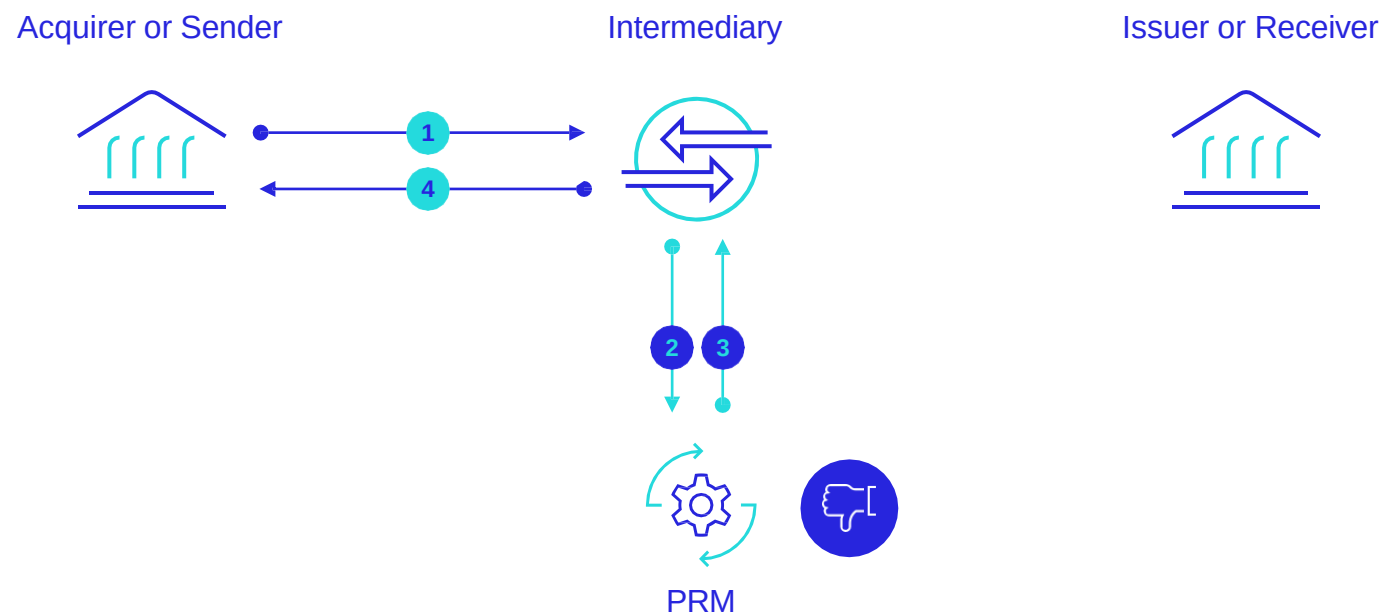
Risk score added as element – issuer/receiver can utilise on its own or enhance existing solution

Add Real-Time Rules

With PRM's Rules Engine + ACI Model Generator

DECLINE

Transaction is sent to intermediary, then assessed in real-time, potentially declining on behalf of member

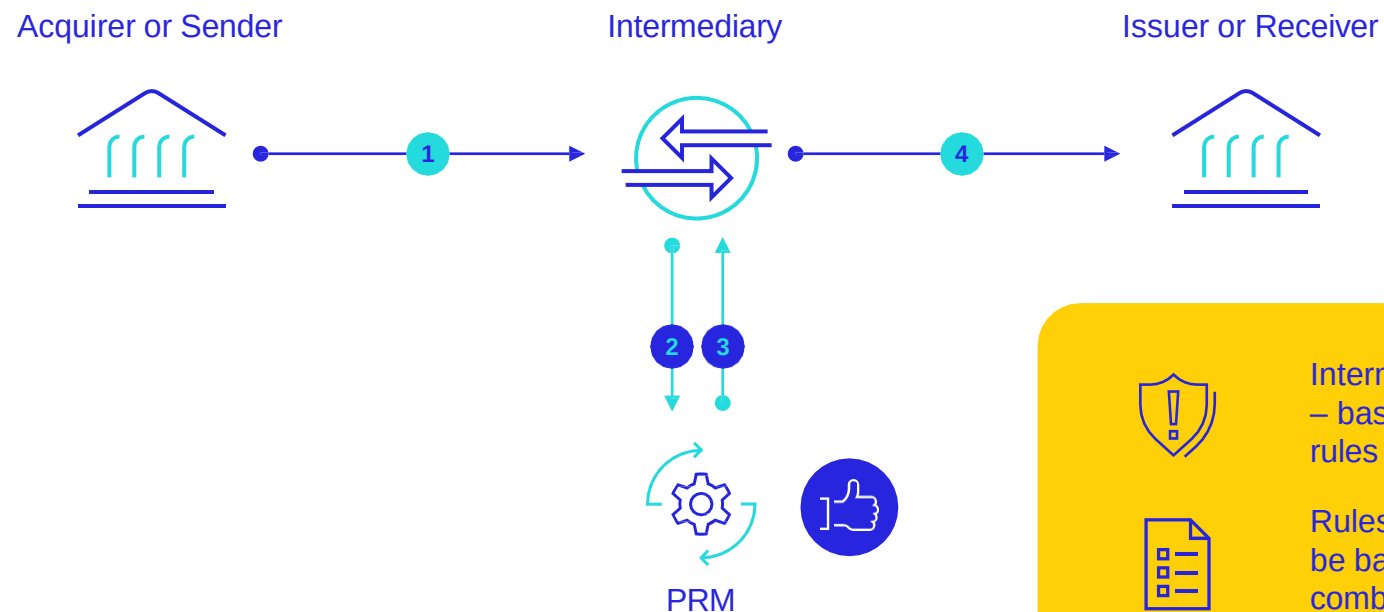


Add Real-Time Rules

With PRM's Rules Engine + ACI Model Generator

APPROVE

Transaction is sent to intermediary, then assessed in real-time, potentially declining on behalf of member



Intermediary can decline on members' behalf – based on SLAs, requirements, requested rules



Rules provide the construct to decline – can be based on SQL, risk score, or a combination



Scoring performed in milliseconds, little impact to SLA

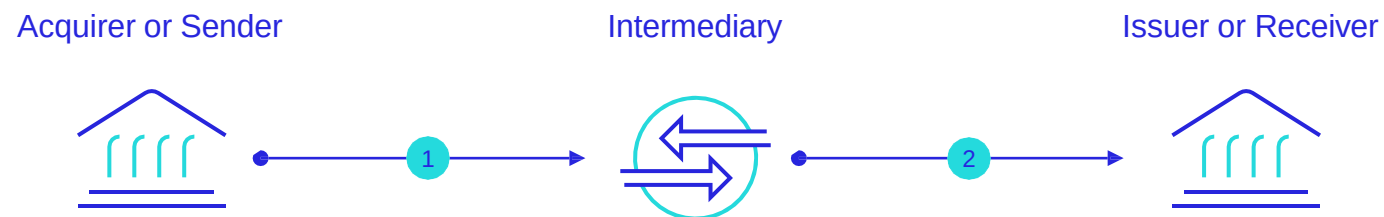
Intermediaries

Multi-Tenancy

Processing Today

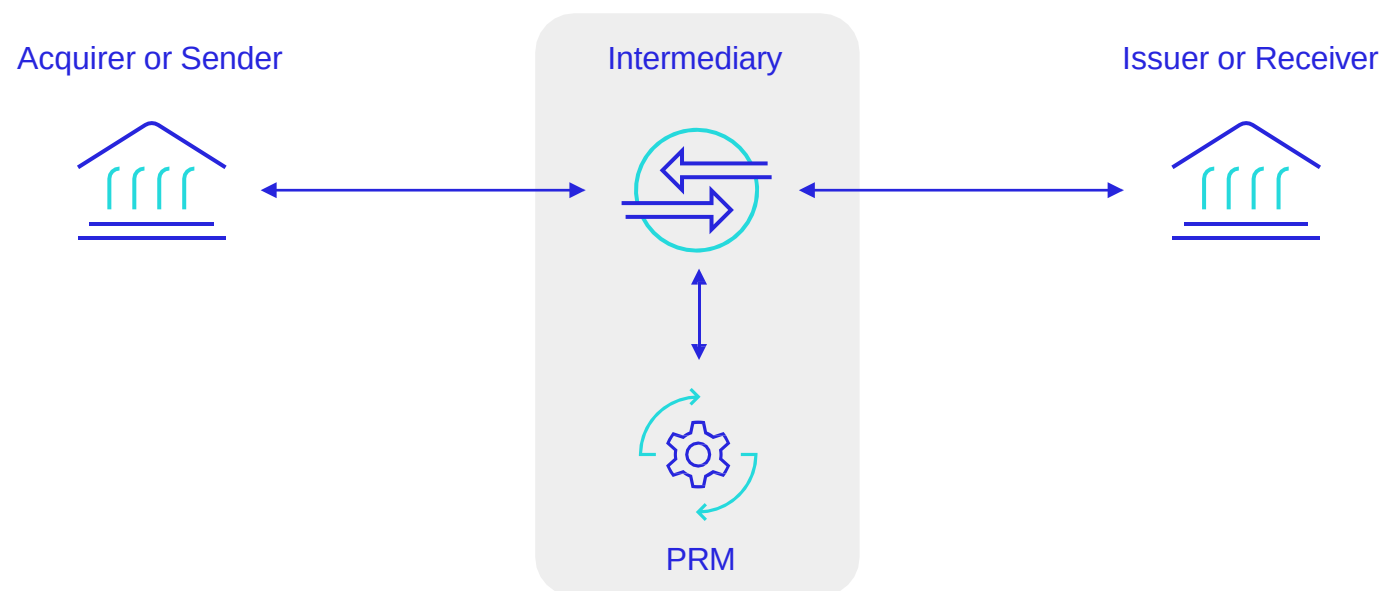
Without fraud protection

‘Standard’ processing has the Sender/Acquirer transaction sent to Intermediary
– if rudimentary checks are passed, the transaction is sent to the Receiver/Issuer



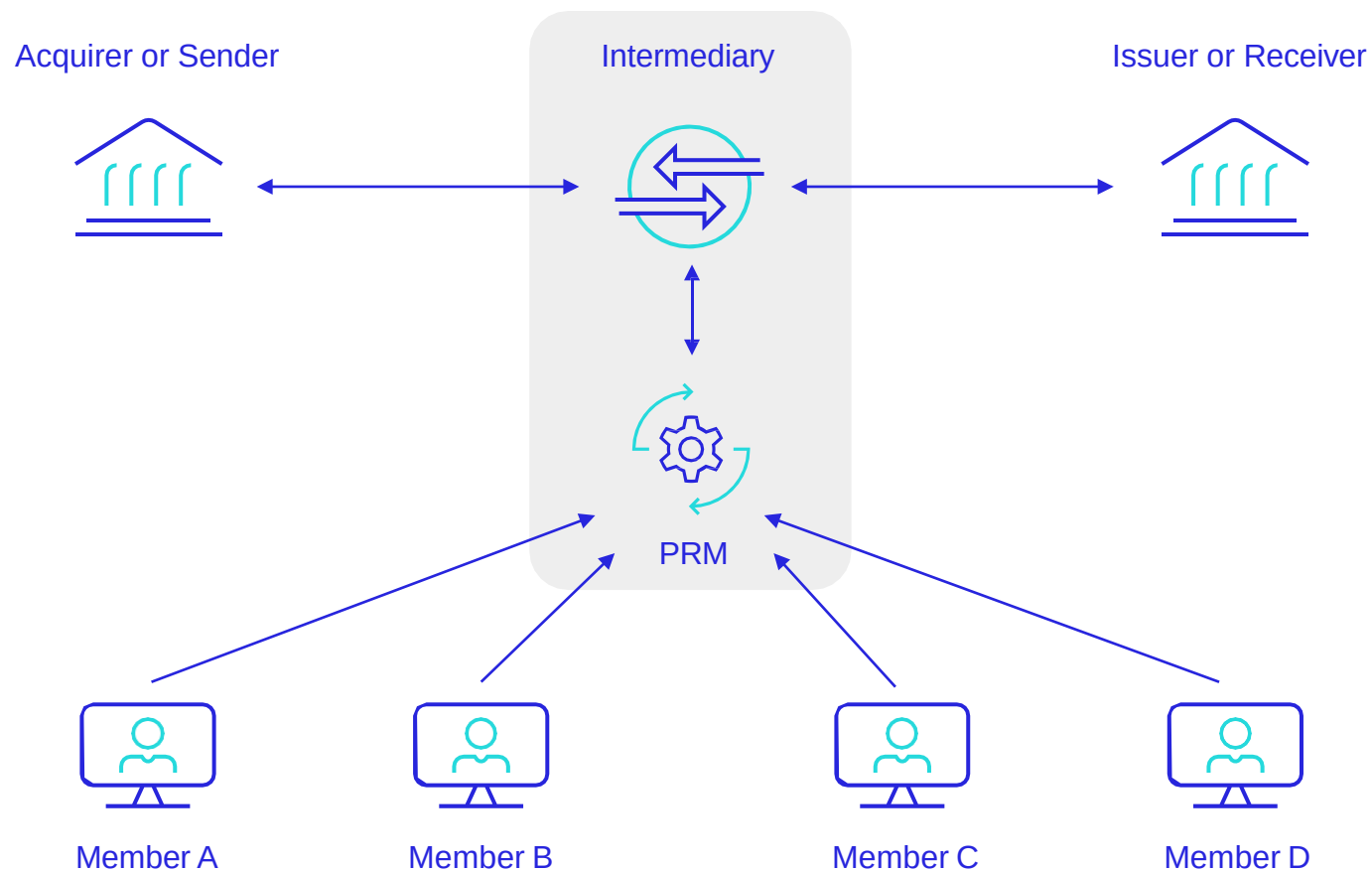
How It Works

Multi-tenancy fraud detection



How It Works

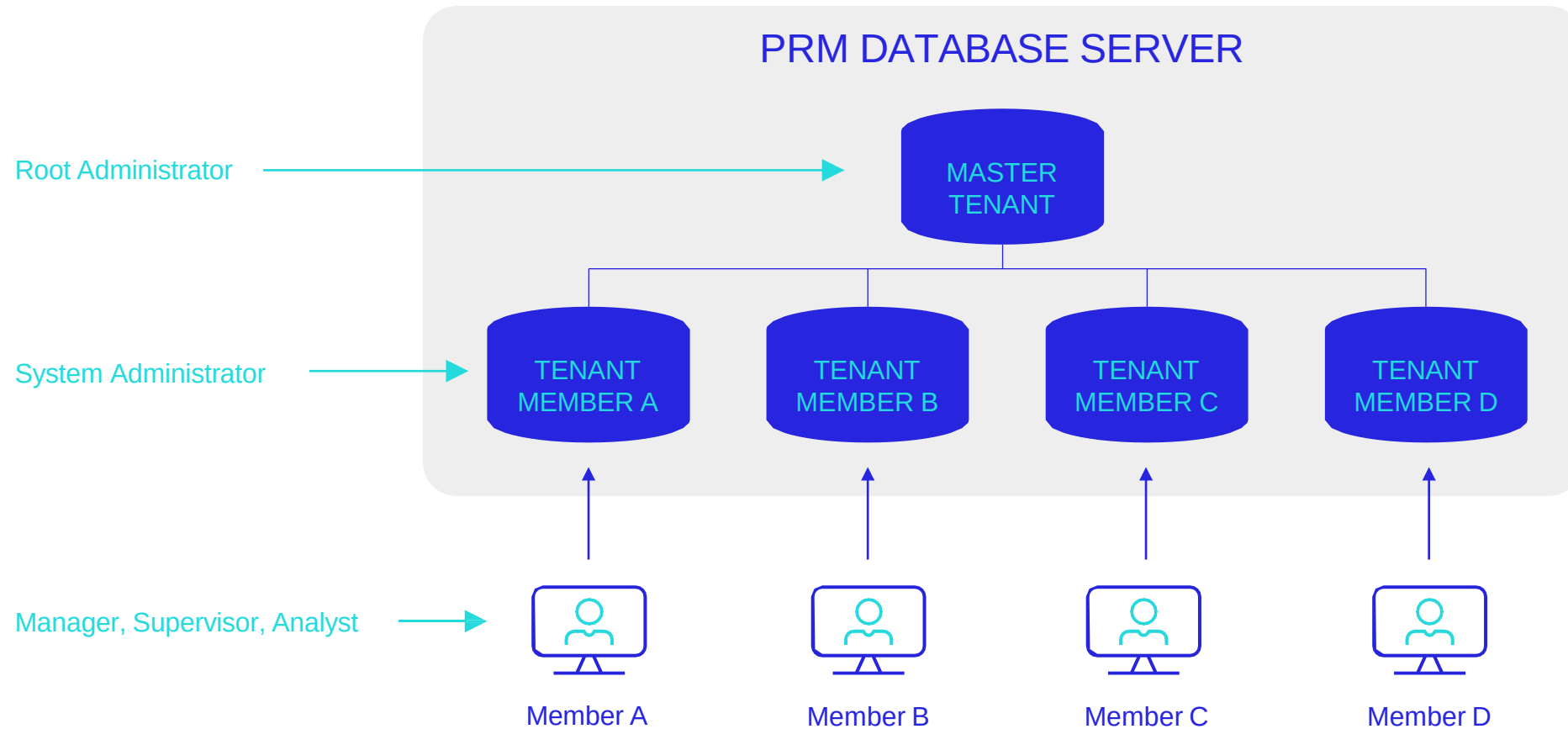
Multi-tenancy fraud detection



- Intermediary feeds all transactions to a single instance of PRM
- Each member has their own database
- One member cannot see the data of any other member
- Members write their own rules, configure their own workflows, work their own alerts

How It Works

Inside the database



Intermediaries

Multi-Tenancy w/ Network Intelligence

Network Intelligence

For Intermediaries & Multi-tenancy



KNOWLEDGE SHARING

- Members subscribe to the community, calculating common signals which are shared 'up' to the central body who publishes the best performers, which are pulled 'down' by the members.



100% ANONYMOUS

- No PII is sent – only the feature and its results (signal) is passed to the central body, who then assesses against all signals and publishes the best performer.



ENHANCED DATA FEEDS MACHINE LEARNING

- Members now have the best performing features available for their models which may include consideration of fraud not yet seen by themselves.

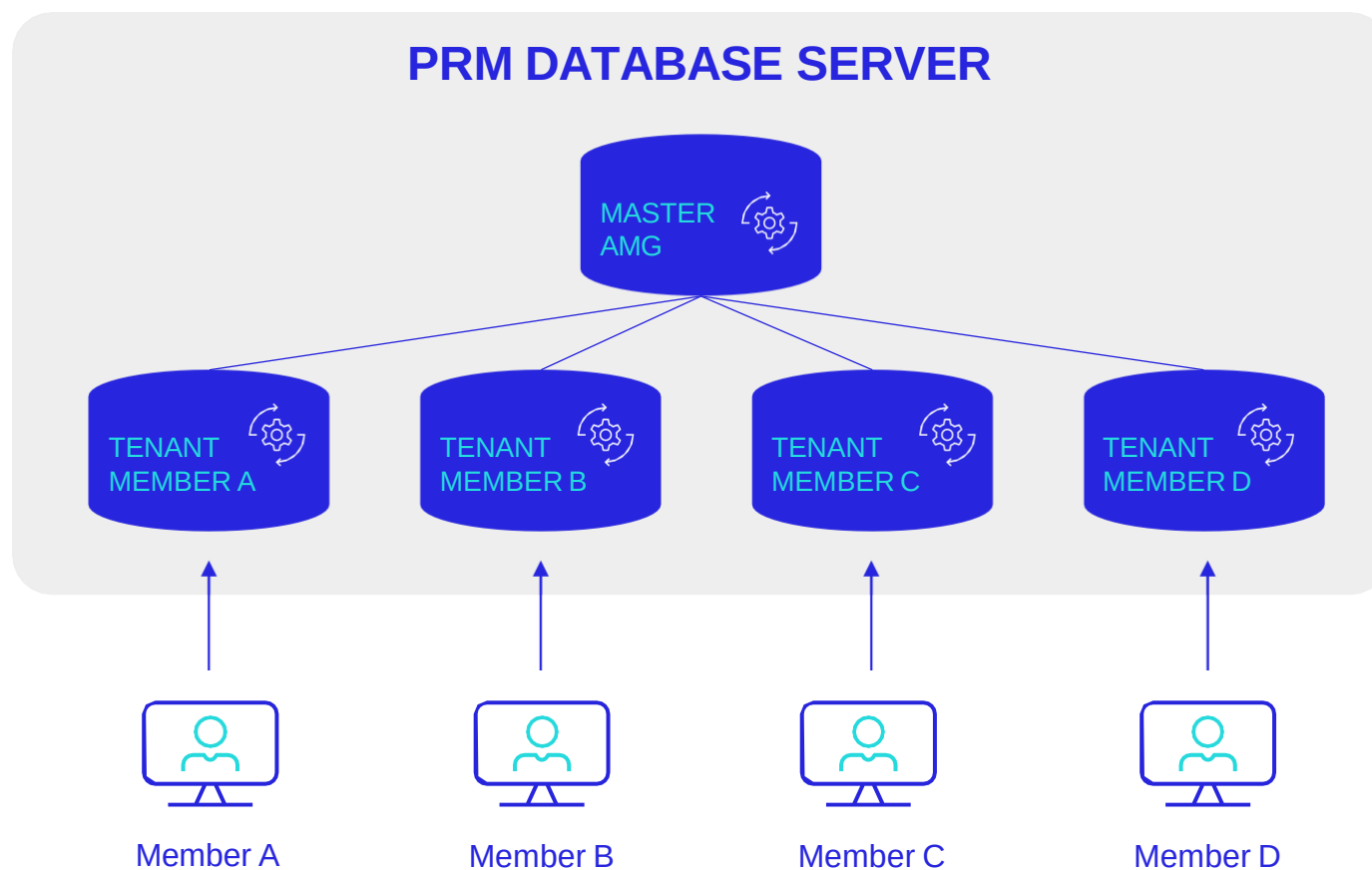


COMPLEMENTS EXISTING FRAUD STRATEGIES

- Members can assess the shared feature against their own data and add to their model(s), continue with their 'local' feature, or create a hybrid.

How It Works

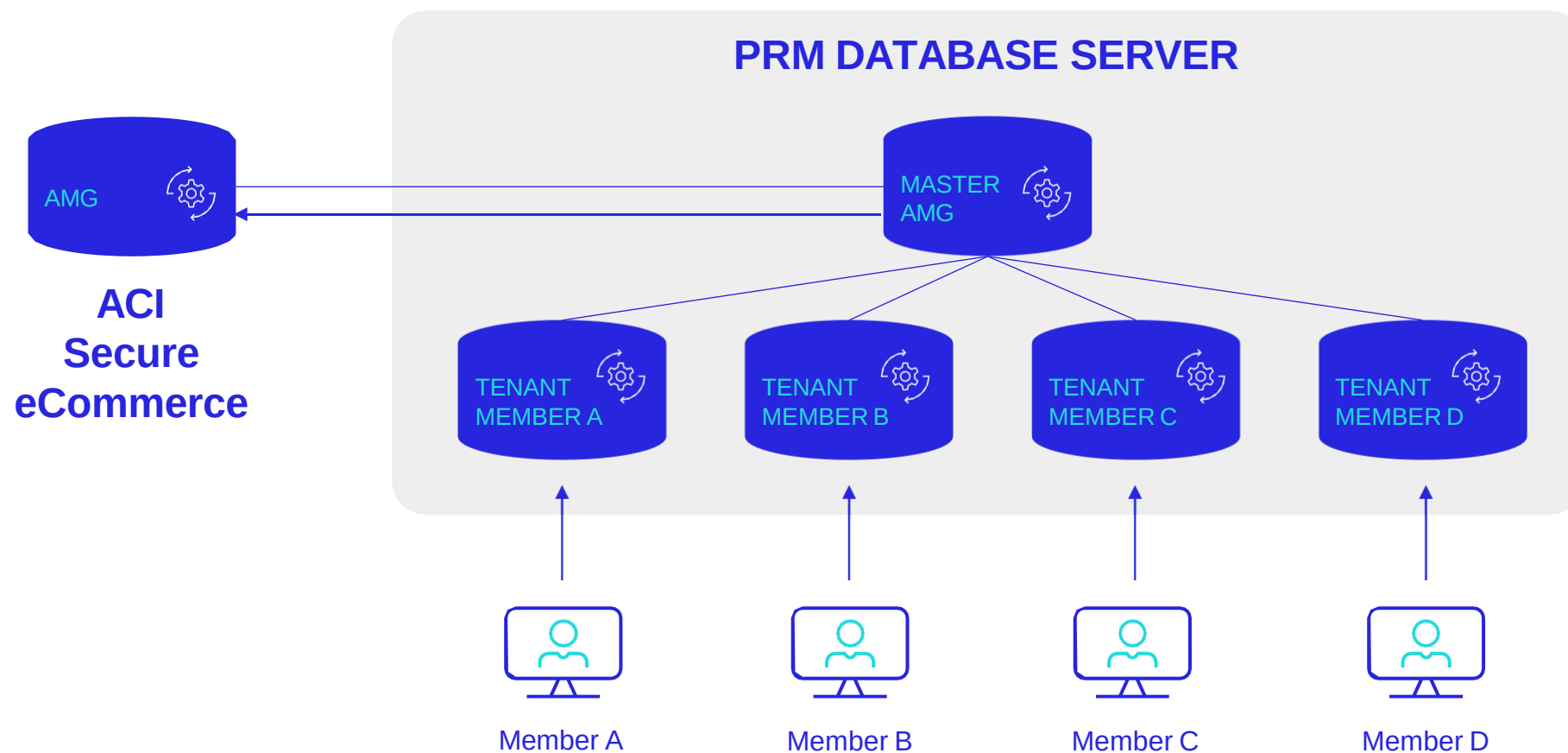
Multi-tenancy w/ Network Intelligence



- Master publishes best practice signals which are loaded by members
- Members assess feature against their own data
- Members can use feature, contributing to a model; ignore; or create hybrid
- Members return results to Master to reassess and determine best performers

How It Works

Multi-tenancy w/ Network Intelligence & ACI



- ACI eCommerce (Global CNP SaaS for online merchants) feeds signals to Master
- Master assesses, makes available to members
- Members assess against their own data, add to model(s) if valuable
- Members return results to Master
- Master (optionally) returns results to ACI

Network Intelligence

From intermediary perspective

- Intermediary could be:
 - Real-Time Payments CI (PayNet, NPP, BI-FAST)
 - Scheme (MasterCard, Visa, JCB, eftpos)
 - Processor (3rd party, PSP, intermediary)
 - Payment gateway
- The process:
 - Intermediary publishes best practice signals
 - Members download and assess against their data, then
 - Add to model
 - Create hybrid
 - Ignore
 - Members publish signal back to intermediary

Shared Community
Cannot be changed. Weight of Evidence based on intermediary dataset

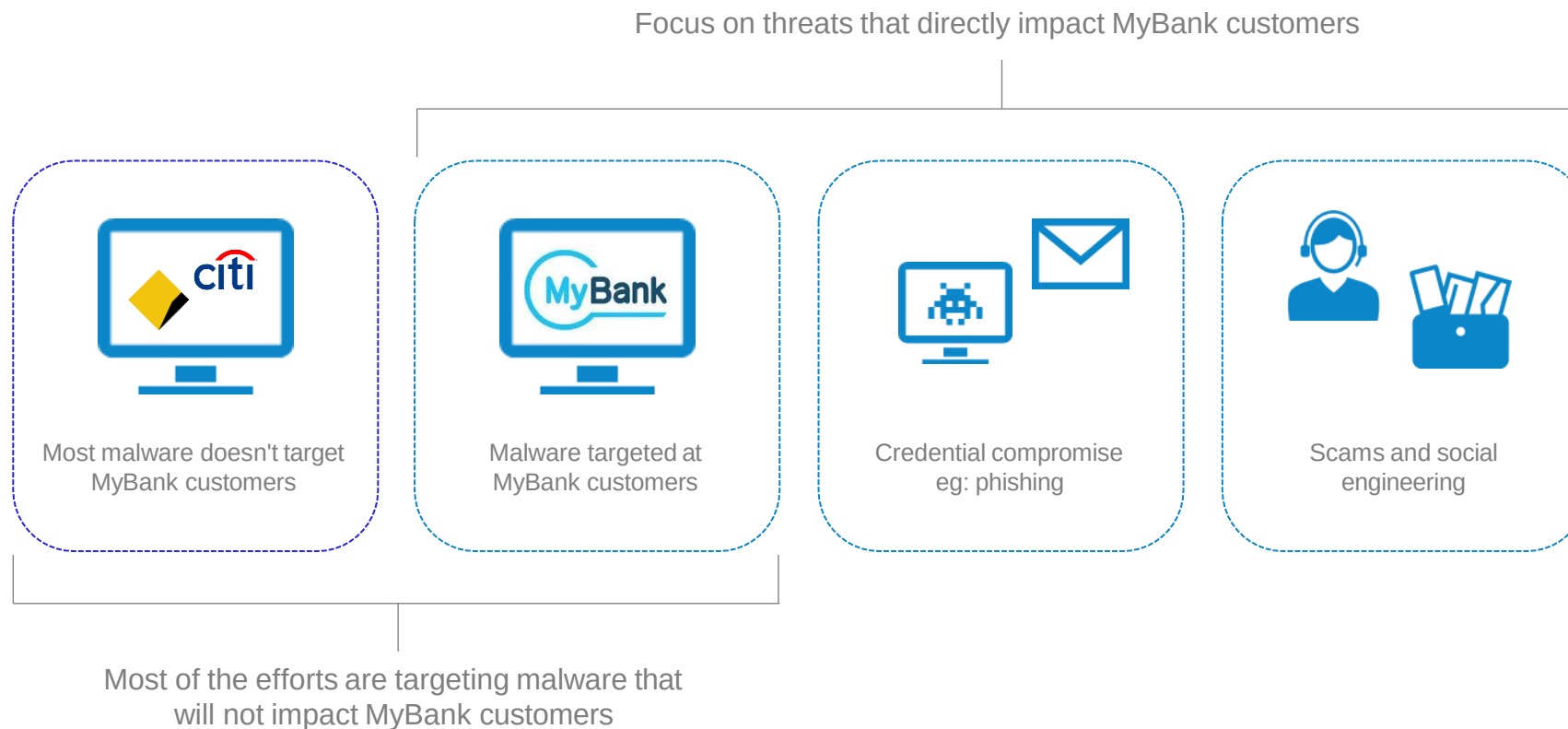
Shared Local
Weight of Evidence based on your dataset

Features									
☐	Name (ID)	Type	%C#	NT-IV	TG-IV	IV	sr%	sk%	
☐	DaysSinceLastFraud	Local	9999	3.647788	-0.873926	4.521714	100	100	
☐	MCC_Habituality	Local	9999	0.340729	-2.695616	3.036345	100	100	
☐	DaysSinceLastFraud	Shared Community	397	0.852158	-0.438309	1.290467	100	100	
☐	DaysSinceLastFraud	Shared Local	397	0.852158	-0.438309	1.290467	100	100	
☐	VelocityAmount	Shared Community	815	0.261069	-0.665128	0.926198	100	100	
☐	BenfordLaw	Shared Community	502	0.462770	-0.195789	0.658558	100	100	
☐	BenfordLaw	Local	9999	0.163518	-0.473747	0.637266	100	100	
☐	Retailer	Local	9999	0.470234	-0.050438	0.520673	100	100	
☐	TerminalName	Local	9999	0.407143	-0.089490	0.496633	100	100	
☐	TerminalZip	Shared Community	919	0.230569	-0.179875	0.410445	100	100	
☐	VeloAmt	Local	9999	0.113472	-0.293238	0.406710	100	100	
☐	VelocityAmount	Shared Local	815	0.108338	-0.262302	0.370640	100	100	
☐	VelocityTxnCnt	Shared Community	574	0.079214	-0.117077	0.196291	100	100	
☐	MCC_AmountGrp	Shared Community	81	0.091704	-0.090503	0.182206	100	100	
☐	MCC_AmountGrp	Shared Local	81	0.067393	-0.076683	0.144076	100	100	
☐	VeloCnt	Local	9999	0.055048	-0.078686	0.133734	100	100	
☐	FraudulentTerminal	Local	9999	0.064066	-0.067264	0.131330	100	100	
☐	BenfordLaw	Shared Local	502	0.068651	-0.047919	0.116569	100	100	
☐	MCCAmtCtg	Local	9999	0.049361	-0.053911	0.103273	100	100	
☐	MCC	Local	9999	0.040356	-0.043654	0.084010	100	100	
☐	TxnAmt	Local	9999	0.038657	-0.041529	0.080186	100	100	

Machine Learning

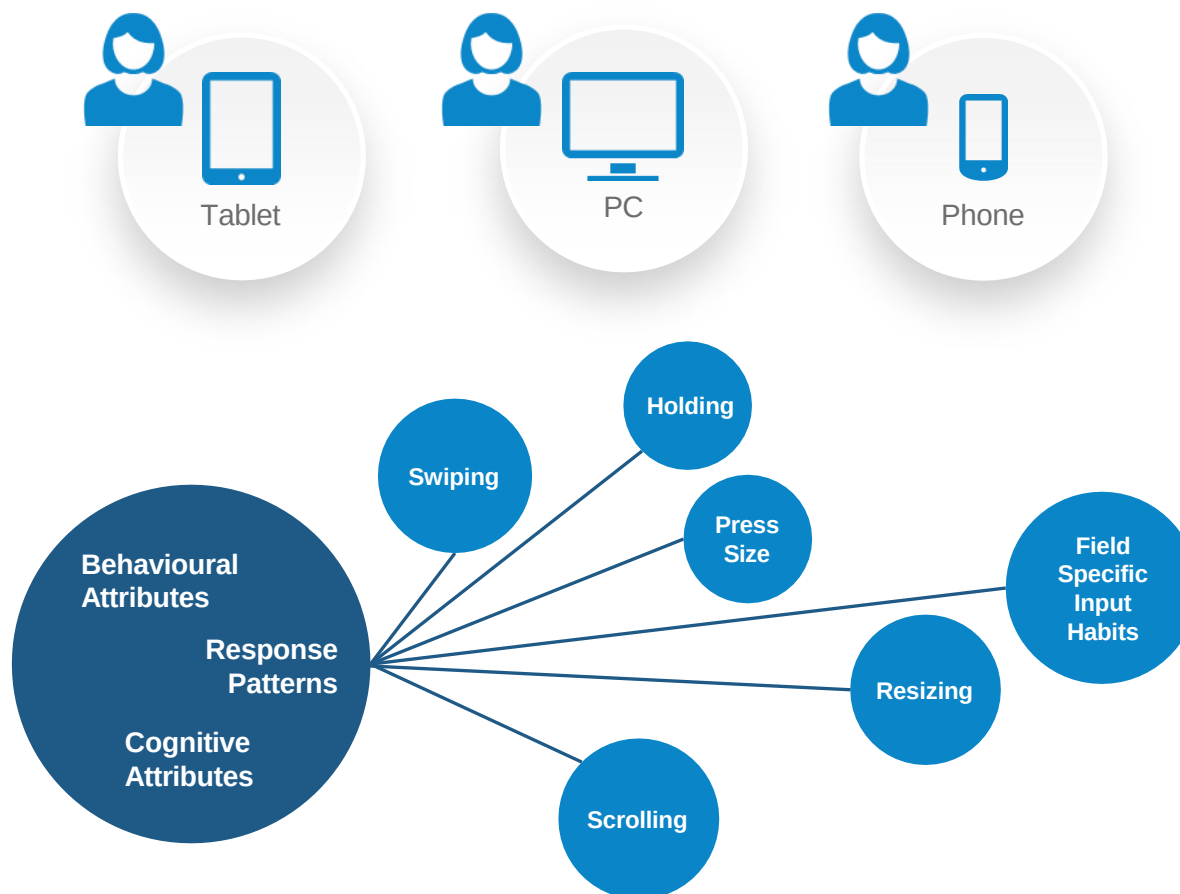
w/ BioCatch

Digital Banking



Behavioural Biometrics

Human-Device Interactions



Interaction Parameters

1000s of Data Elements

Behavioural

How does the user interact with the device?

Cognitive

How does the user interact with the specific application?

Response

Assesses subtle UI changes

Attributes

2000+ Data Elements

Behavioural

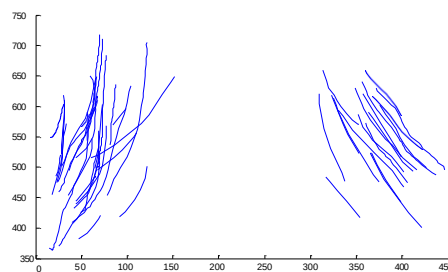
- How the user holds a mobile device, touch, and interact with it
- Scroll or swipe?
- Right finger or left?
- Finger or thumb?
- Specific thumb motion
- Measures mouse motion parameters eg: speed, acceleration, & curvature

Cognitive

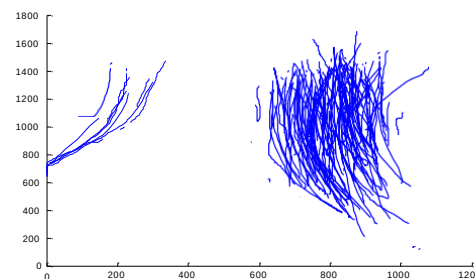
- How the user interacts with specific application
- Moving between fields – tab or mouse?
- Submit button – click with mouse or ENTER?
- Type correction – delete or backspace?
- Hold delete or press multiple times?
- How are date fields entered?
- Auto-complete? How?

Behavioural Biometrics

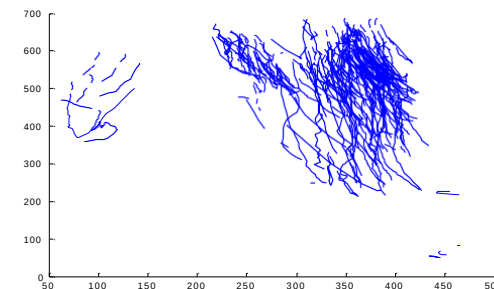
Tablet Scrolling



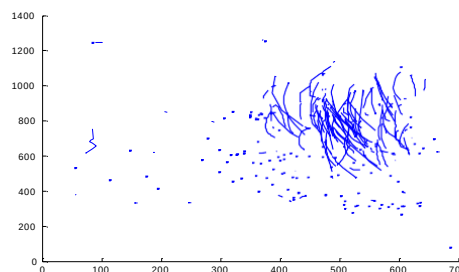
User scrolls with left & right finger



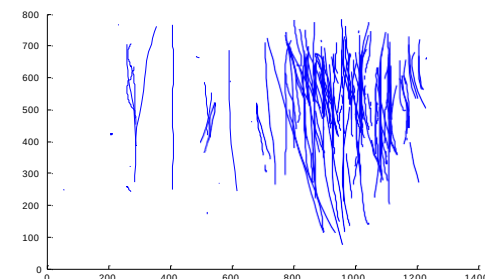
User scrolls mostly with right finger



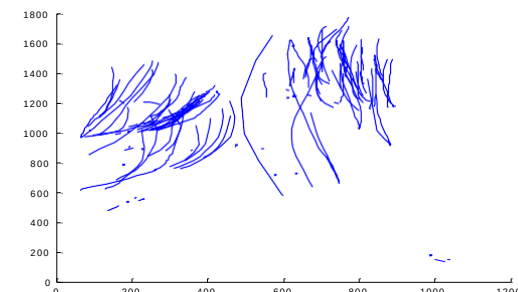
Same, but a shaky hand



Damaged screen, scrolling sticks



Vertical scroll



Right finger, left thumb

Behavioural Biometrics

In Practice



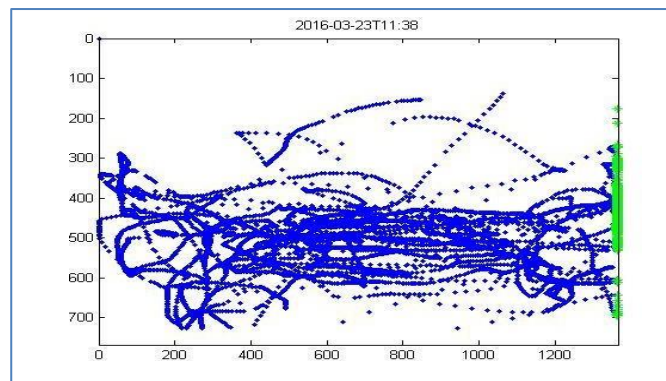
Usual location



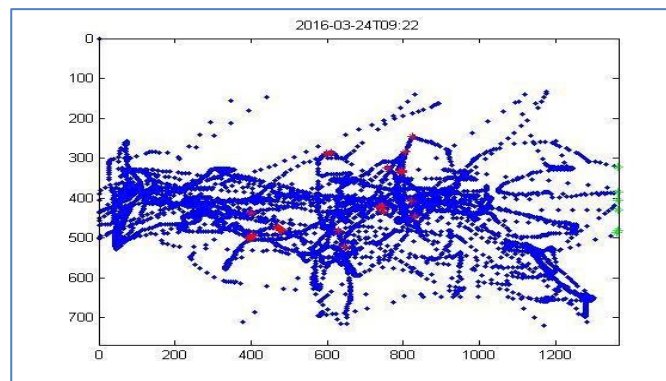
Trusted device



Previous SWIFT transfers



Genuine User
Prefers scroll bar



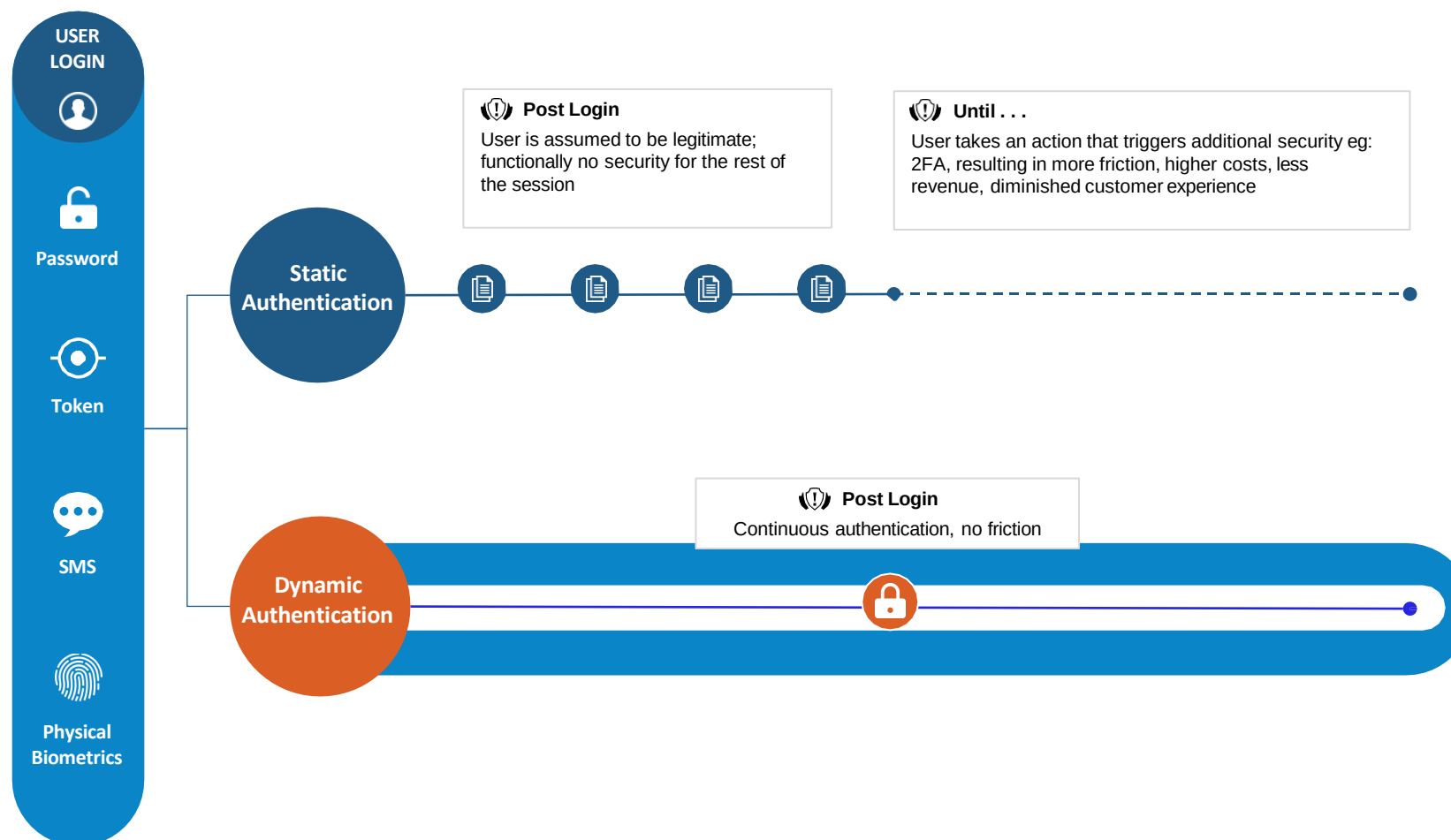
Fraudster
Prefers mouse wheel

Device Interactions

Behaviour	Fraudster	Genuine
Toggling Between Applications	Multiple	Never
Country Selection	Keyboard	Mouse
Currency Selection	Keyboard	Keyboard
Reference Text	Pasted	Typed

Behavioural Biometrics

Real-Time, Continuous, Frictionless



Behavioural Biometrics

Driven By Machine Learning



Analyse large datasets to deduce fraud patterns & develop detection parameters



Algorithms are designed, using an array of parameters



Additional datasets analysed to test effectiveness of proposed model(s)



Parallel testing of models



Deployment



Models periodically calibrated with confirmed fraud files

Machine Learning

Targeted Models



- Risk score based on the behavioural biometrics
- "Yes/No" indicators:
 - Malware, RAT, Bot?
- Data fields the customer requires:
 - IP, Device ID, location
 - Any of the behavioural biometric data elements

- Works differently as the customer is not known eg:
 - PII pasted
 - Alt+Tab
 - Pausing

Machine Learning

More Than A Score

Score Performance*	
False Rejection Rate	5.8%
False Acceptance Rate	5.5%
Equal Error Rate	~1.0%

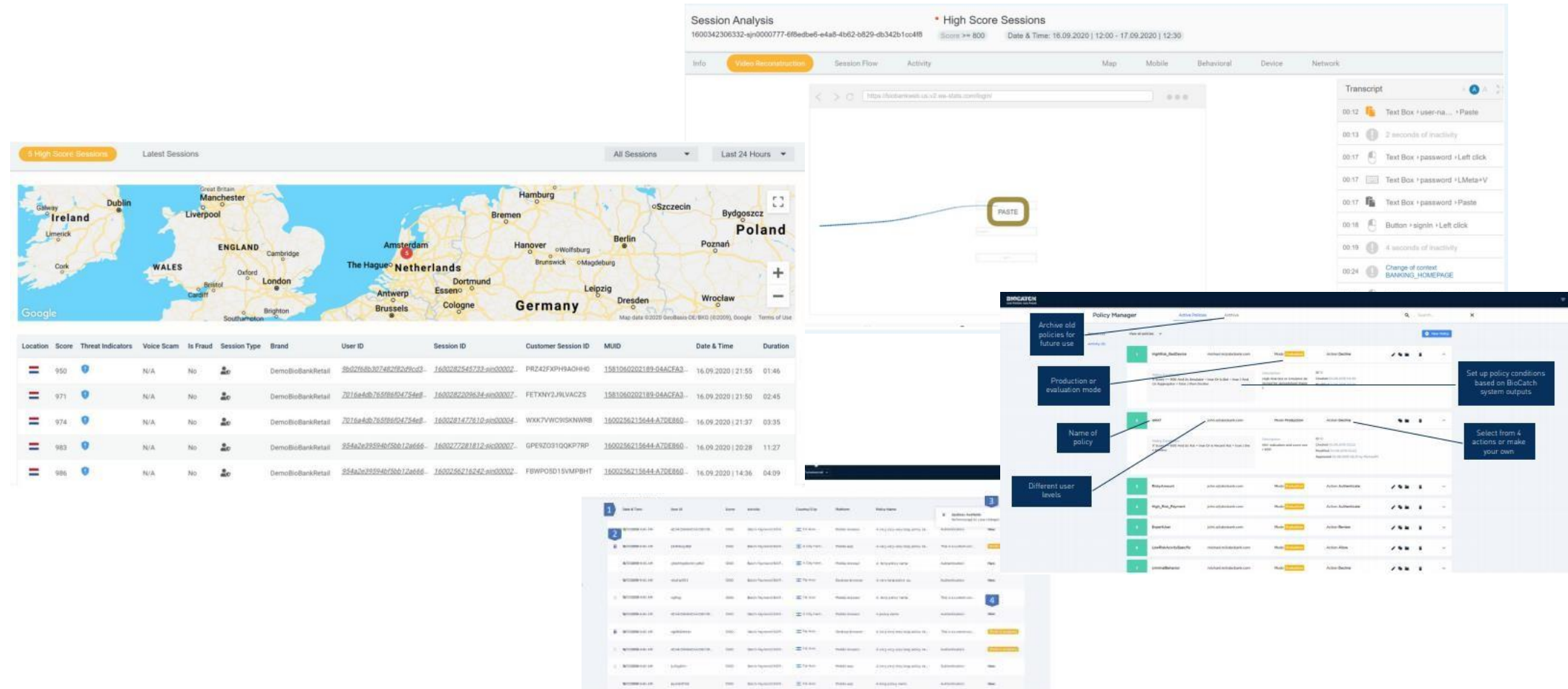
* European customer: 100,253 online sessions

Standard Device & Network Data

Malware?	Location
RAT?	OS
Bot?	OS language
Emulator?	Screen orientation
IP Address	Time zone
Device ID	...

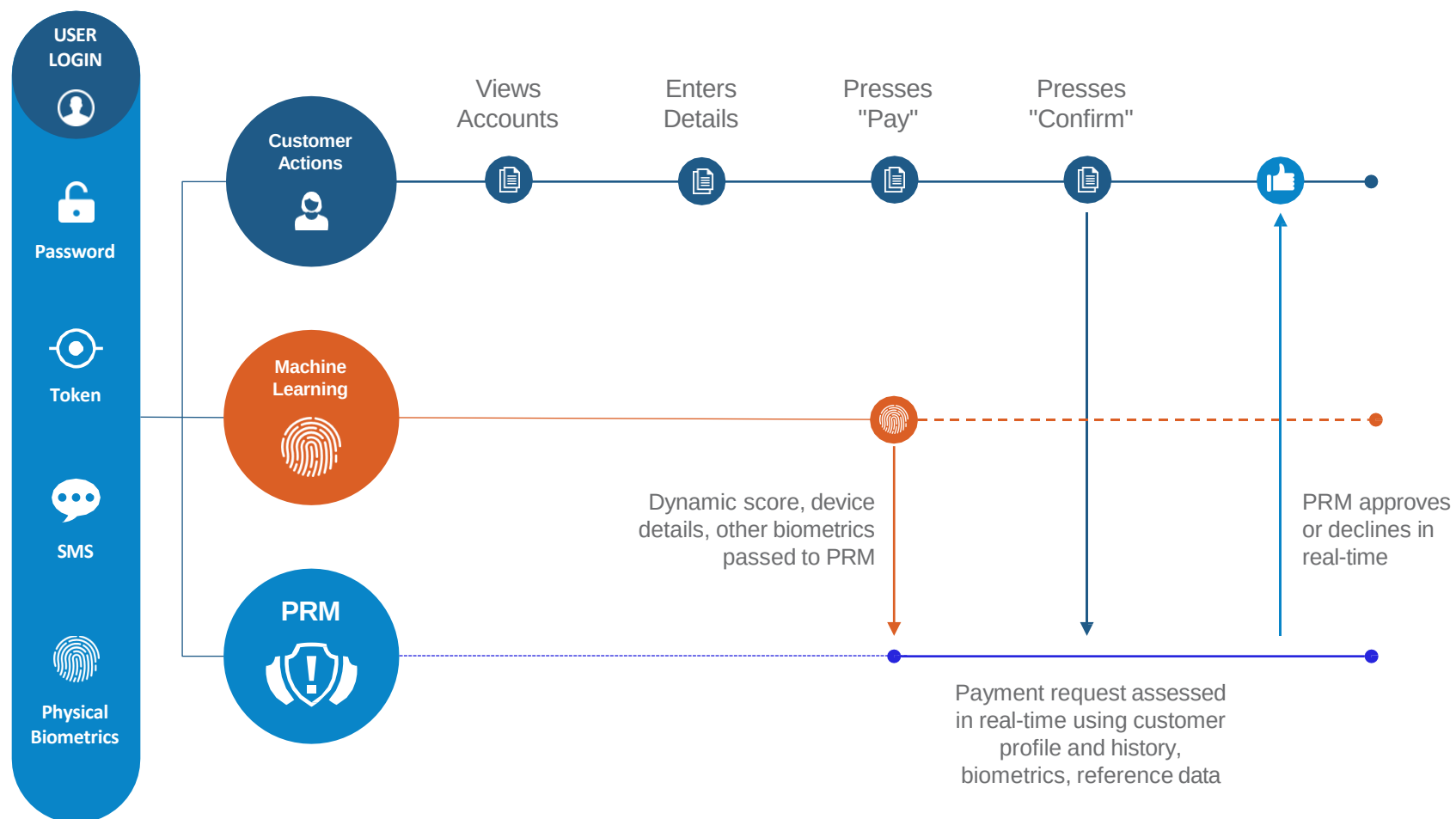
Data Available	
RISK SCORE	0-1000
altTab	0-32767
autofill_in_email	True/False
autofill_in_phone_number	True/False
autofill_in_last_name	True/False
count_segmented_typing_DOB	0-32767
count_segmented_typing_phone	0-32767
ctrlTab	0-32767
ctrlV	0-32767
ctrlX	0-32767
...	...

BioCatch Screens



How It Works

Real-Time, Continuous, Frictionless

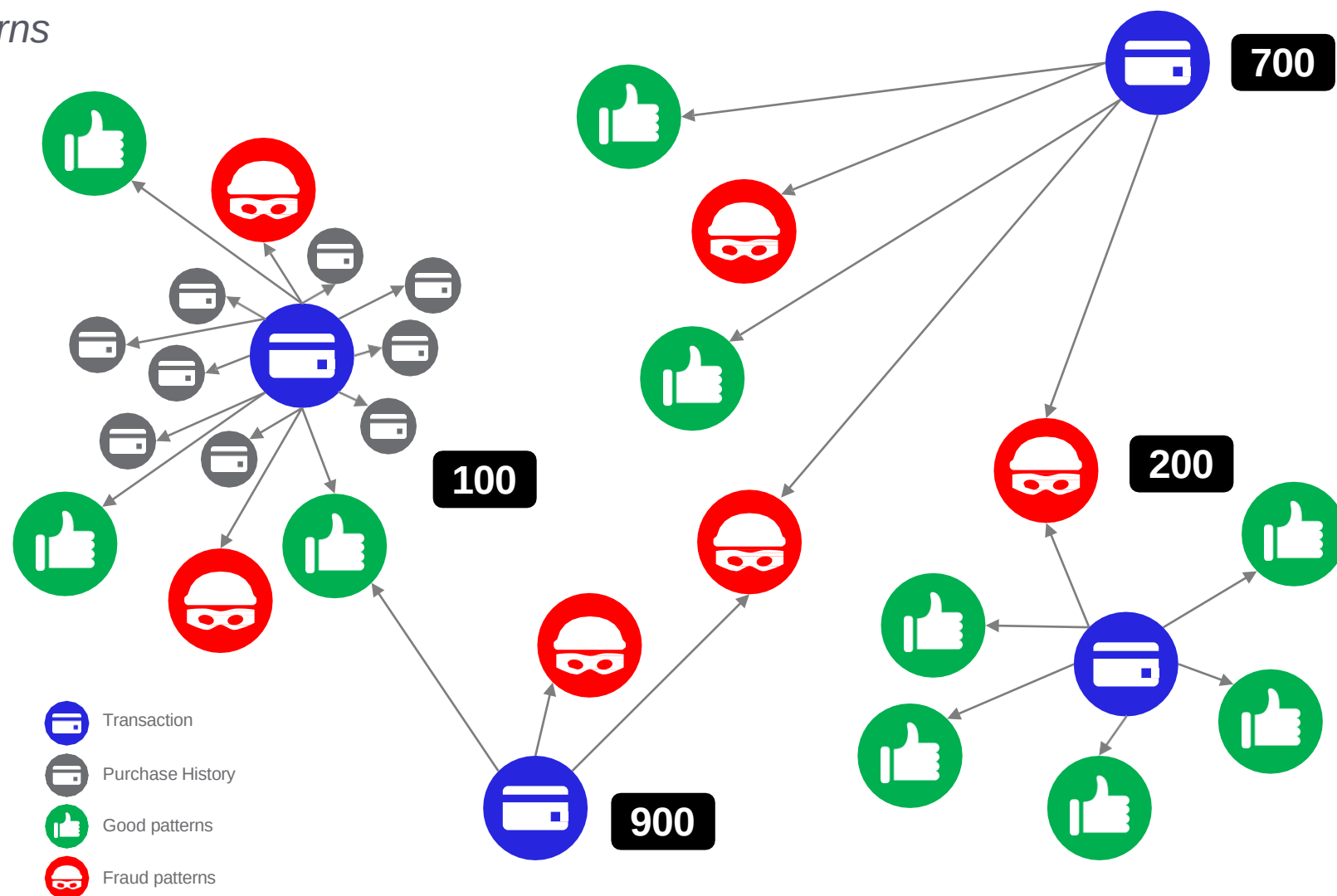


Machine Learning

w/ PRM Scoring Engine (PSE)

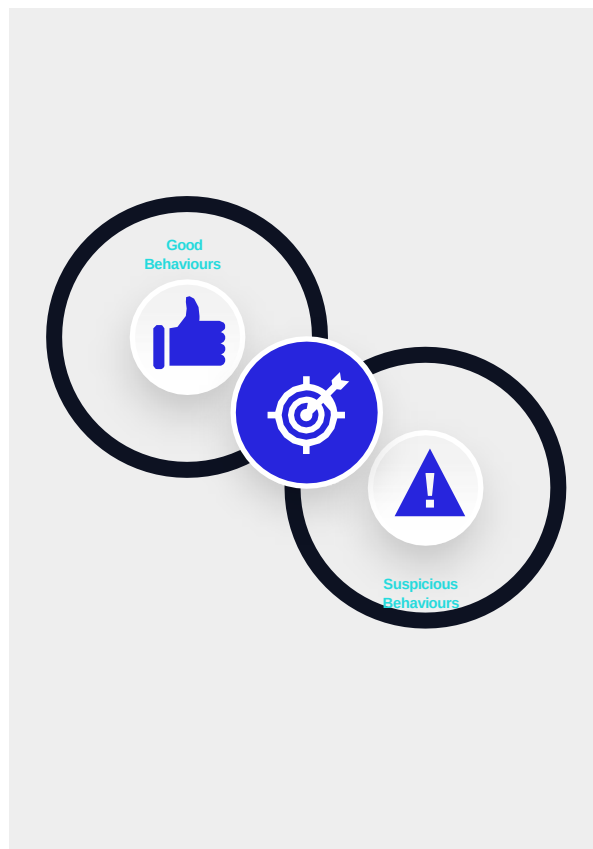
Custom Model

Features & patterns



Custom Model

Technology overview



How does the technology work?

- Develop the **features** that describe the data
- Features are **aggregates** or **derivations** from the raw transaction data
- Identify **similarities** to, and **deviations** from, genuine behaviour
- Typical custom model has 25-50, non-linear, multi-dimensional features representing **good** and **suspicious** behaviours
- Best algorithms for the job – neural network, sparse kernel method, dimensionality expansion method

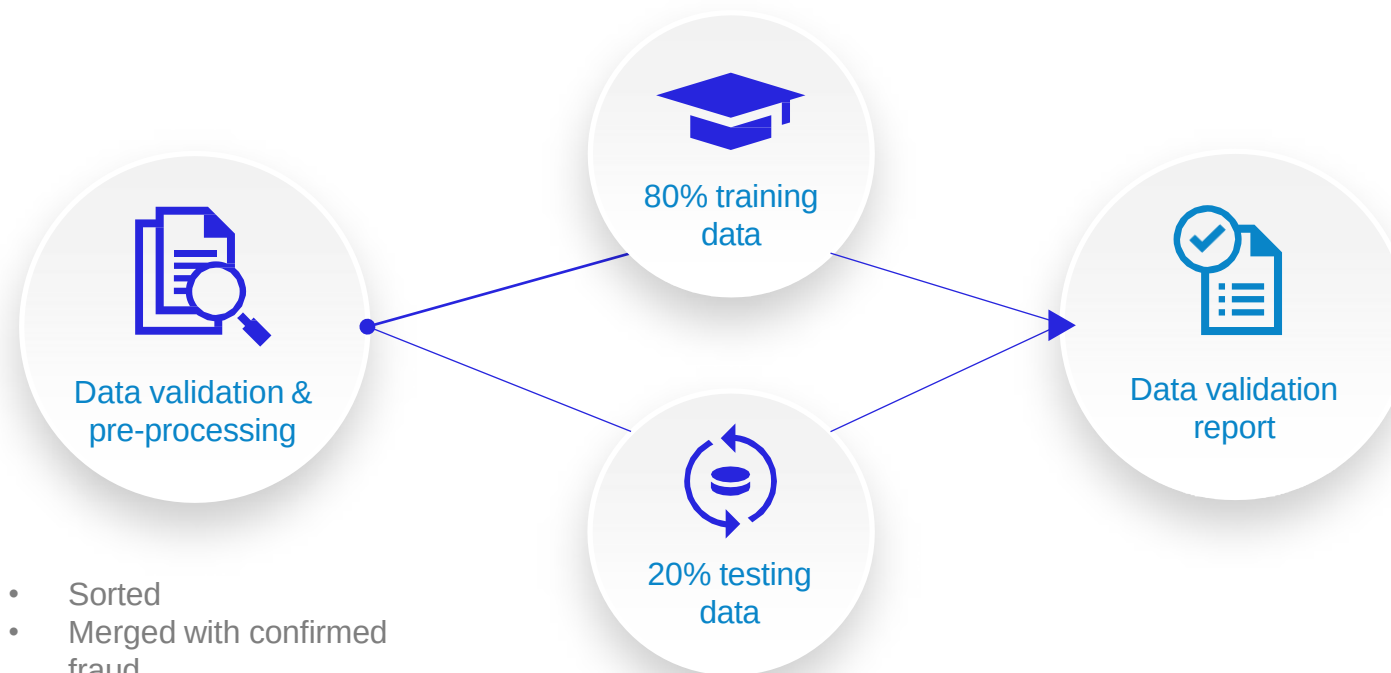
Custom Model

Data requirements



Custom Model

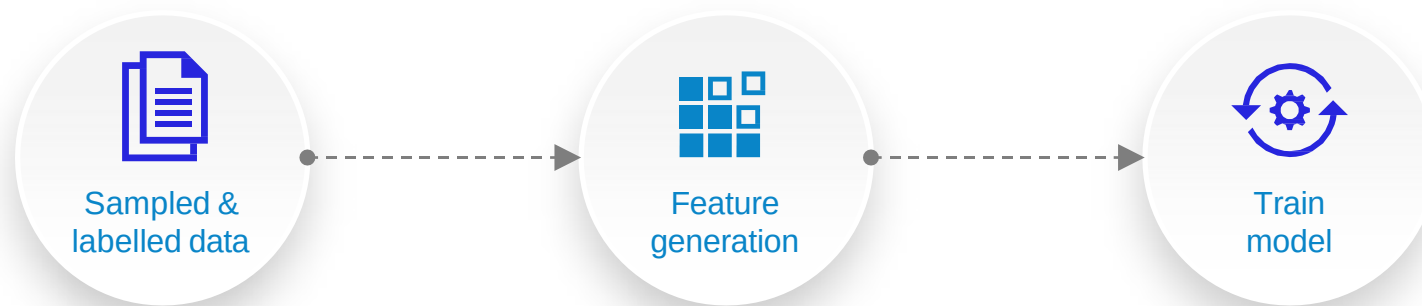
Sampling, labelling, validation



- Sorted
- Merged with confirmed fraud
- Labelled

Custom Model

Feature generation & training



IDENTIFY:

- Risk patterns
- Categorised averages
- Segmentation patterns (conditional probability)
- Association patterns (probabilities from compound variables)
- Account information patterns

BUILD:

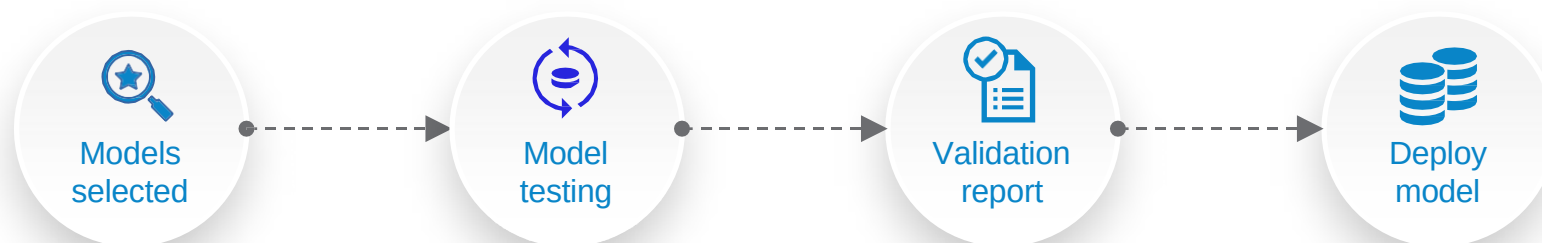
- Combine different fields from transaction
- Combine information from different transactions
- Similarities and deviations from genuine customer behaviour

TRAINING:

- 40-50 features developed for customer
- Best 20-30 from the model
- 100-200 training runs

Custom Model

Model finalised



SELECTION:

- Scoring accuracy
- Early detection
- Savings / loss avoidance

TESTING:

- Tested on 20% holdout data
- Best performed model selected

REPORT:

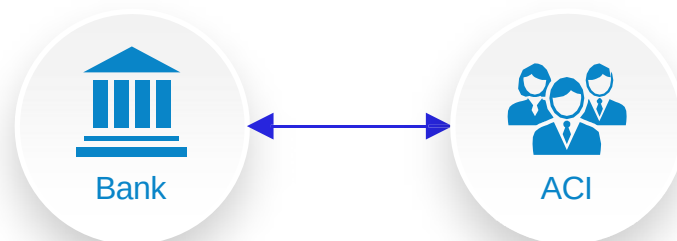
- Data validation report produced
- Report provided to customer

GO-LIVE:

- Deliver model
- Deploy model

Model Refresh

Every quarter



Model Refresh:

- Daily feed of transactions to ACI
- Daily feed of marked frauds to ACI
- ACI delivers a new model every quarter

360° View

Enterprise Fraud

One System, Any Channel



360° View

Through Dimensions



	Card	2020-09-15 11:39	\$99.99	Patagonia	San Diego	USA	192.168.1.1	Purchase	985	SCA
	Internet	2020-09-15 07:21	\$147.63				192.168.2.1	Transfer	456	
	Branch	2020-09-14 17:58	\$200.00					Withdrawal	823	
	Card	2020-09-14 11:39	\$99.99	Kona Cycles	Bellingham	USA	192.168.1.1	Purchase	766	SCA
	Logon	2020-09-14 11:30	N/A				192.168.2.1		899	OTP
	Card	2020-09-13 17:00	\$200.00	ANZ ATM	Wellington	NZ		Withdrawal	589	
	Card	2020-09-12 11:38	\$123.42	Thai Thai	Bangkok	THA		Purchase	786	
	Internet	2020-09-11 07:11	\$79.99	My Telco	Ocean Grove	AUS	192.168.1.1	BillPay	878	
	Card	2020-09-10 11:20	\$220.45	Costco	Moorabbin	AUS		Purchase	985	
	Card	2020-09-10 11:05	\$99.99	Pizza Palace	Ocean Grove	AUS		Purchase	985	

360° View

Through Dimensions



	Card	2020-09-15 11:39	\$99.99	Patagonia	San Diego	USA	192.168.1.1	Purchase	985	SCA
	Card	2020-09-14 11:39	\$45.50	Kona Cycles	Bellingham	USA	192.168.1.1	Purchase	766	SCA
	Card	2020-09-13 17:00	\$200.00	ANZ ATM	Wellington	NZ		Withdrawal	589	
	Card	2020-09-12 11:38	\$123.42	Thai Thai	Bangkok	THA		Purchase	786	
	Card	2020-09-10 11:20	\$220.45	Costco	Moorabbin	AUS		Purchase	985	
	Card	2020-09-10 11:05	\$99.99	Pizza Palace	Ocean Grove	AUS		Purchase	985	

360° View

Through Dimensions



	Card	2020-09-15 11:39	\$99.99	Patagonia	San Diego	USA	192.168.1.1	Purchase	985	SCA
	Card	2020-09-14 11:22	\$45.509	Kona Cycles	Bellingham	USA	192.168.1.1	Purchase	766	SCA
	Internet	2020-09-11 07:11	\$79.99	My Telco	Ocean Grove	AUS	192.168.1.1	BillPay	878	

360° View

Drill-Down

- Click Retailer ID:
 - All transactions performed by your cards at t

All Transactions 1							Marked Transactions 2	Alerted Transactions 3	Enhanced Profiling 4	Standard Profiles 5	Linked Accounts 6
☐	Transaction Date and Time	Retailer ID	Amount 1	NRT Score	MCC/SIC	Terminal ID					
☐	2020-04-29 17:59:18	RTL004560045412AA458955	200.00		92 7995	TERM145788WER644					
☐	2020-04-25 17:59:08	RTL004560045412AA458955	200.00		92 7995	TERM145788WER644					
☐	2020-04-25 17:59:07	RTL004560045412AA458955	200.00		7995	TERM145788WER644					
☐	2020-04-25 17:59:06	RTL004560045412AA458955	200.00		7995	TERM145788WER644					
☐	2020-04-25 17:59:04	RTL004560045412AA458955	200.00		7995	TERM145788WER644					
☐	2020-04-24 17:59:04	RTL004560045412AA458955	200.00		7995	TERM145788WER644					
☒	2020-04-24 17:49:04	RTL004560045412AA458955	500.00		815 7995	TERM145788WER644					

- Same applies for corporates:
 - For the Corporate ID, all transactions dis
 - Click Account, only transactions for that a

All Transactions 1										Marked Transactions 2	Alerted Transactions 3	Enhanced Profiling 4	Standard Profiles 5	Page Size 50
☐	Transaction Date and Time	Customer ID	Amount 1	NRT Score	NRT Reason Code 1	PAN	Terminal ID	Terminal Name and Location						
☐	2020-06-18 12:24:50	3456789000000000...	21,149.99		129	4077 411122*****5568	TERM145788WER644	TBD.com						
☐	2020-06-18 12:19:28	3456789000000000...	149.99		47	4077 411122*****5568	TERM145788WER644	TBD.com						
☐	2020-06-18 12:13:10	3456789000000000...	149.99		46	4077 411122*****5568	TERM145788WER644	TBD.com						
☐	2020-06-17 15:54:02	3456789000000000...	149.99		408	4077 411122*****5568	TERM145788WER644	TBD.com						
☐	2020-06-17 15:51:56	3456789000000000...	149.99		76	4077 411122*****5564	TERM145788WER644	TBD.com						
☐	2020-06-17 15:50:03	3456789000000000...	149.99		77	4077 411122*****5564	TERM145788WER644	TBD.com						
☐	2020-06-17 15:44:26	3456789000000000...	200.00		103	4077 411122*****5564	TERM145788WER644	TBD.com						
☐	2020-06-17 15:29:27	3456789000000000...	200.00		104	4077 411122*****5564	TERM145788WER644	TBD.com						
☐	2020-06-17 15:22:58	3456789000000000...	200.00		462	4077 2020-0*****388Z	TERM145788WER644	TBD.com						
☐	2020-06-17 15:18:20	3456789000000000...	200.00		101	4077 411122*****5564	TERM145788WER644	TBD.com						
☒	2020-06-02 12:29:01	3456789000000000...	299.99		46	4077 411122*****5564	TERM145788WER644	TBD.com						
☐	2020-06-01 12:29:01	3456789000000000...	299.99		530	4077 411122*****5564	TERM145788WER644	TBD.com						
☐	2020-05-12 12:29:01	3456789000000000...	6,000.00		92	4077 411122*****5564	TERM145788WER644	TBD.com						
☐	2020-05-02 12:29:01	3456789000000000...	600.00		92	4077 411122*****5564	TERM145788WER644	TBD.com						
☐	2020-05-01 12:29:01	3456789000000000...	600.00		92	4077 411122*****5564	TERM145788WER644	TBD.com						
☐	2020-05-01 12:27:01	3456789000000000...	600.00		3234	4077 411122*****5564	TERM145788WER644	TBD.com						
☐	2020-04-29 17:59:18	3456789000000000...	200.00		92	4077 411122*****5569	TERM145788WER644	TBD.com						

Rules Engine

Rules Engine

Flexible SQL-based rule editor

- Rules can reference:
 - Transaction data
 - Non-monetary data
 - Demographic data (if available)
 - Reference / watchlist data
- Real-Time or Near Real-Time
- Rule Scoring
 - Score of -999 to 999 can be included
 - Score is computed after all the NRT rules have

The screenshot shows the 'Rule Standard Editor(Edit)' window. The rule is named 'Same Day, Different IPs'. The description is 'Uses new 3DS2 fields. Same card, same merchant, more than 2 different IP addresses in a 24 hour period.' The action level is set to 'Card'. The entrance criteria is '@SD_TRAN_CDE = "01"'. The SQL syntax is 'EXISTS (SELECT 1 FROM SHORTWINDOW WHERE SD_PAN = @SD_PAN AND SD_RET_ID = @SD_RET_ID AND DD_DATE >= DATEADD(HOUR, -24, @DD_DATE) HAVING COUNT(DISTINCT SD_CLIENT_ID) > 2)'. The rule is configured to create a matching queue, with an alert color of green and an alert category of 'None'. The rule score is 0. The rule is not set for auto action, response required, or email notification.

Rule Standard Editor(Edit)

*Name: Same Day, Different IPs

Description: Uses new 3DS2 fields. Same card, same merchant, more than 2 different IP addresses in a 24 hour period.

*Action Level: Card

Entrance Criteria: @SD_TRAN_CDE = "01"

SQL Syntax: EXISTS (SELECT 1 FROM SHORTWINDOW WHERE SD_PAN = @SD_PAN AND SD_RET_ID = @SD_RET_ID AND DD_DATE >= DATEADD(HOUR, -24, @DD_DATE) HAVING COUNT(DISTINCT SD_CLIENT_ID) > 2)

Buttons: Variables, Operators, Functions, SQL, Aggregate, Clear

Buttons: Transaction, Demographic, Variables, Operators, Functions, SQL, Aggregate, Clear

Buttons: Test Rule, Syntax Check, Team Only, Save, Save As, Cancel, Help

Rules Engine

Flexible SQL-based rule editor

The screenshot displays the 'Rule Standard Editor(Edit)' window. At the top, the title bar reads 'Rule Standard Editor(Edit)'. Below the title bar, there are several configuration options: 'Create matching queue' (selected), '*Alert Color:' (yellow), 'Assign Alert Category' (unselected), '*Alert Category:' (dropdown), and 'Rule Score' (0). The '*Name:' field contains 'Late Night Corporate Transaction, Large Amount'. The '*Alert Property:' dropdown is set to 'Consume and Discard Alerts'. A list of variables is shown, including 'Action Level (@ACTION_LEVEL_ID)', 'Action Level ID (@ACTION_LEVEL_KEY)', 'Batch Release Date (@DD_ACH_BATCH_RELEASE_DAT_T)', 'Acq lchg Settle Date (@DD_ACQ_ICHG_SETL_DAT)', 'AP Date (@DD_APDATE)', 'Last Emboss (@DD_CRD_EMBOSS)', 'Last CRV (@DD_CRD_LAST_CRV)', 'Last Issue (@DD_CRD_LAST_ISS)', 'Last Status Chng (@DD_CRD_LAST_STAT_CHNG)', 'Creditor Alias Last Update (@DD_CREDITOR_ALIAS_LAST)', 'Creditor Alias Register Date (@DD_CREDITOR_ALIAS_REC)', 'Cust Txn Date (@DD_CUST_TRAN_DAT)', 'Txn Date (@DD_DATE)', 'Expire Date (@DD_EXP_DAT)', 'Date Fraud IP Flag (@DD_IP_FRAUD_FLG)', 'Date Mal IP Flag (@DD_IP_MAL_FLG)', 'Date Mal Dev Flag (@DD_MAL_DEVICE_FLG)', and 'Partition Date (@DD_PARTITION_DATE)'. The 'SQL Syntax:' window is open, showing the following SQL code:

```
1 DATEPART(hour,@DD_DATE) >= '01'
2 AND DATEPART(hour,@DD_DATE) <= '05'
3
4 EXISTS (SELECT * FROM SHORTWINDOW WHERE @ACTION_LEVEL_KEY = SD_UAN
5
6 AND @MD_TRAN_AMT1 > 100000)
```

 The bottom of the window contains buttons for 'Test Rule', 'Syntax Check', 'Team Only', 'Save', 'Save As', 'Cancel', and 'Help'.

Rule Standard Editor(Edit)

*Name: Late Night Corporate Transaction, Large Amount

*Alert Property: Consume and Discard Alerts

Variables

- Action Level (@ACTION_LEVEL_ID)
- Action Level ID (@ACTION_LEVEL_KEY)
- Batch Release Date (@DD_ACH_BATCH_RELEASE_DAT_T)
- Acq lchg Settle Date (@DD_ACQ_ICHG_SETL_DAT)
- AP Date (@DD_APDATE)
- Last Emboss (@DD_CRD_EMBOSS)
- Last CRV (@DD_CRD_LAST_CRV)
- Last Issue (@DD_CRD_LAST_ISS)
- Last Status Chng (@DD_CRD_LAST_STAT_CHNG)
- Creditor Alias Last Update (@DD_CREDITOR_ALIAS_LAST)
- Creditor Alias Register Date (@DD_CREDITOR_ALIAS_REC)
- Cust Txn Date (@DD_CUST_TRAN_DAT)
- Txn Date (@DD_DATE)
- Expire Date (@DD_EXP_DAT)
- Date Fraud IP Flag (@DD_IP_FRAUD_FLG)
- Date Mal IP Flag (@DD_IP_MAL_FLG)
- Date Mal Dev Flag (@DD_MAL_DEVICE_FLG)
- Partition Date (@DD_PARTITION_DATE)

SQL Syntax:

```
1 DATEPART(hour,@DD_DATE) >= '01'
2 AND DATEPART(hour,@DD_DATE) <= '05'
3
4 EXISTS (SELECT * FROM SHORTWINDOW WHERE @ACTION_LEVEL_KEY = SD_UAN
5
6 AND @MD_TRAN_AMT1 > 100000)
```

Test Rule Syntax Check Team Only Save Save As Cancel Help

Rules Engine

ACI Proactive Risk Manager

- Test and evaluate rules before deployment:

Rule Standard Editor(Edit)

*Name: Large Transaction Amount

*Alert Property: Consume and Discard Alerts

Description: Single transaction greater than 100000

*Action Level: Card

Entrance Criteria:

SQL Syntax: @MD_TRAN_AMT1 > 100

Test Rule

Rule Tester

Rule Tester Dataset

Start date: Apr 28, 2020

Range of days from start date: 10

Number of transactions in dataset: 24,684

Execute Test

DETAIL table:

Included Mark Types:

- ☒ Select All/Deselect All
- ☒ Acct. Takeover
- ☒ Counterfeit
- ☒ Fraud App
- ☒ Lost
- ☒ Merchant
- ☒ MOTO
- ☒ N-R-I
- ☒ Other
- ☒ Stolen

Results for current test:

Results for previous test:

Alerted transactions that are marked (hits):

Show Hits

Alerted Transactions that are not marked (false positives):

Show False Positive

Export Close Help

Rules Engine

ACI Proactive Risk Manager

Rule Tester

Rule Tester Dataset: Start date: Apr 28, 2020
Range of days from start date: 10
Number of transactions in dataset: 24,684

DETAIL table:

ExecuteTest

Total number of transactions belonging to BusinessTeam Card portfolios: 14,076
Total number of marked transactions processed against the rule: 510

Included Mark Types: Results for current test: Results for previous test:

Transaction Demographic Variables Operators Functions SQL Aggregate

SQL Syntax:

@MD_TRAN_AMT1 > 500

Test Rule Syntax Check Team Only Save Save As Cancel

Alerted Transactions that are not marked (false positives):

Column	Operator	Discrete value	User-defined value	User-defined value
Txn Date/Time				
NRT Score				
NRT R1				
NRT R2				
NRT R3				
Amount 1				
Msg Type				
Txn Co				

May 5, 2020 12:2... 402.79 0210 01
May 5, 2020 12:2... 1,005.32 0210 1454
May 5, 2020 12:2... 586.07 0210 2214

Export Close Help

Rule Tester

Rule Tester Dataset: Start date: Apr 28, 2020
Range of days from start date: 10
Number of transactions in dataset: 24,684

DETAIL table:

ExecuteTest

Total number of transactions belonging to BusinessTeam Card portfolios: 14,076
Total number of marked transactions processed against the rule: 510

Included Mark Types: Results for current test: Results for previous test:

Completed: Jul 06, 2020 10:31 PM
Time elapsed: 00:00:00
Alerts: 4,829

	Total	% of Alerts
Marked:	179	3.7
Unmarked:	4,650	96.2
Overlap by Card rule:	178	3.6
Overlap by non Card rule:	22	0.4
Marked and overlap by		

Completed: Jul 06, 2020 10:28 PM
Time elapsed: 00:00:01
Alerts: 9,900

	Total	% of Alerts
Marked:	388	3.9
Unmarked:	9,512	96.0
Overlap by Card rule:	387	3.9
Overlap by non Card rule:	88	0.8
Marked and overlap by		

Alerted transactions that are marked (hits):

Column	Operator	Discrete value	User-defined value	User-defined value
Txn Date/Time				
NRT Score				
NRT R1				
NRT R2				
NRT R3				
Amount 1				
Msg Type				
Txn Co				

May 1, 2020 11:5... 1,061.02 0210 2214
May 1, 2020 11:5... 768.49 0210 2214
May 1, 2020 11:5... 659.93 0210 01
May 1, 2020 12:0... 617.28 0210 1454

Alerted Transactions that are not marked (false positives):

Column	Operator	Discrete value	User-defined value	User-defined value
Txn Date/Time				
NRT Score				
NRT R1				
NRT R2				
NRT R3				
Amount 1				
Msg Type				
Txn Co				

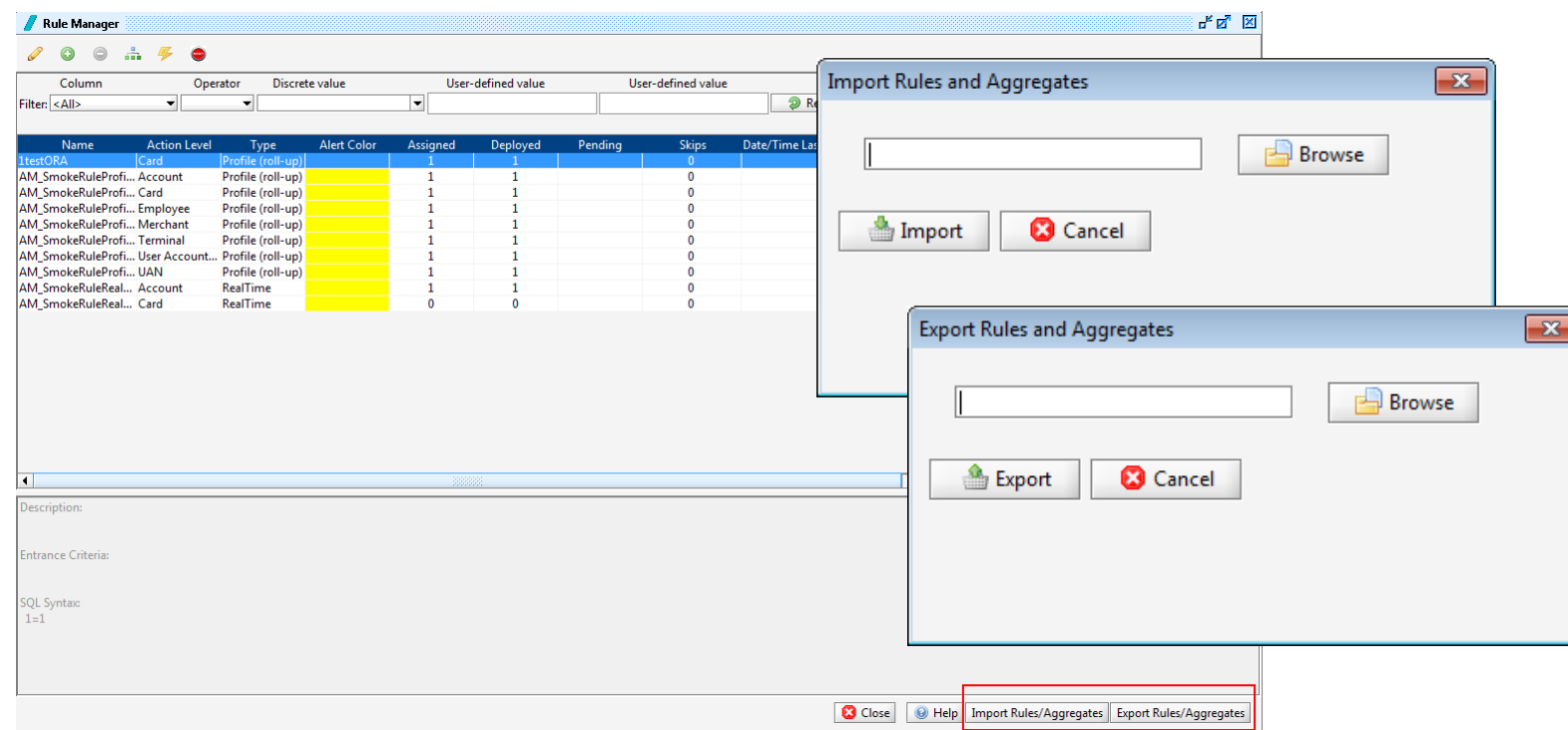
May 5, 2020 12:2... 1,005.32 0210 1454
May 5, 2020 12:2... 586.07 0210 2214
May 5, 2020 12:2... 755.27 0210 2214

Export Close Help

Rules Engine

Import/Export Rules & Aggregates

- Export Rules and Aggregates to specified location, re-import as needed
- Expedites promotion to other environments (UAT, Prod) & troubleshooting



Rule Scoring

Rule Scoring

Computing

- Rule Scoring is available for all Action Levels
- When rules are added, a Rule Score of -999 to 999 can be included
- For each transaction passed to PRM, a Rule Score will be computed for each Action Level to which the transaction is assigned
- The Rule Score will be computed after all NRT Rules have processed, being the sum of the score of each triggered NRT Rule, for each assigned Action Level
- A Rule Score field will be present for each Action Level – if the Action Level was not assigned, it will be *NULL*
- The Rule Score field will be shown in the UI in the Transaction, Summary and Mark history tabs, and available in Rule Tester

Rule Scoring

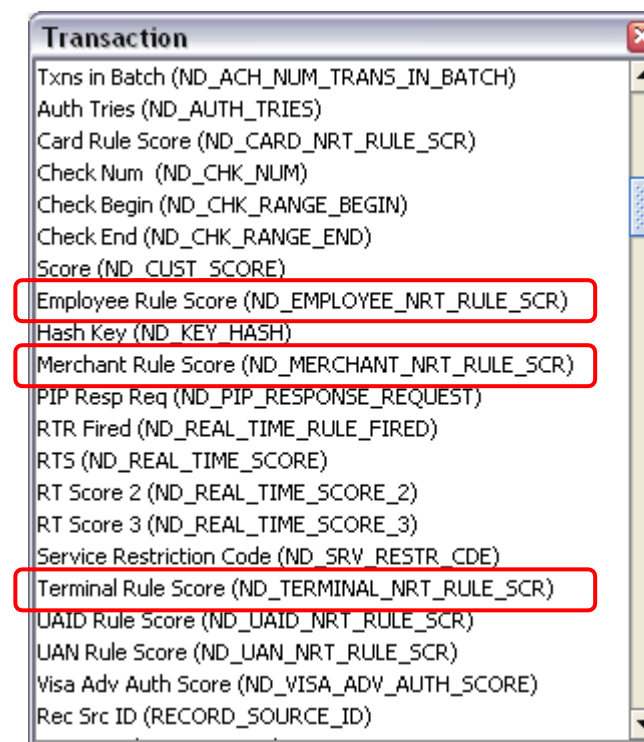
Computing

- During rules processing, the cumulative Rule Score is held in the variable @RULES_SCORE
- When each rule is triggered, the score from that rule is added to @RULES_SCORE
- At the end of rule processing, the value of @RULE_SCORE, for each Action Level, is updated as follows for the Detail, Fraud Detail, ShortWindow, and RT ShortWindow tables:
 - ND_ACCOUNT_NRT_RULE_SCR
 - ND_CARD_NRT_RULE_SCR
 - ND_EMPLOYEE_NRT_RULE_SCR
 - ND_MERCHANT_NRT_RULE_SCR
 - ND_TERMINAL_NRT_RULE_SCR
 - ND_UAID_NRT_RULE_SCR
 - ND_UAN_NRT_RULE_SCR

Rule Scoring

Using Rule Score in Rules

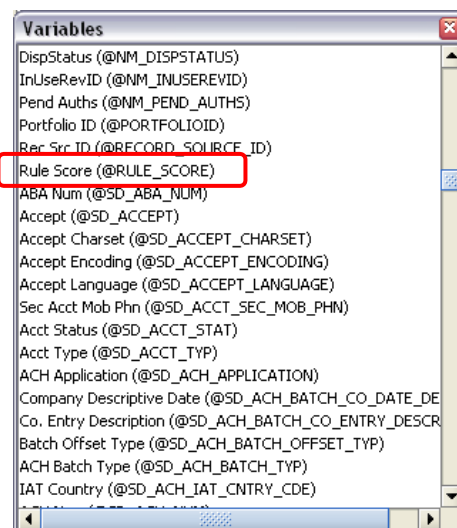
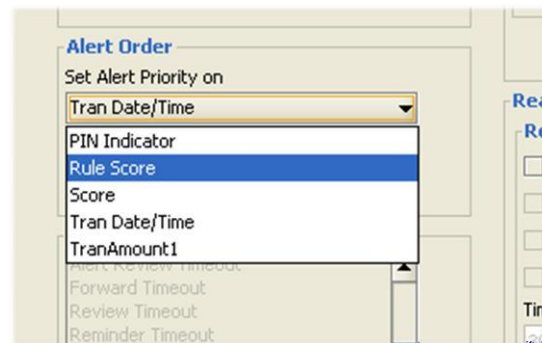
- These fields are available for RT & NRT rules, aggregates, & Enhanced Profiling
- To be clear, the Rule Score for the current transaction will not be available until rules processing for that transaction is complete
- The benefit of the Rules Score is when assessing past transactions, not the current



Rule Scoring

Queueing using Rule Score

In the Action Level Editor, one of the options for Alert Order > Set Alert Priority on is Rule Score



By writing specific SQL in the Queue Editor (handy for creating a high priority queue when a very high Rule Score results as multiple rules have triggered)

Rule Scoring

Rule Fired Display

- When a user double-clicks a transaction on the Review screen, what is displayed is determined by the RuleFiredDisplay parameter in the System Config:

0	=	Nothing
1	=	Rule name
2	=	Rule description
3	=	Rule name & rule description
4	=	Rule name & score
5	=	Rule name, rule description, & score
6	=	Category / queue
7	=	All (rule name, rule score, rule description, category / queue)
- '7' needs to be the parameter to include the Rule Score

Profiling

Standard & Peer Profiles

Profiles

Standard & Peer

- Standard & Peer Profiles record counts, totals and averages for each Dimension ID for the time period defined.

ACI DASHBOARD SEARCH REVIEW PENDING ALERTS REPORTS FORWARDS & REMINDERS English prn Logout

403504*****2307 - Card

Alert, Retrieved from - Catch-All Queue (Consume and Discard Alerts) Marked • 403504*****2307 • VC • 2020-06-22 14:54:14 • 681.82

All Transactions Marked Transactions Alerted Transactions Enhanced Profiling **Standard Profiles** Linked Accounts

1 - 50 of 89 Page Size 50

Transaction Date and Time	Mark Type	Amount 1	NRT Score	NRT Score 2	NRT Score 3	Retailer ID	MCC/SIC	T
2020-06-22 14:54:14		681.82	658	825	876		6011	U
2020-06-22 14:48:55		444.51	654	807	874		6011	U
2020-06-22 12:51:55		1,262.93	304	488	338		6011	U
2020-06-22 12:15:21	Counterfeit	246.25				MR33309991	3424	
2020-06-21 14:03:23		299.67				MR41036359	3424	
2020-06-20 13:41:59		198.43	658	841	827		6011	
2020-06-20 11:28:13		977.07				MR45615405	0777	

Profiles

Standard & Peer

- Time periods for the profiles differ across Dimension IDs, the Card Dimension ID profile (31 Day) is shown here:

All Transactions Marked Transactions Alerted Transactions Enhanced Profiling Standard Profiles Linked Accounts						
Daily 3 Day 31 Day Peer Daily Peer 3 Day Peer 31 Day						
1 - 2 < > Page Size 50 ↺						
☐	Date	PAN	Average Transactions	Average Debit Transactions	Average Sum of Debit Transactions	Average Credit Transactions
☐	2020-06-22 00:00:00	4035044559342307	0.4516	0.0000	0.00	0.0000
☐	2020-06-17 00:00:00	4035044559342307	0.3226	0.0000	0.00	0.0000

- Profiles available vary across Dimension IDs

Profile Example

Card Dimension ID, Daily Profile

- Transactions
- Deposits
 - Debit Transactions
 - Sum of Deposits
- Withdrawals
 - Credit Transactions
 - Sum of Credit Transactions
- Incoming Transfers
 - ATM Transactions
 - ATM Transactions Amounts
- Outgoing Transfers
 - Sum of ATM Deposits
- Sum of Incoming Transfers
- Sum of Outgoing Transfers
 - POS Transactions
 - POS Transactions Amount

- eCommerce Transactions
- Sum of eCommerce Transactions
- MOTO Transactions
- Sum of MOTO Transactions
- Card Present Transactions
- Sum of Card Present Transactions
- Manual Entry
- Sum of Manual Entry

Profiles

Standard & Peer

Dimension ID	Daily	3 Day	Weekly	Monthly	Quarterly	Peer Daily	Peer 3 Day	Peer Weekly	Peer Monthly	Peer Quarterly	Deposit History
Card	X	X		X		X	X		X		
Customer ID	X	X		X		X	X		X		
Merchant	X		X	X	X	X		X	X	X	X
Terminal	X	X	X			X	X	X			
Account	X	X		X							
UAID	X	X		X							
Employee	X		X	X	X	X		X	X		

- Peer Profiles are sub-groups of Standard Profiles, determined through configuration e.g. separation of Customer and Corporate transactions

Profiling

Mark Profiles

Mark Profiles

For Non-Actionable Dimension IDs

- There are three non-actionable Dimension IDs:
 - Device ID
 - IP Address
 - Payee



- The Mark Profiles record information for transactions that have been marked, for this non-actionable Dimension ID, for the selected period (1, 3 or 7 days):
 - Transactions
 - Sum of transactions
 - Average number of transactions
 - Average sum of transactions
 - Total sum of transactions *(for 3 and 7 day periods)*

Profiling

Custom Profiles

Profiles

Custom Profiles

- Custom (Enhanced) Profiles allow users to create their own behavioural profiles.

The screenshot displays the ACI Review interface. The top navigation bar includes 'DASHBOARD', 'SEARCH', 'REVIEW' (highlighted), 'PENDING ALERTS', 'REPORTS', and 'FORWARDS & REMINDERS'. The right side shows 'English', a user icon 'prm', and 'Logout'. Below the navigation bar, a breadcrumb trail reads: 'Alert, Retrieved from - Catch-All Queue (Consume and Discard Alerts) Marked • 403504*****2307 • VC • 2020-06-22 14:54:14 • 681.82'. The main content area has tabs for 'All Transactions', 'Marked Transactions', 'Alerted Transactions', 'Enhanced Profiling' (highlighted with a dashed orange box), 'Standard Profiles', and 'Linked Accounts'. Below the tabs, a table displays transaction data. The table has columns: Transaction Date and Time, Mark Type, Amount 1, NRT Score, NRT Score 2, NRT Score 3, Retailer ID, MCC/SIC, and a status column. The data rows show various transactions, including one marked 'Counterfeit'.

Transaction Date and Time	Mark Type	Amount 1	NRT Score	NRT Score 2	NRT Score 3	Retailer ID	MCC/SIC	
2020-06-22 14:54:14		681.82	658	825	876		6011	U
2020-06-22 14:48:55		444.51	654	807	874		6011	U
2020-06-22 12:51:55		1,262.93	304	488	338		6011	U
2020-06-22 12:15:21	Counterfeit	246.25				MR33309991	3424	
2020-06-21 14:03:23		299.67				MR41036359	3424	
2020-06-20 13:41:59		198.43	658	841	827		6011	
2020-06-20 11:38:13		077.07				MR45615405	0777	

Custom Profiles

100% Configurable

- Custom profiles allow for any two data elements (or more) to be joined and activity for that linkage to be recorded
- For example, **Customer ID + Beneficiary**:
 - Creates EP_CUSTOMER_BENEFICIARY table in the database
 - Adds a row for each unique combination of the profile
 - In this case, we are recording every beneficiary paid by this customer
- For each profile, the following is calculated:
 - Count
 - Sum
 - Smallest amount
 - Largest amount
 - First time use
 - Last time use
- Thus, we are now recording:
 - Every beneficiary paid by the customer
- And for each beneficiary:
 - When they were first paid
 - When they were last paid
 - How many times they were paid
 - The total paid
 - Smallest amount paid
 - Largest amount paid

Custom Profiles

100% Configurable

Enhanced Profile Editor (Add)

***Name:** CUSTOMER_BENEFICIARY

Description: Customer ID + Beneficiary

***Shortwindow table:** SHORTWINDOW

***Cycle:** Weekly

***Cycle date field:** Txn Date (SHORTWINDOW.DD_DATE)

***Number of cycles to retain:** 52

***Primary profile field:** UAN (SHORTWINDOW.SD_UAN)

Optional profile field, substring or combination: SHORTWINDOW.SD_BEN_WIR_ACCT_IDENT

***Sum field:** Amount 1 (SHORTWINDOW.MD_TRAN_AMT1)

Estimated number of rows per cycle:

Summing negative amounts

- ☒ Use negative amount
- ☐ Use absolute value

Annotations:

- How often to run: Points to the *Cycle dropdown (Weekly).
- How long to keep: Points to the *Number of cycles to retain input (52).
- Date field: Points to the *Cycle date field dropdown (Txn Date (SHORTWINDOW.DD_DATE)).
- Primary field = Customer ID (UAN): Points to the *Primary profile field dropdown (UAN (SHORTWINDOW.SD_UAN)).
- Secondary field = Beneficiary: Points to the Optional profile field dropdown (SHORTWINDOW.SD_BEN_WIR_ACCT_IDENT).
- Value field: Points to the *Sum field dropdown (Amount 1 (SHORTWINDOW.MD_TRAN_AMT1)).

Custom Profiles

User Interface

Select Enhanced Profiling

Review Marked • 540356*****2497 • MC • 2020-06-22 14:48:53 • 901.07

All Transactions Marked Transactions Alerted Transactions **Enhanced Profiling** Standard Profiles Linked Accounts

1 - 37 Page Size 50

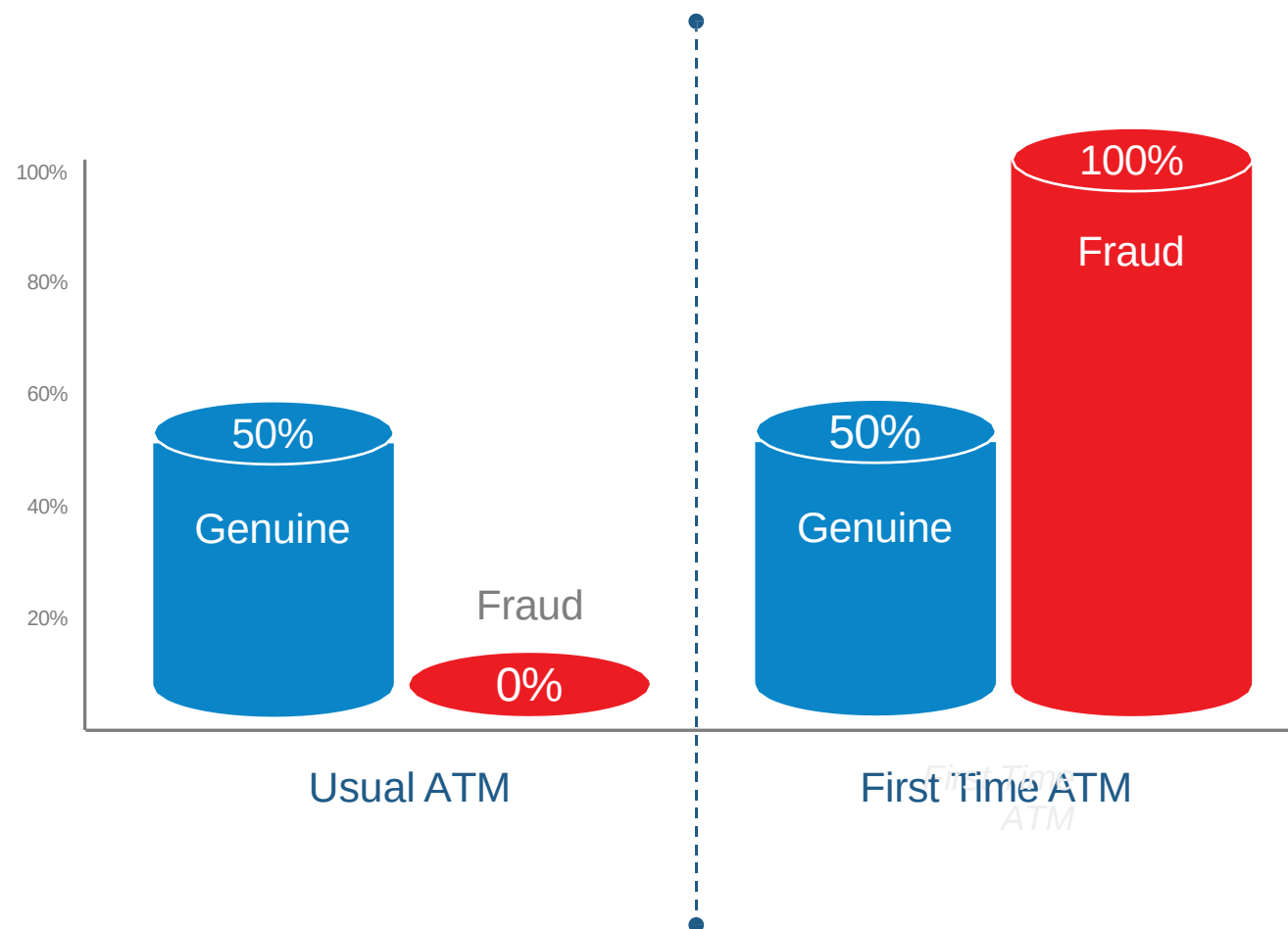
Profile Type	Name	Primary Field	Primary Value	Optional Field	Optional Value	Cycle	Date Field
Positive	CARD_MCC	SHORTWINDOW.SD_PAN	5403565225272497	Shortwindow.SD_RETL_SIC_CDE	9877	Monthly	SHORTWINDOW.DD_DATE
Positive	CARD_MCC	SHORTWINDOW.SD_PAN	5403565225272497	Shortwindow.SD_RETL_SIC_CDE	9777	Monthly	SHORTWINDOW.DD_DATE
Positive	CARD_MCC	SHORTWINDOW.SD_PAN	5403565225272497	Shortwindow.SD_RETL_SIC_CDE	6552	Monthly	SHORTWINDOW.DD_DATE
Positive	CARD_MCC	SHORTWINDOW.SD_PAN	5403565225272497	Shortwindow.SD_RETL_SIC_CDE	3424	Monthly	SHORTWINDOW.DD_DATE
Positive	CARD_MCC	SHORTWINDOW.SD_PAN	5403565225272497	Shortwindow.SD_RETL_SIC_CDE	3225	Monthly	SHORTWINDOW.DD_DATE
Positive	CARD_MCC	SHORTWINDOW.SD_PAN	5403565225272497	Shortwindow.SD_RETL_SIC_CDE	2232	Monthly	SHORTWINDOW.DD_DATE
Positive	CARD_MCC	SHORTWINDOW.SD_PAN	5403565225272497	Shortwindow.SD_RETL_SIC_CDE	1322	Monthly	SHORTWINDOW.DD_DATE
Positive	CARD_ATM_DOM	SHORTWINDOW.SD_PAN	5403565225272497	Shortwindow.SD_TERM_ID	ATM943763	Yearly	SHORTWINDOW.DD_DATE
Positive	CARD_ATM_DOM	SHORTWINDOW.SD_PAN	5403565225272497	Shortwindow.SD_TERM_ID	ATM870415	Yearly	SHORTWINDOW.DD_DATE

Tracking Card + MCC

Tracking Card + ATM ID + Domestic

Custom Profiles

ATM Use Case



Deposit History

Deposit History

Merchant Dimension IDs

Deposit History Daily 7 Day 30 Day 90 Day Peer Daily Peer 7 Day Peer 30 Day Peer 90 Day

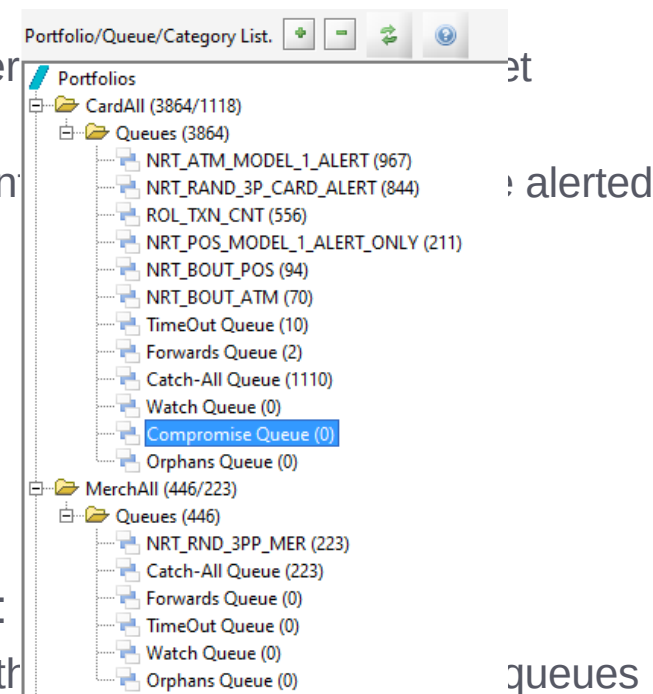
- Available for the Merchant Dimension ID
- Provides the following columns:
 - Date
 - Count of deposits
 - Count of transactions (*excludes deposits, refunds, retrieval requests or chargebacks, other non-monetary transactions or failed authorizations*)
 - Sum of deposits
 - Sum of transactions
 - Average deposit amount (*sum of deposits ÷ count of deposits*)
 - Average transaction amount (*sum of transactions ÷ count of transactions*)

Point of Compromise

Point of Compromise

CPP/POC Functionality

- The principle behind PRM's Point of Compromise (POC) functionality:
 - Multiple cards, with marked fraud, have performed transactions at the same merchant / terminal / period
 - All other cards that have performed transactions at the same terminal / merchant / period are alerted
- Parameters:
 - Time period
 - Number of marked accounts
 - Card Present and/or CNP
 - Merchants or Terminals
- *PP_GENPOCALERTS* procedure automatically runs at regular intervals:
 - Alerts are allocated to the Compromise queue which is assigned to analysts in the system



Point of Compromise

Compromise Queue

- Marked cards (that triggered the alert)

POC Alert, Retrieved from Compromise Queue POC_TERM596F • WAJlaZap

Demographic Data

Terminal Data

Terminal Information

Terminal ID	Type	Status	Owner ID	Acquirer Bank Name	Name and Location	Owner	Terminal Open Date	Cash Refresh Date
POC_TERM596F		68363295	CVnpwnt	cofEaCE	WAJlaZap	8041902523HYWwWfRYB	2004-08-14	2019-09-04 02:17:37

Service Date	Camera Service Date	Update Date and Time	Email	City	State/Province	Postal Code	Country
2019-09-03 20:36:37		2019-09-03 20:01:37	Levi.Terence30@acworldwide.com	Hong Kong	HI	447580	USA

Monetary Capacity: 158,542.76 Update Type: A

Marked Cards

Cards at Risk

Subject ID	Subject Type	Card Plastic Type	Alert Status	Mark Types (Start Date to To...
P3400 *****2622	Card	AM	Alerted	Acct. Takeover
P3400 *****4519	Card	AM	Alerted	Lost
P3400 *****0249	Card	MC	Not Alerted	Stolen
P3400 *****3520	Card	VS	Not Alerted	Lost

- Cards at Risk (those that transacted at same terminal/terminal with the same window)

Marked Cards

Cards at Risk

Blocked	Subject ID	Subject Type	Card ...	Alert Status	Mark Type...	Amount	Num...	Last Transaction Date
☐	P3400 *****2622	Card	AM	Alerted	Acct. Takeo...	22	1	2019-09-03 02:19:56
☐	P3400 *****0144	Card	MC	Alerted		22	1	2019-09-03 02:19:56
☒	P3400 *****1017	Card	AM	Not Alerted		22	1	2019-09-03 02:19:56
☐	P3400 *****4519	Card	AM	Alerted	Lost	22	1	2019-09-03 02:19:56
☐	P3400 *****0249	Card	MC	Not Alerted	Stolen	22	1	2019-09-03 02:19:56
☐	P3400 *****1099	Card	AM	Not Alerted		22	1	2019-09-03 02:19:56
☐	P3400 *****3520	Card	VS	Not Alerted	Lost	22	1	2019-09-03 02:19:56
☐	P3400 *****2739	Card	MC	Not Alerted		22	1	2019-09-03 02:19:56

List Management

List Management

Reference Data

- Positive or negative lists
- Load any type of data into the Reference Data table – watchlists, accounts, VIPs
- Data loaded via batch file from PRM
- All data is referenceable from within

Reference Data Manager

Load Data / Lookup Data

Filter: Reference Value 3 < Discrete value User-defined value User-defined value Reset

Reference ID	Reference Value	Reference Value 1	Reference Value 2	Reference Value 3	Reference Value 4	Reference Value 5
44123456786671	Compromised	Track Data	1/6/2017	5/6/2017	5812	Chipolte
44123456786672	Compromised	Track Data	1/7/2017	5/7/2017	5812	Chipolte
44123456786673	Compromised	Track Data	1/8/2017	5/8/2017	5812	Chipolte
44123456786674	Compromised	Track Data	1/9/2017	5/9/2017	5812	Chipolte
44123456786675	Compromised	Track Data	1/10/2017	5/10/2017	5812	Chipolte
44123456786676	Compromised	Track Data	1/11/2017	5/11/2017	5812	Chipolte
44123456786677	Compromised	Track Data	1/12/2017	5/12/2017	5812	Chipolte
44123456786678	Compromised	Track Data	1/13/2017	5/13/2017	5812	Chipolte
44123456786679	Compromised	Track Data	1/14/2017	5/14/2017	5812	Chipolte
BIINS	017916	PRIVATE LABEL CARD	OZON CLUB	DEBIT	RUS	643
BIINS	028037	PRIVATE LABEL CARD	BASHKOMSNAABANK	DEBIT	RUS	643
BIINS	100196	PRIVATE LABEL CARD	EDY/RAKUTEN	DEBIT	JPN	392
BIINS	180000	JCB	JCB CO. LTD.	CREDIT	JPN	392
BIINS	180001	JCB	JCB CO. LTD.	CREDIT	JPN	392
BIINS	180002	JCB	JCB CO. LTD.	CREDIT	JPN	392
BIINS	180003	JCB	JCB CO. LTD.	CREDIT	JPN	392
BIINS	180004	JCB	JCB CO. LTD.	CREDIT	JPN	392
BIINS	180005	JCB	JCB CO. LTD.	CREDIT	JPN	392
BIINS	180006	JCB	JCB CO. LTD.	CREDIT	JPN	392
BIINS	180007	JCB	JCB CO. LTD.	CREDIT	JPN	392
BIINS	180008	JCB	JCB CO. LTD.	CREDIT	JPN	392
BIINS	180009	JCB	JCB CO. LTD.	CREDIT	JPN	392
BIINS	180010	JCB	JCB CO. LTD.	CREDIT	JPN	392
BIINS	180011	JCB	JCB CO. LTD.	CREDIT	JPN	392
BIINS	180012	JCB	JCB CO. LTD.	CREDIT	JPN	392
BIINS	180013	JCB	JCB CO. LTD.	CREDIT	JPN	392
BIINS	180014	JCB	JCB CO. LTD.	CREDIT	JPN	392
BIINS	180015	JCB	JCB CO. LTD.	CREDIT	JPN	392
BIINS	180016	JCB	JCB CO. LTD.	CREDIT	JPN	392

Input File Information

Field Delimiter: ☒ Comma (,) ☐ Tilde (~)

Ignore First Line?: ☒ No ☐ Yes

Number of fields per record:

Number of lines per file:

Download file:

File:

Record Specifications

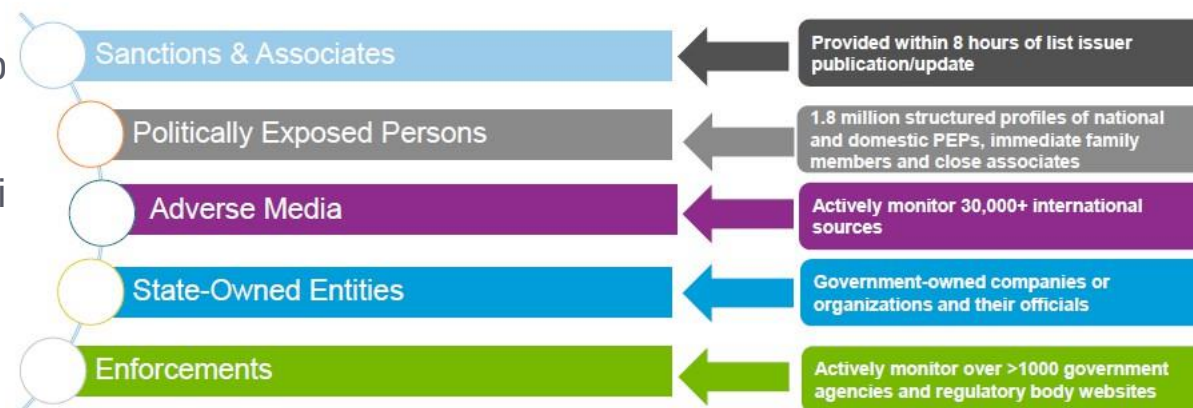
	Position	Fixed Values
Reference ID	☐	☐
Reference Value	☐	☐
Reference Value 1	☐	☐
Reference Value 2	☐	☐
Reference Value 3	☐	☐
Reference Value 4	☐	☐
Reference Value 5	☐	☐

List Management

With Lexis Nexus



- Verified partner of ACI & PRM
- Compliance screening:
 - Sanctions & associates
 - Provided within 8 hours of list/issuer publication/up
 - Politically Exposed Persons
 - 1.8 million structured profiles of national & domesti
 - Adverse media
 - Actively monitor 30,000+ international sources
 - State-owned entities
 - Government-owned companies or organisations ar
 - Enforcements
 - Active monitoring > 1,000 government agencies & regulatory body websites



Lexis Nexus

WorldCompliance Operations



- > 500 researchers
- 9 research centres
- 59 languages
- Geopolitical knowledge
- Cultural and linguistic knowle



Mark Detail Table

Mark Detail Table

Permanent Record of Marks

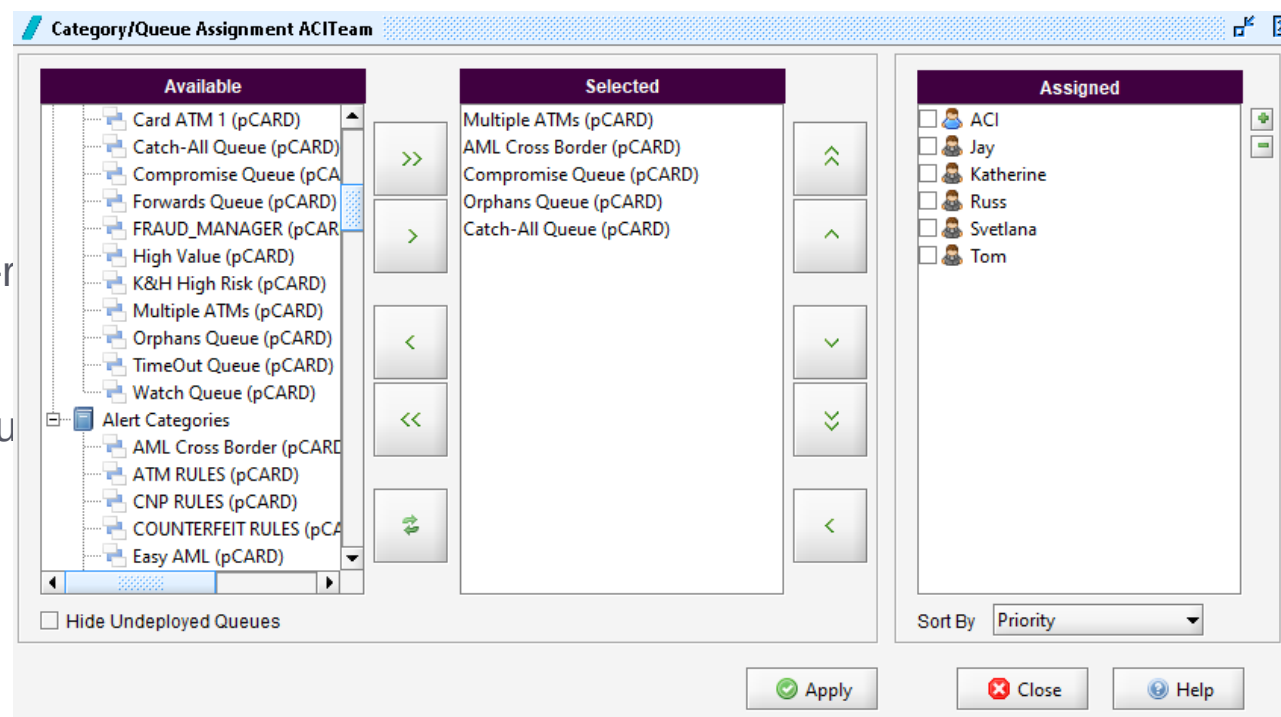
- All transactions confirmed to be fraud or identified as suspicious must be marked as such in PRM
- Marking transactions copies these transactions to the Mark Detail table
- This table serves as a permanent record of all marked transactions (those not marked will aged off after 90-180 days)
- Queries can be run against the Mark Detail table to determine key characteristics of fraudulent and suspicious transactions:
 - High risk countries
 - High risk MCC
 - High risk merchants or ATMs

Workflow Management

Workflow Management

Queues & Categories

- Create and manage workflow:
 - Granular queues (1:1 relationship to rules)
 - High-level categories (more like buckets)
- Assign & prioritize queues for each analyst
- Prioritise within queue eg: by amount, funds-at-risk
- Monitor queue status in real-time
- System queues “Orphans” & “Catch-All” to ensure coverage



Workflow Management

Robot Process Automation

- RPA through PRM's Auto-Actions
- When creating a rule, users can select up to 3 Auto-A
 - Block
 - Mark
 - Watch
 - Ice
 - Call
 - Memo
 - Response Required
- Each entry can have text recorded
- Every action is audited and visible to users (with acco

Auto Action

Block

Auto Action	Block
User	Block
Aggregate	Watch
Block Code	Mark
	Ice
	Memo
	Response Required

OK

Cancel

Help

Enter action description

+ Add Auto Action

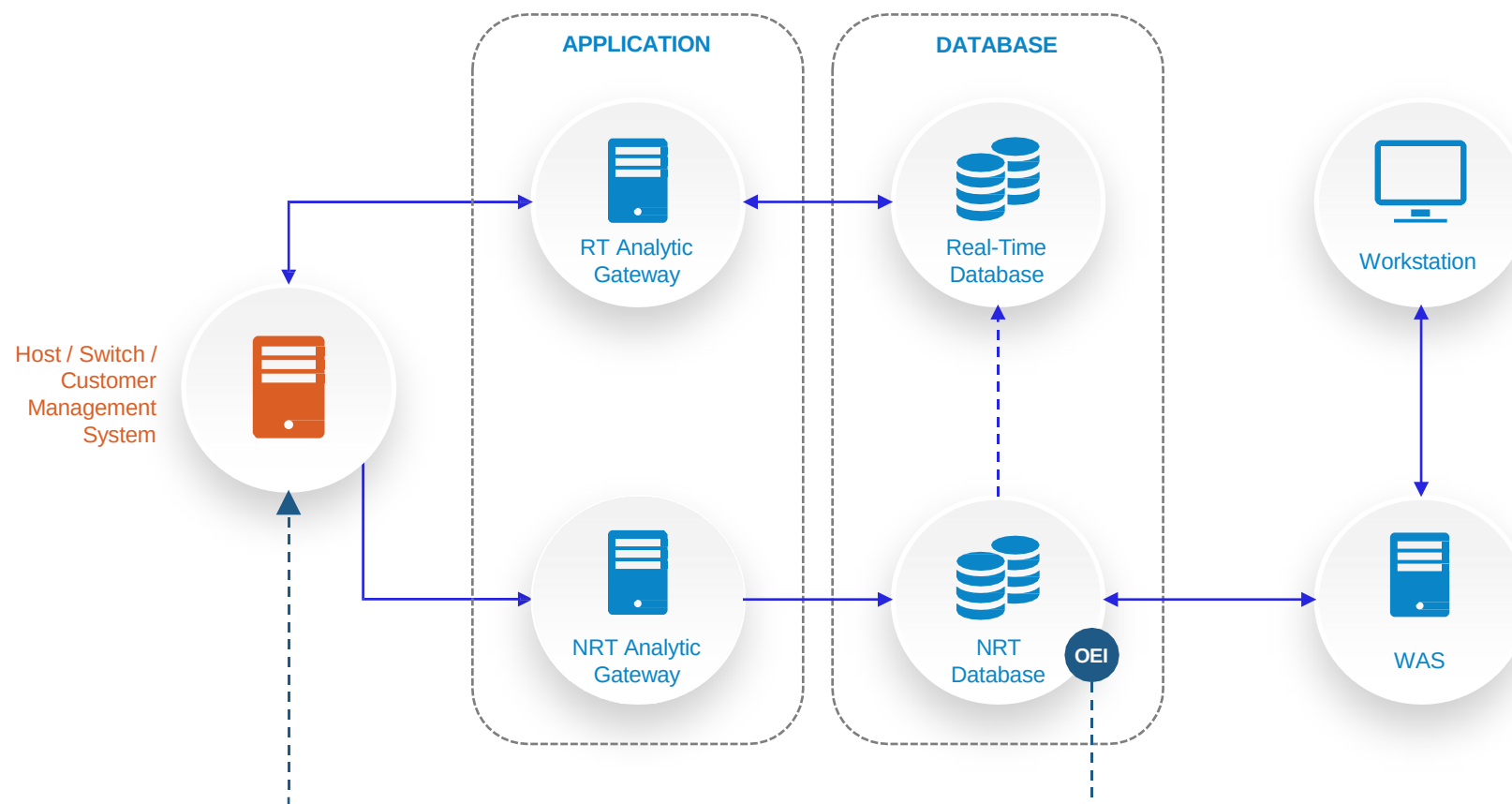
External Messaging

Online Extract Interface

- The Online Extract Interface (OEI) module is used to notify an external system or request actions eg:
 - Analyst manually blocks card in PRM
 - Event inserted into EMM (External Message Management) table
 - OEI automatically extracts new event and sends message to host requesting a block
 - Host responds with success or otherwise
 - If analyst subsequently unblocks card, a remove block message will be sent

Architecture

With OEI



Travel Notifications

Role & Access Management

User Access & Control

- PRM requires that a username and password be configured for all users
 - Supports the use of complex passwords, password aging, and password histories to prevent unauthorized users from using an ID by guessing the password.
- Custom roles
 - Greater granularity
 - Role templates
- Session timeout
 - Configurable inactivity timeout
- Mask Sensitive data
 - PAN, Account, etc
- Secure locations for extracting data and printing

User Management

The screenshot displays the 'Security' application window with the 'Users' tab selected. On the left, a tree view shows the organizational structure: 'Users' (expanded) contains 'AutoTeam', 'Admins', and 'BusinessTeam'. 'BusinessTeam' contains 'CardAll (Card)', 'MerchAll (Merchant)', 'AccountALL (Account)', 'UANALL (UAN)', 'TerminalALL (Terminal)', 'UAIDALL (User Account ID)', 'EmpALL (Employee)', 'amg', 'Manager', 'prm' (highlighted), and 'Supervisor'. The main area shows the configuration for user 'prm':

- *Name: prm
- *Password: [masked] with a 'Change Password' button.
- *Full Name: Mark Southby
- Phone: [empty field]
- E-Mail: southby@outlook.com
- Notes: [empty text area]
- Role: AllPermissionNoAdmin (dropdown menu)
- Buttons: 'Clear Failed Password' and 'Clear Failed Override'.
- Status: Last Login: 3/21/21 11:22:44 PM, Failed Password Attempts: 0, Shortwindow Group 1 Failed Override Attempts: 0.
- Buttons: 'Save' (green checkmark) and 'Cancel' (red X).

At the bottom of the window are 'Help' and 'Close' buttons.

Role Management

Functions Available

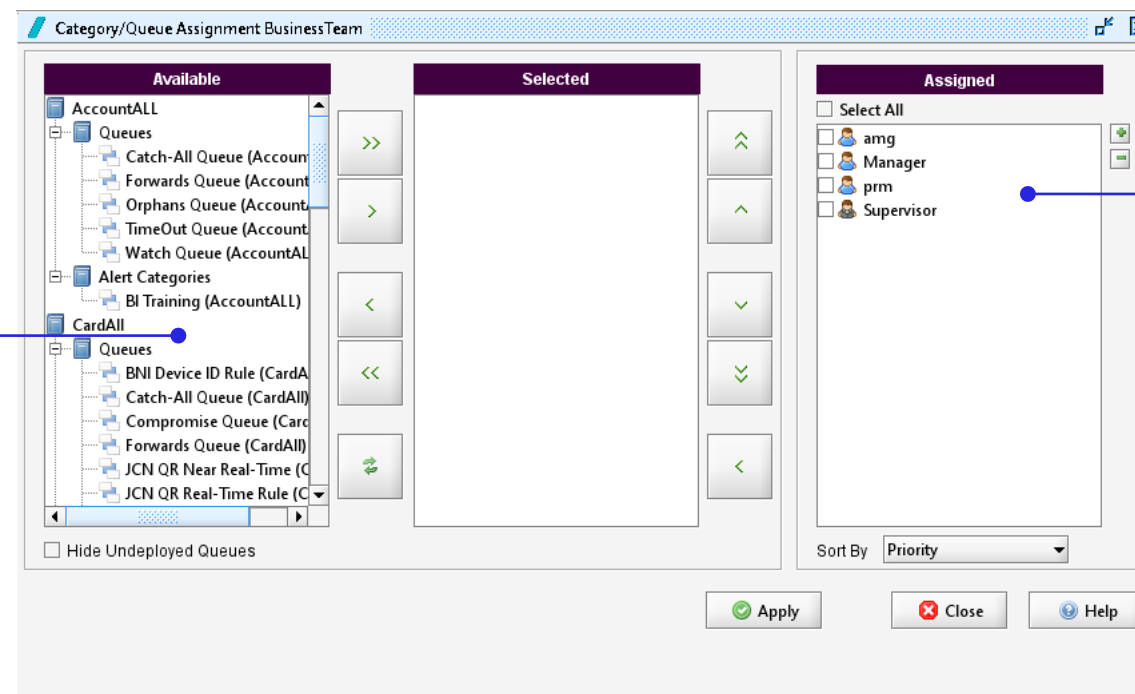
Scratchpad

Roles

The screenshot displays the 'Role Configuration' window. It features three main panes: 'Available' on the left, 'Selected' in the center, and 'Assigned' on the right. The 'Available' pane contains a tree view of functions, including 'File', 'Review', 'Workflow', and 'Config'. The 'Selected' pane is currently empty. The 'Assigned' pane lists various roles with checkboxes, such as 'AllPermissionNoAdmin', 'AllPermissions', 'AlmostALL', 'Root Administrator', 'System Administrator', 'Manager', 'Team Administrator', 'Supervisor', 'Analyst', and 'AutoUser'. A 'Templates' dropdown menu is located at the top, and a 'Delete' button is next to it. At the bottom, there is a 'Template Name' field with the value 'Senior_Analyst' and a 'Save' button. Other buttons at the bottom include 'Add Role', 'Order Roles', 'Apply', 'Close', and 'Help'. Blue lines with dots point from the text labels to specific elements: 'Functions Available' points to the 'Available' pane, 'Scratchpad' points to the 'Config' section in the 'Available' pane, and 'Roles' points to the 'Assigned' pane.

Queue Management

Available
Queues



User
Assignment

Operations

Tasks

Batch Jobs

- To assist DBAs determine the organisation's PRM database strategy, and support operations, a *Database Operation Manual* is provided
- A table in the manual provides an overview of procedures required, and frequency they should be run

PROCEDURE	FREQUENCY
PP_ACCOUNTTIMEOUT	Every 10 minutes
PP_CLEANUP_DECISION	Nightly
PP_PARTITION_DECISION	Nightly
PP_COPY_DETAIL_TO_FRAUDDetail and PP_GENPOCALERTS	Frequency is dependent upon business needs. PP_COPY_DETAIL_TO_FRAUDDetail should be run prior to PP_GENPOCALERTS and, therefore, it is recommended that these procedures be contained in the same job.
PP_ACTIVE_AL_REPORT	Nightly
*PP_CARD_PROFILES_JOB	Nightly
*PP_MER_PROFILES_JOB	Nightly
*PP_UAN_PROFILES_JOB	Nightly
*PP_TERM_PROFILES_JOB	Nightly
*PP_EMPLOYEE_PROFILES_JOB	Nightly
*PP_UAID_PROFILES_JOB	Nightly
*PP_UAN_PROFILES_JOB	Nightly
*PP_ACCT_PROFILES_JOB	Nightly
*PP_IP_PROFILES_JOB	Nightly
*PP_DEVICE_PROFILES_JOB	Nightly
*PP_PAYEE_PROFILES_JOB	Nightly
PP_ENH_PROFILE_CALC	Every 10 minutes if enhanced profiles are being used
PP_ENH_PROFILE_CLEANUP	Nightly
PP_TCR_STATISTIC	Nightly
PP_DASHB_RULE_PERF_CALC	Frequency is dependent upon business needs. Applicable only for users with A&R interface
PP_DASHB_AP_CALC	Frequency is dependent upon business needs. Applicable only for users with A&R interface

Tasks

Profile Jobs

- Authorised users are able to manage and review the standard profile jobs via user interface
- Jobs can be run in parallel as they insert into different tables
- User and system actions that affect job processing are audited

Standard Profile Rules Job Audit						Page 1 of 8275	
Column	Operator	Discrete value	User-defined value	User-defined value	Reset	Items On Page:100	
Filter: C:All							
User Id	Workstation	Action Details	Date	Message	Job Id		
0	Cou2prmdemo	Shutdown job threads	Nov 12, 2020 6:27:00 AM	Standard Profiles Job PRM90 - Shutdown job for action level Card and date 11/11/20 12:00 AM	0		
0	Cou2prmdemo	Shutdown job threads	Nov 12, 2020 6:27:00 AM	Standard Profiles Job PRM90 - Shutdown job for action level UAN and date 11/11/20 12:00 AM	0		
0	Cou2prmdemo	Finish job	Nov 12, 2020 6:27:00 AM	Standard Profiles Job PRM90 - Set status to FINISHED for job with action level id 1, date 11/11/20 12:00 AM, proc...	0		
0	Cou2prmdemo	Finish job	Nov 12, 2020 6:27:00 AM	Standard Profiles Job PRM90 - Set status to FINISHED for job with action level id 3, date 11/11/20 12:00 AM, proc...	0		
0	Cou2prmdemo	Finish job	Nov 12, 2020 6:27:00 AM	Standard Profiles Job PRM90 - Finished job, 0 keys processed successfully, 0 failed, 0 exceptions encountered.	0		
0	Cou2prmdemo	Finish job	Nov 12, 2020 6:27:00 AM	Standard Profiles Job PRM90 - Finished job, 0 keys processed successfully, 0 failed, 0 exceptions encountered.	0		
0	Cou2prmdemo	Run rules	Nov 12, 2020 6:27:00 AM	Standard Profiles Job PRM90 - Set status to RUNNING_RULES for job with action level id 1, date 11/11/20 12:00 A...	0		
0	Cou2prmdemo	Run rules	Nov 12, 2020 6:27:00 AM	Standard Profiles Job PRM90 - Set status to RUNNING_RULES for job with action level id 3, date 11/11/20 12:00 A...	0		
0	Cou2prmdemo	Shutdown job threads	Nov 12, 2020 6:27:00 AM	Standard Profiles Job PRM90 - Shutdown job for action level Merchant and date 11/11/20 12:00 AM	0		
0	Cou2prmdemo	Start job	Nov 12, 2020 6:27:00 AM	Standard Profiles Job PRM90 - Start job for Card - 11/11/20 12:00 AM	0		
0	Cou2prmdemo	Finish job	Nov 12, 2020 6:27:00 AM	Standard Profiles Job PRM90 - Finished job, 0 keys processed successfully, 0 failed, 0 exceptions encountered.	0		
0	Cou2prmdemo	Finish generating profile data	Nov 12, 2020 6:27:00 AM	Standard Profiles Job PRM90 - Set status to FINISHED_GEN for job with action level id 1, date 11/11/20 12:00 AM...	0		
0	Cou2prmdemo	Finish generating profile data	Nov 12, 2020 6:27:00 AM	Standard Profiles Job PRM90 - Set status to FINISHED_GEN for job with action level id 3, date 11/11/20 12:00 AM...	0		
0	Cou2prmdemo	Run rules	Nov 12, 2020 6:27:00 AM	Standard Profiles Job PRM90 - Set status to RUNNING_RULES for job with action level id 2, date 11/11/20 12:00 A...	0		
0	Cou2prmdemo	Start job	Nov 12, 2020 6:27:00 AM	Standard Profiles Job PRM90 - Start job for Merchant - 11/11/20 12:00 AM	0		
0	Cou2prmdemo	Start job	Nov 12, 2020 6:27:00 AM	Standard Profiles Job PRM90 - Start job for UAN - 11/11/20 12:00 AM	0		
-1	COV2PRMDEMO	Profile Debug	Nov 12, 2020 6:27:00 AM	Deleted 0 records from UAN_31_DAYS_PEER_PROFILE table.	-1		
-1	COV2PRMDEMO	Profile Debug	Nov 12, 2020 6:27:00 AM	Finished Card 31 Days Peer Profile.	-1		
-1	COV2PRMDEMO	Profile Debug	Nov 12, 2020 6:27:00 AM	Finished Copy Profile data into temporary tables 12020-11-11	-1		

Standard Profile Rules Job Status													Page 1 of 2	
Column	Operator	Discrete value	User-defined value	User-defined value	Reset	Items On Page:50								
Filter: C:All														
Id	Action level	Date	Start time	End time	Status	Recommendation	Successful	Failed	Process ID	Requested by process id	Counters last updated on	Running Rules		
97	Card	Nov 11, 2020	Nov 12, 2020 6:24:00 AM	Nov 12, 2020 6:24:00 AM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 12, 2020 6:24:00 AM	Nov 12, 2020 6:24:00 AM	RUNNING_RULES		
96	Account	Nov 11, 2020	Nov 12, 2020 12:00:00 AM	Nov 12, 2020 12:00:00 AM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 12, 2020 12:00:00 AM	Nov 12, 2020 12:00:00 AM	FINISHED		
95	Terminal	Nov 11, 2020	Nov 12, 2020 12:00:00 AM	Nov 12, 2020 12:00:00 AM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 12, 2020 12:00:00 AM	Nov 12, 2020 12:00:00 AM	FINISHED		
94	UAN	Nov 11, 2020	Nov 12, 2020 6:24:00 AM	Nov 12, 2020 6:24:00 AM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 12, 2020 6:24:00 AM	Nov 12, 2020 6:24:00 AM	RUNNING_RULES		
93	Merchant	Nov 11, 2020	Nov 12, 2020 6:24:00 AM	Nov 12, 2020 6:24:00 AM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 12, 2020 6:24:00 AM	Nov 12, 2020 6:24:00 AM	RUNNING_RULES		
92	Terminal	Nov 10, 2020	Nov 11, 2020 12:00:00 AM	Nov 11, 2020 12:00:00 AM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 11, 2020 12:00:00 AM	Nov 11, 2020 12:00:00 AM	FINISHED		
91	Account	Nov 10, 2020	Nov 11, 2020 12:00:00 AM	Nov 11, 2020 12:00:00 AM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 11, 2020 12:00:00 AM	Nov 11, 2020 12:00:00 AM	FINISHED		
90	Card	Nov 10, 2020	Nov 11, 2020 11:57:00 PM	Nov 11, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 11, 2020 11:57:00 PM	Nov 11, 2020 11:57:00 PM	RUNNING_RULES		
89	UAN	Nov 10, 2020	Nov 11, 2020 11:57:00 PM	Nov 11, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 11, 2020 11:57:00 PM	Nov 11, 2020 11:57:00 PM	RUNNING_RULES		
88	Merchant	Nov 10, 2020	Nov 11, 2020 11:57:00 PM	Nov 11, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 11, 2020 11:57:00 PM	Nov 11, 2020 11:57:00 PM	RUNNING_RULES		
87	Card	Nov 9, 2020	Nov 10, 2020 11:57:00 PM	Nov 10, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 10, 2020 11:57:00 PM	Nov 10, 2020 11:57:00 PM	RUNNING_RULES		
86	Terminal	Nov 9, 2020	Nov 10, 2020 12:00:00 AM	Nov 10, 2020 12:00:00 AM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 10, 2020 12:00:00 AM	Nov 10, 2020 12:00:00 AM	FINISHED		
85	Account	Nov 9, 2020	Nov 10, 2020 12:00:00 AM	Nov 10, 2020 12:00:00 AM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 10, 2020 12:00:00 AM	Nov 10, 2020 12:00:00 AM	FINISHED		
84	Merchant	Nov 9, 2020	Nov 10, 2020 11:57:00 PM	Nov 10, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 10, 2020 11:57:00 PM	Nov 10, 2020 11:57:00 PM	RUNNING_RULES		
83	UAN	Nov 9, 2020	Nov 10, 2020 11:57:00 PM	Nov 10, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 10, 2020 11:57:00 PM	Nov 10, 2020 11:57:00 PM	RUNNING_RULES		
82	Terminal	Nov 8, 2020	Nov 9, 2020 12:00:00 AM	Nov 9, 2020 12:00:00 AM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 9, 2020 12:00:00 AM	Nov 9, 2020 12:00:00 AM	FINISHED		
81	Card	Nov 8, 2020	Nov 9, 2020 11:57:00 PM	Nov 9, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 9, 2020 11:57:00 PM	Nov 9, 2020 11:57:00 PM	RUNNING_RULES		
80	UAN	Nov 8, 2020	Nov 9, 2020 11:57:00 PM	Nov 9, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 9, 2020 11:57:00 PM	Nov 9, 2020 11:57:00 PM	RUNNING_RULES		
79	Account	Nov 8, 2020	Nov 9, 2020 12:00:00 AM	Nov 9, 2020 12:00:00 AM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 9, 2020 12:00:00 AM	Nov 9, 2020 12:00:00 AM	FINISHED		
78	Merchant	Nov 8, 2020	Nov 9, 2020 11:57:00 PM	Nov 9, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 9, 2020 11:57:00 PM	Nov 9, 2020 11:57:00 PM	RUNNING_RULES		
77	Card	Nov 7, 2020	Nov 8, 2020 11:57:00 PM	Nov 8, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 8, 2020 11:57:00 PM	Nov 8, 2020 11:57:00 PM	RUNNING_RULES		
76	UAN	Nov 7, 2020	Nov 8, 2020 11:57:00 PM	Nov 8, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 8, 2020 11:57:00 PM	Nov 8, 2020 11:57:00 PM	RUNNING_RULES		
75	Merchant	Nov 7, 2020	Nov 8, 2020 11:57:00 PM	Nov 8, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 8, 2020 11:57:00 PM	Nov 8, 2020 11:57:00 PM	RUNNING_RULES		
74	Terminal	Nov 7, 2020	Nov 8, 2020 12:00:00 AM	Nov 8, 2020 12:00:00 AM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 8, 2020 12:00:00 AM	Nov 8, 2020 12:00:00 AM	FINISHED		
73	Account	Nov 7, 2020	Nov 8, 2020 12:00:00 AM	Nov 8, 2020 12:00:00 AM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 8, 2020 12:00:00 AM	Nov 8, 2020 12:00:00 AM	FINISHED		
72	Merchant	Nov 7, 2020	Nov 8, 2020 11:57:00 PM	Nov 8, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 8, 2020 11:57:00 PM	Nov 8, 2020 11:57:00 PM	RUNNING_RULES		
71	Card	Nov 6, 2020	Nov 7, 2020 11:57:00 PM	Nov 7, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 7, 2020 11:57:00 PM	Nov 7, 2020 11:57:00 PM	RUNNING_RULES		
70	UAN	Nov 6, 2020	Nov 7, 2020 11:57:00 PM	Nov 7, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 7, 2020 11:57:00 PM	Nov 7, 2020 11:57:00 PM	RUNNING_RULES		
69	Merchant	Nov 6, 2020	Nov 7, 2020 11:57:00 PM	Nov 7, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 7, 2020 11:57:00 PM	Nov 7, 2020 11:57:00 PM	RUNNING_RULES		
68	Terminal	Nov 6, 2020	Nov 7, 2020 12:00:00 AM	Nov 7, 2020 12:00:00 AM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 7, 2020 12:00:00 AM	Nov 7, 2020 12:00:00 AM	FINISHED		
67	Card	Nov 5, 2020	Nov 6, 2020 11:57:00 PM	Nov 6, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 6, 2020 11:57:00 PM	Nov 6, 2020 11:57:00 PM	RUNNING_RULES		
66	UAN	Nov 5, 2020	Nov 6, 2020 11:57:00 PM	Nov 6, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 6, 2020 11:57:00 PM	Nov 6, 2020 11:57:00 PM	RUNNING_RULES		
65	Merchant	Nov 5, 2020	Nov 6, 2020 11:57:00 PM	Nov 6, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 6, 2020 11:57:00 PM	Nov 6, 2020 11:57:00 PM	RUNNING_RULES		
64	Account	Nov 5, 2020	Nov 6, 2020 12:00:00 AM	Nov 6, 2020 12:00:00 AM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 6, 2020 12:00:00 AM	Nov 6, 2020 12:00:00 AM	FINISHED		
63	Terminal	Nov 4, 2020	Nov 5, 2020 12:00:00 AM	Nov 5, 2020 12:00:00 AM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 5, 2020 12:00:00 AM	Nov 5, 2020 12:00:00 AM	FINISHED		
62	Card	Nov 4, 2020	Nov 5, 2020 11:57:00 PM	Nov 5, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 5, 2020 11:57:00 PM	Nov 5, 2020 11:57:00 PM	RUNNING_RULES		
61	UAN	Nov 4, 2020	Nov 5, 2020 11:57:00 PM	Nov 5, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 5, 2020 11:57:00 PM	Nov 5, 2020 11:57:00 PM	RUNNING_RULES		
60	Merchant	Nov 4, 2020	Nov 5, 2020 12:00:00 AM	Nov 5, 2020 12:00:00 AM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 5, 2020 12:00:00 AM	Nov 5, 2020 12:00:00 AM	FINISHED		
59	Account	Nov 4, 2020	Nov 5, 2020 11:57:00 PM	Nov 5, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 5, 2020 11:57:00 PM	Nov 5, 2020 11:57:00 PM	RUNNING_RULES		
58	Terminal	Nov 4, 2020	Nov 5, 2020 11:57:00 PM	Nov 5, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 5, 2020 11:57:00 PM	Nov 5, 2020 11:57:00 PM	RUNNING_RULES		
57	Card	Nov 3, 2020	Nov 4, 2020 12:00:00 AM	Nov 4, 2020 12:00:00 AM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 4, 2020 12:00:00 AM	Nov 4, 2020 12:00:00 AM	FINISHED		
56	UAN	Nov 3, 2020	Nov 4, 2020 11:57:00 PM	Nov 4, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 4, 2020 11:57:00 PM	Nov 4, 2020 11:57:00 PM	RUNNING_RULES		
55	Merchant	Nov 3, 2020	Nov 4, 2020 11:57:00 PM	Nov 4, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 4, 2020 11:57:00 PM	Nov 4, 2020 11:57:00 PM	RUNNING_RULES		
54	Account	Nov 3, 2020	Nov 4, 2020 12:00:00 AM	Nov 4, 2020 12:00:00 AM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 4, 2020 12:00:00 AM	Nov 4, 2020 12:00:00 AM	FINISHED		
53	Terminal	Nov 3, 2020	Nov 4, 2020 11:57:00 PM	Nov 4, 2020 11:57:00 PM	FINISHED	No profile rules are deployed. Creat...	0	0	PRM90	Nov 4, 2020 11:57:00 PM	Nov 4, 2020 11:57:00 PM	RUNNING_RULES		
52	Card	Nov 3, 2020	Nov 4, 2020 11:57:00 PM	Nov 4, 2020 11:57:00 PM	FINISHED	No profile rules								

Data Extensibility

Data Extensibility

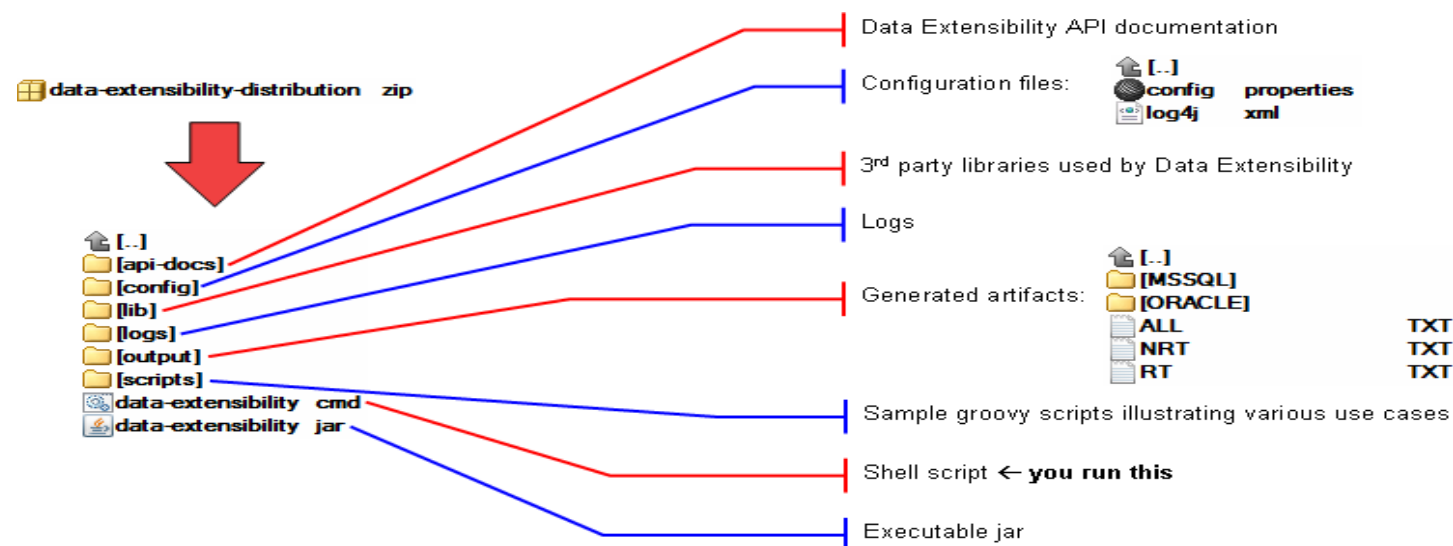
Beyond the Default Message Formats

- Extend the default message layouts
 - Add existing PRM fields
 - Add new fields
 - Or create new layouts
 - Use new fields in rules / aggregates / profiles / models
 - Position new demographic fields in the UI
 - Display new fields on review screen (Detail & Summary tabs)
 - Map Dimension IDs to the new layouts
 - Upgrade simply, custom layouts are carried forward
- Easy to deploy
 - Simple scripting language
- 
- The Groovy logo is a stylized, bubbly, and playful font in white with a thick black outline. It is set against a blue five-pointed star with a black outline. The word 'Groovy' is written in a cursive-like, rounded font.
- Different extensions per tenant if required

Data Extensibility

Customer Requirements

- PRM 9.0.2 or above
- JRE 8+
- The DE distribution package
- Knowledge:
 - Familiarity with the PRM database model
 - Basic structure programming & debugging
 - Basic object orientated programming
 - Basic Groovy scripting language



```
Administrator: C:\Windows\System32\cmd.exe - data-extensibility.cmd
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

d:\data-extensibility>data-extensibility.cmd
Usage: data-extensibility <path-to-groovy-script>
Press any key to continue . . .
```

Data Extensibility

Process Flow

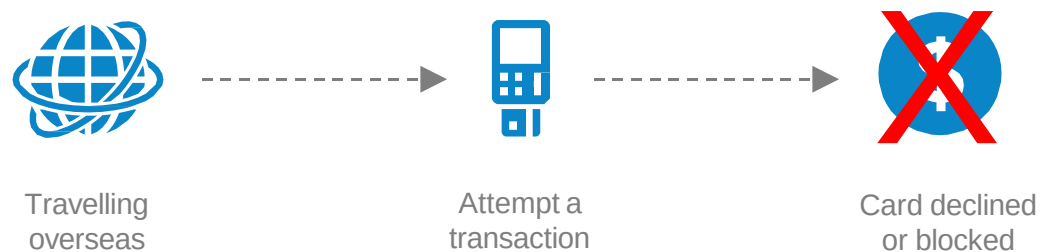
- DE is a module in PRM
- It has its own software that interprets Groovy scripts and produces output of ANT scripts
- ANT scripts are used by the PRM installer to create the SQL scripts that make the database changes



Mobile Alerting

MoneyGuard from our partner Spectrum

The Poor Customer Experience



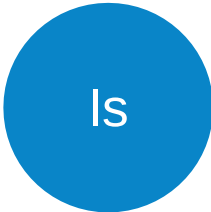
- Leaves a message on your home phone
- Sends one-way SMS
- Does nothing



- Rings the bank whilst roaming
- Manually authenticates
- Uses another card

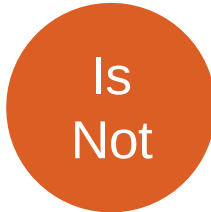
What is MoneyGuard?

And What It's Not



Is

- An interactive messaging platform that can connect back-end systems to a range of mobile channels
- Fully integrated with PRM
- Can interact via in-app notifications, SMS, email, IVR
- An enterprise wide customer contact solution



Is
Not

- An SMS gateway – it connects to those
- A single purpose messaging solution
- Just for fraud alert management
- Difficult to integrate

Why Mobile Alerting?



Customer Experience Is Paramount

Customer is immediately notified of an issue with a payment – quick response ensures transaction is accepted with minimal friction



Gain Additional Intelligence

Information garnered from a positive response can prevent future false positives eg: customer had not advised they were travelling to Bali



Time Is Of The Essence

Immediate customer feedback can prevent future losses - a "No" response can block card automatically; mass compromise – significant delays



Customer Preferred

Push notifications and SMS interaction are preferred by customers to calls and authentication questions; more trusted than email



Reduction In Costs

Operator contact is estimated \$6 to \$10 per call – SMS is cents, with push notifications negligible

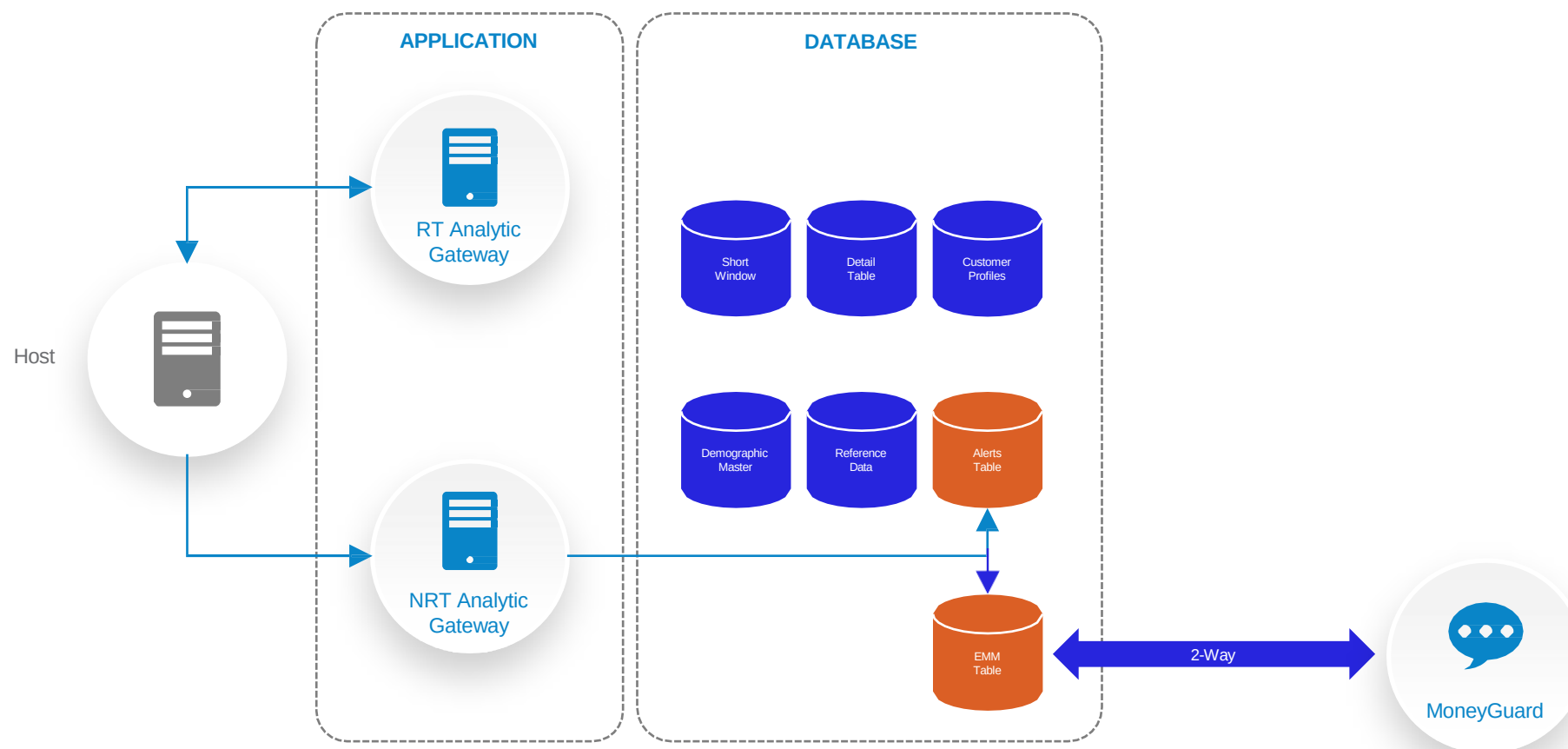


Reduce Fraud Losses

Immediate notifications reduce the point of detection; fraud ruleset / strategy can be "opened up"

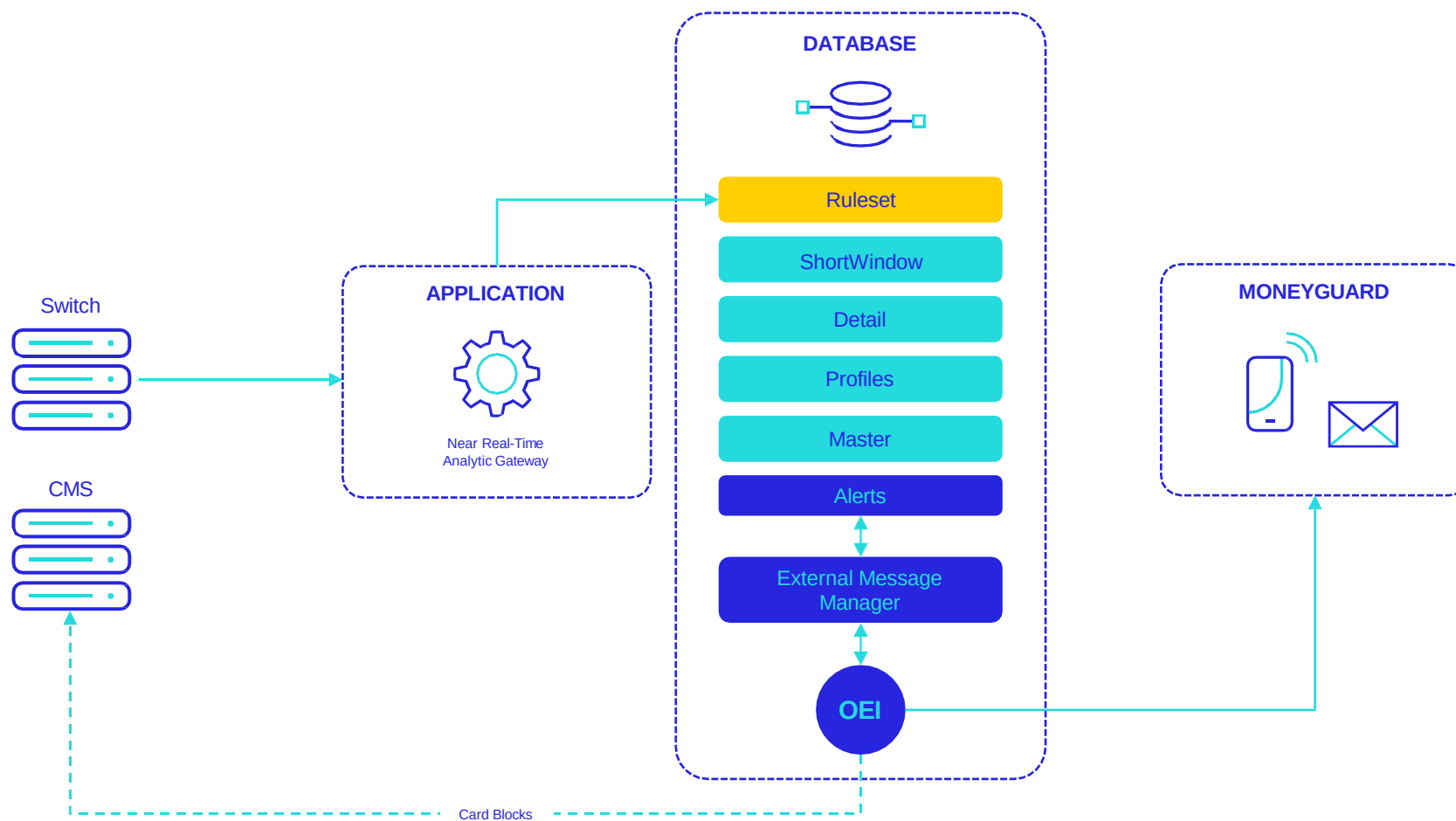
MoneyGuard

How it works



MoneyGuard

How It Works



PRM Integration

Controlled from PRM UI

- Advise MoneyGuard of specific PRM events through Notify Auto-Actions:
 - Alert
 - Decline
 - Block
- Specify notification type:
 - Push
 - SMS
 - Email
- Available for Card, UAN, Account and UAID Action Levels

Auto Action

Notify

Auto Action: Notify

User: none

Aggregate: none

Notification Type: none

Contact Strategy: none, Call, Determined by the external system, Email, Send Push, Send SMS

Enter action description

OK, Cancel, Help

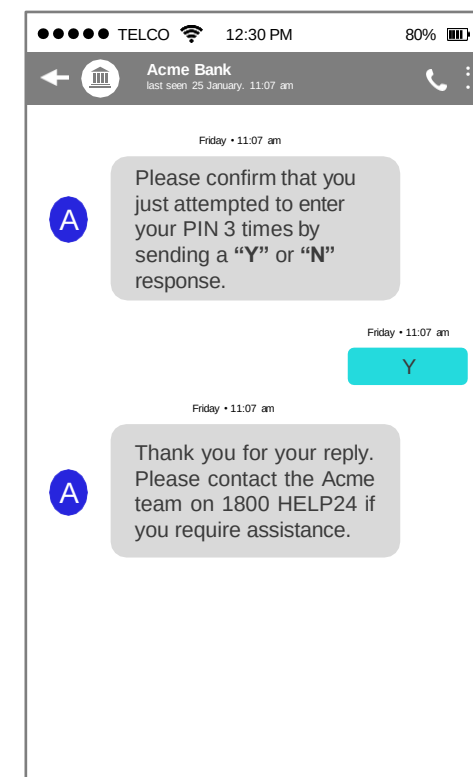
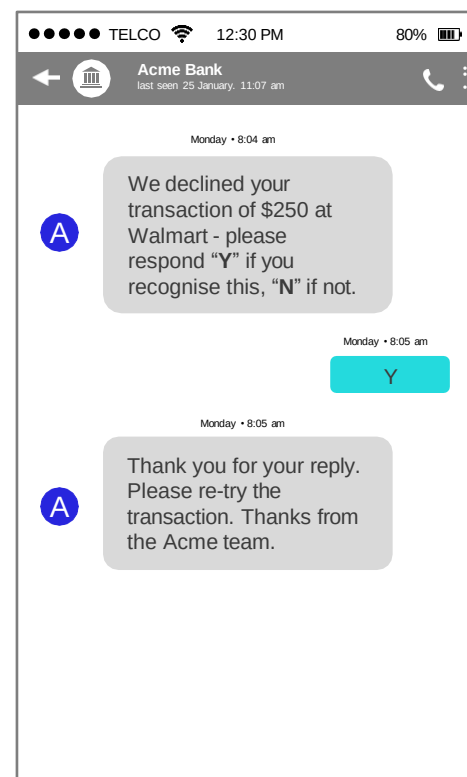
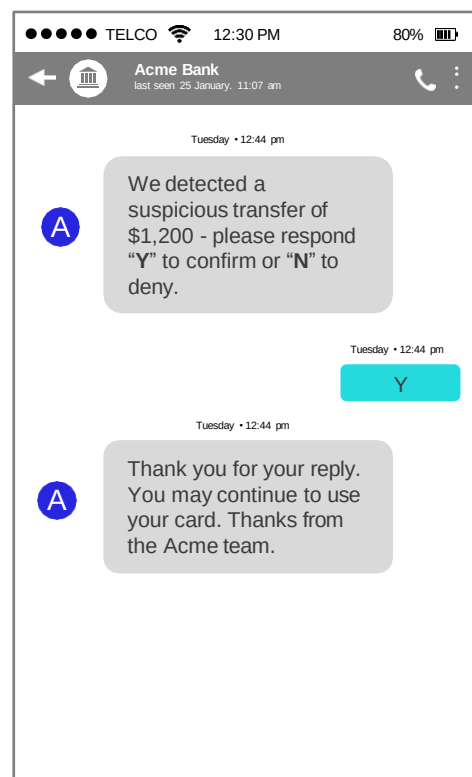
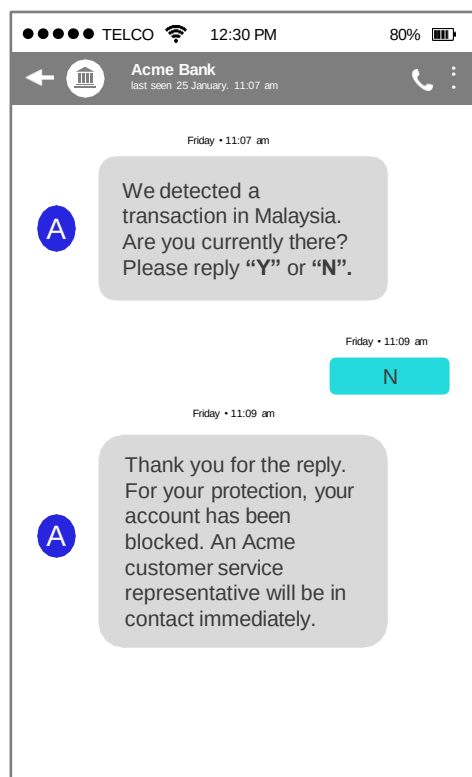
Add Auto Action

PRM Integration

Provides Automation

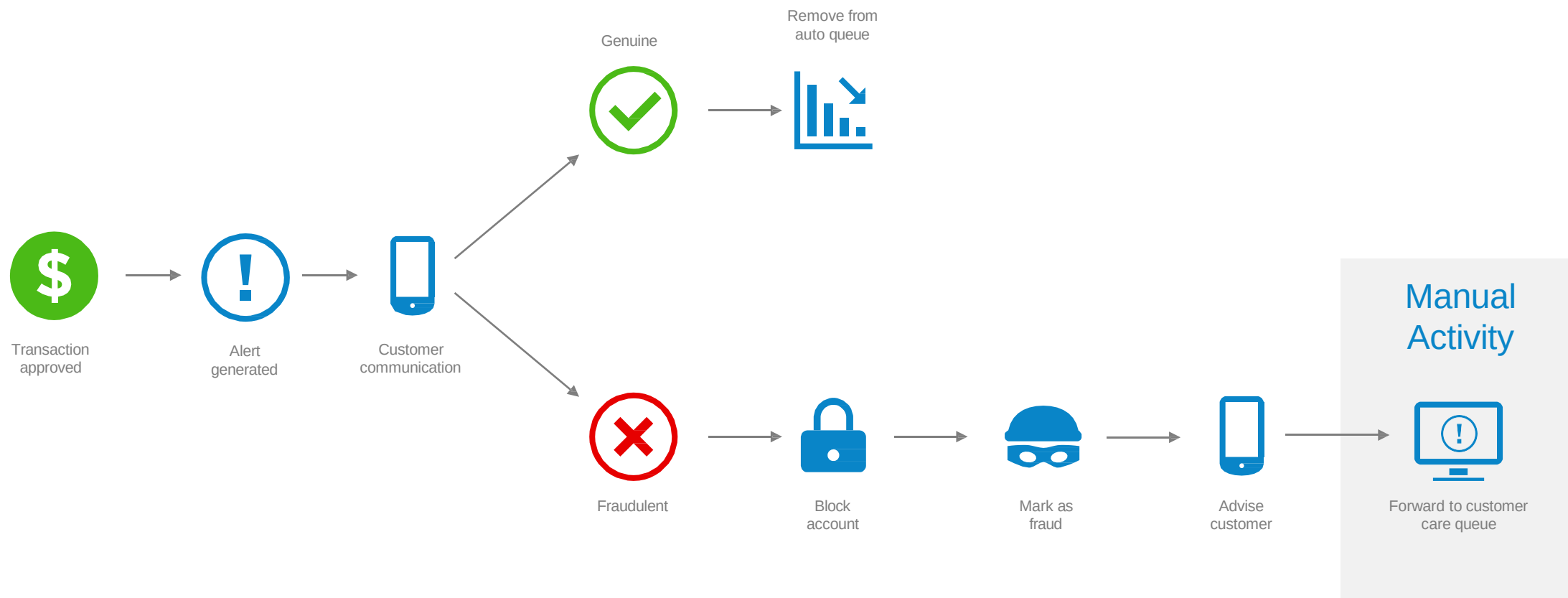
- Notify Auto-Action rule fires and details are sent to MoneyGuard:
 - Rule ID that fired
 - Key transaction fields
 - Real-time disposition eg: Approve, Decline, Hold, Step-Up
- MoneyGuard looks up contact details:
 - Confirm or deny via in-app notification / SMS
 - Dialogue is configurable and varies based on the rule type that fired
 - Response received from cardholder

Configurable Dialogue



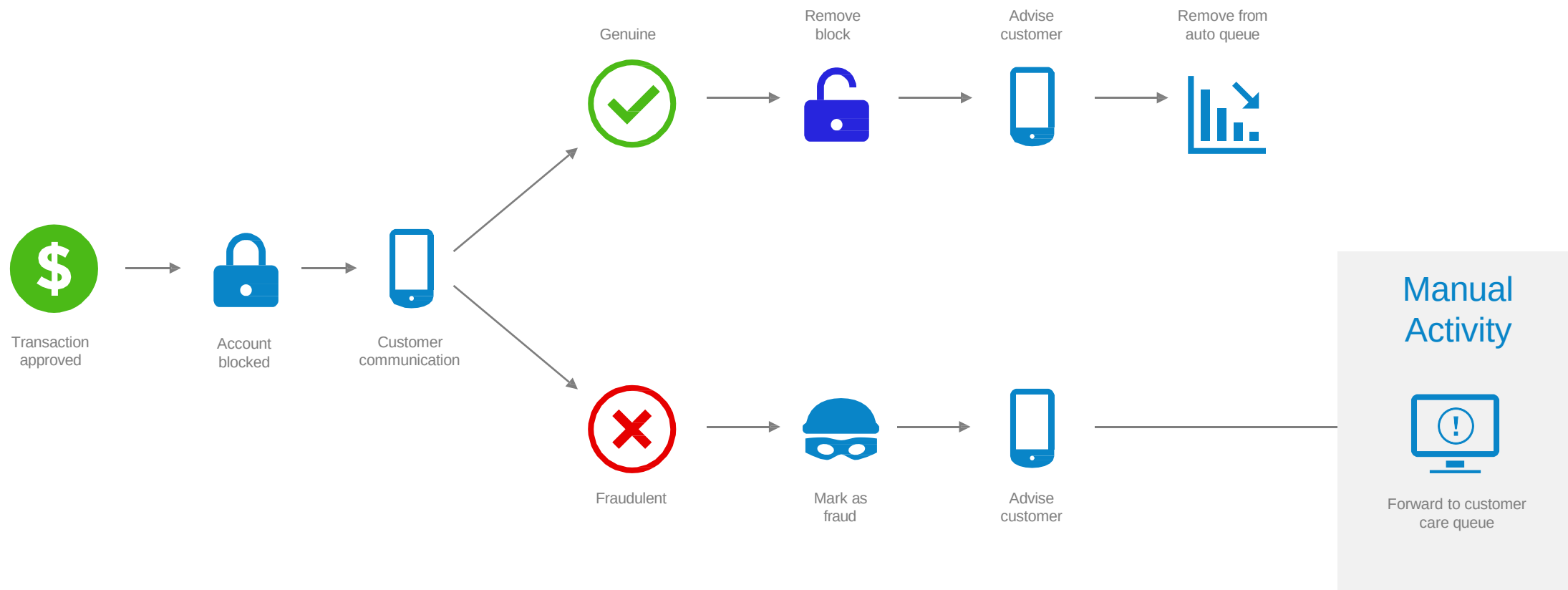
MoneyGuard Use Cases

Transaction Approved, Alert Generated



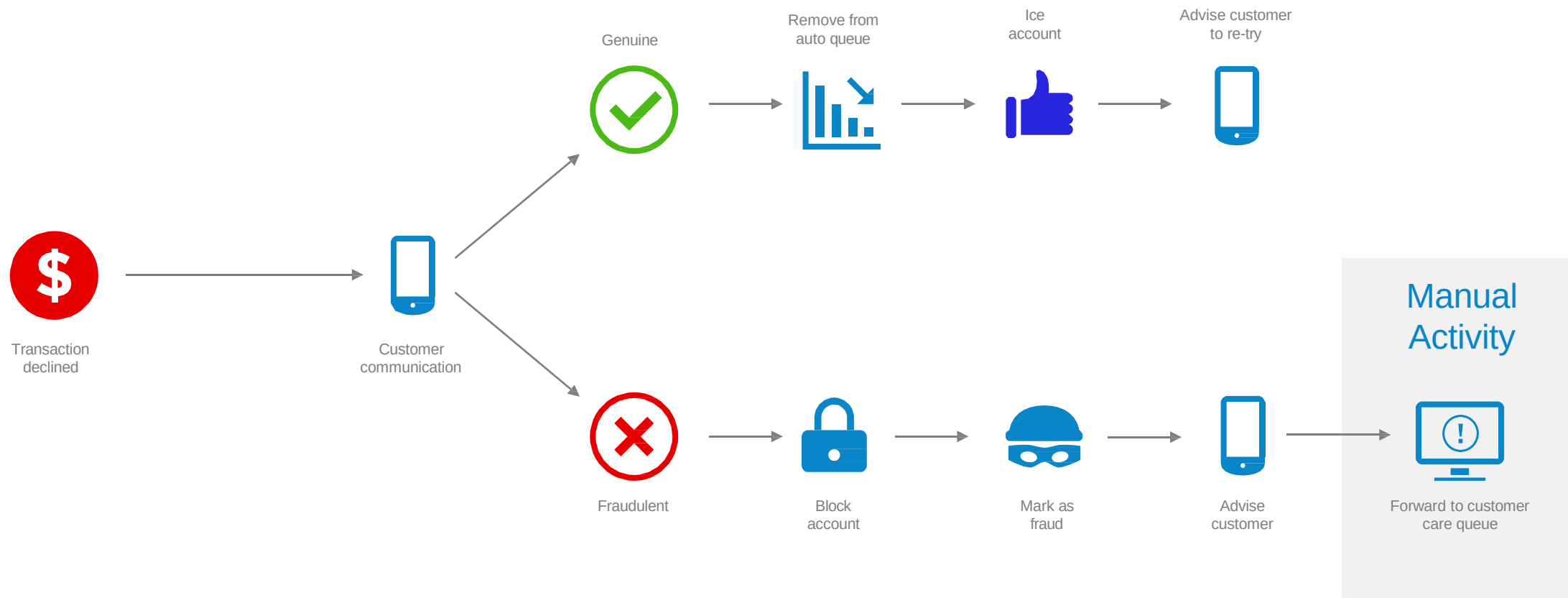
MoneyGuard Use Cases

Transaction Approved, Account Blocked



MoneyGuard Use Cases

Transaction Declined



PRM UI

All actions recorded

ACI DASHBOARD SEARCH REVIEW PENDING ALERTS REPORTS FORWARDS & REMINDERS English prm Logout

557125*****4314 - Card

Alert, Retrieved from - NRT_ATM_MODEL_1_ALERT (Consume and Discard Alerts) Marked • 557125*****4314 • MC • 2020-06-22 14:42:24 • 503.71 Help

Demographic Data Action History

Action Date and Time	Transaction Date and Time	User Name	Action Name	Action Details	Most Recent Transaction
2020-07-13 03:26:11	20200622 14:42:24.000	Prm Business User 1	Alerted	- NRT_ATM_MODEL_1_ALERT	
2020-06-16 08:15:47	20200616 08:15:47.343	AutoUser1	Marked		2020-06-14 08:04:58
2020-06-02 14:49:00	20200602 14:49:00.580	AutoUser1	Marked		2020-05-29 14:28:45
2020-05-29 11:29:26	20200529 11:29:26.587	AutoUser1	Marked		2020-05-27 08:05:22
2020-04-30 15:15:45	20200430 15:15:45.270	AutoUser1	Marked		2020-04-25 13:44:22

1 - 5 Page Size 50

All Transactions Marked Transactions Alerted Transactions Enhanced Profiling Standard Profiles Linked Accounts 1 - 50 of 94 Page Size 50

Transaction Date and Time	Retailer ID	Amount 1	NRT Score	MCC/SIC	Terminal ID	Terminal Name and Location	Terminal Country	Ten
2020-06-22 14:42:24		503.71	641	6011	ATM183277	910 FIRST AVENUE	USA	IL
2020-06-22 14:37:43		1,140.57	639	6011	ATM843782	136 SIXTH STREET	USA	MT
2020-06-22 12:17:42	MR71726146	72.41		8790	POS71388421	447 LIBERTY STREET		
2020-06-22 12:13:24		467.39	642	6011	ATM158292	430 FIRST STREET	USA	IA
2020-06-21 14:21:49	MR21987214	445.88		2232	POS68136361	829 FIRST AVENUE		
2020-06-21 14:15:36		806.61	648	6011	ATM056297	833 THIRD AVENUE	USA	MA
2020-06-21 14:04:02		909.88	646	6011	ATM327882	63 THIRD AVENUE	USA	ME
2020-06-21 13:57:57		202.66	618	6011	ATM787422	972 SECOND AVENUE	USA	MA
2020-06-21 13:39:53		13.46	319	6011	ATM968233	341 FIRST STREET	USA	ME
2020-06-20 13:46:04		1,296.87	650	6011	ATM681754	576 SIXTH STREET	USA	OR
2020-06-20 12:24:35		0.00	510	6011	ATM691571	446 SIXTH STREET	USA	AR

In-App Notifications

Examples of push alerts



Transaction confirmation

Did you request a transfer of
50 USD
to **Mark Southby**

Please press Yes to confirm or No if this was not you

YES

NO

Comments (optional)





Thank you for identifying this transaction as fraudulent

Your card has been blocked and you can no longer use it

Please contact the bank if you need assistance:

Domestic

Overseas

Or leave a message for the fraud team:

Submit message

TD Fraud Alerts

Powered by MoneyGuard



Interac Flash®



Shop online and worldwide
with Visa Debit



Safe and secure



TD Fraud Alerts

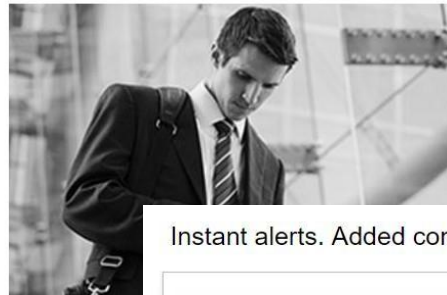
TD Fraud Alert texts, right to your phone.

Get TD Fraud Alerts – we will instantly¹ send you text messages notifying you if we detect suspicious activity made with your TD Access Card for your personal banking accounts.

These alerts will include information about the transaction and ask you to confirm it. You can respond to the texts quickly and easily by replying with a "Y" or "N". Based on your response, we will unblock or block your card immediately.

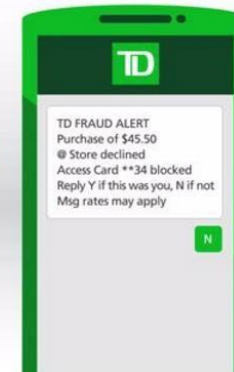
TD will never ask you to reply with any personal information via text.

- **Convenient.** Receive TD Fraud Alert text messages alerting you about suspicious activity - you can respond using your mobile phone.
- **Be in control.** If you receive a TD Fraud Alert, you can reply by text message to confirm the transaction and avoid the inconvenience of an unnecessarily blocked card.
- **Fast.** Your card is blocked or unblocked instantly, with minimal disruption to your day.



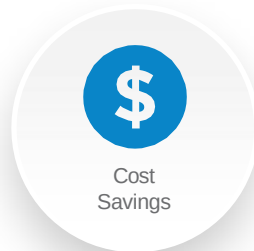
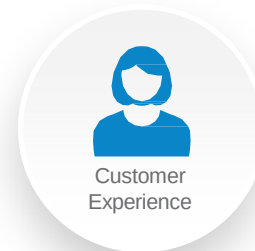
Instant alerts. Added convenience with TD Fraud Alerts

Reply **N** to the



ROI

Four value propositions



ROI

MoneyGuard with PRM

\$8



Is the **average** cost to the **bank** for a **human** to make the **call**

50%



Is the level of **automation** typically achieved within **3-6 months**

70%

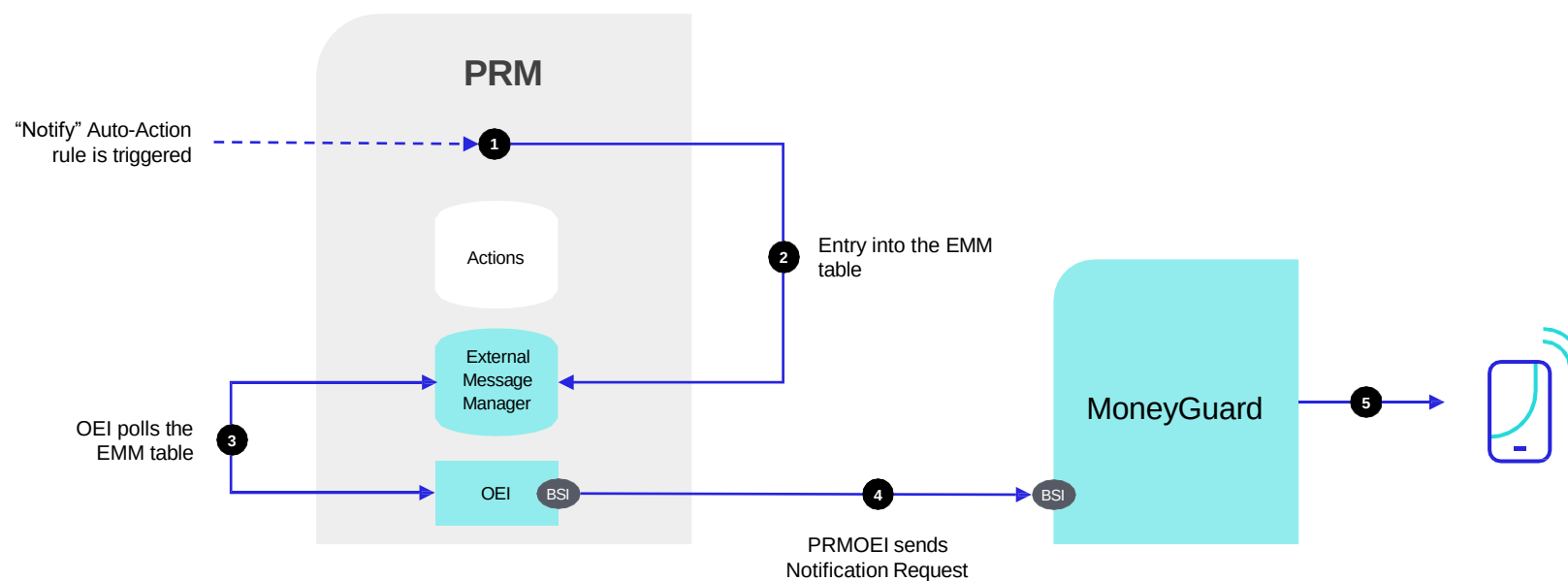


Is the best **customer response** rate seen

- Not to mention:
 - Reduction in Point of Block rate (quicker customer response)
 - Fewer alerts (customer response is considered in rules and suppresses future alerts)
 - Increased fraud detection, potentially opening up rules / alerts due to automation
 - One customer up to 70% automation; goal is 90%
 - Fully integrated with PRM

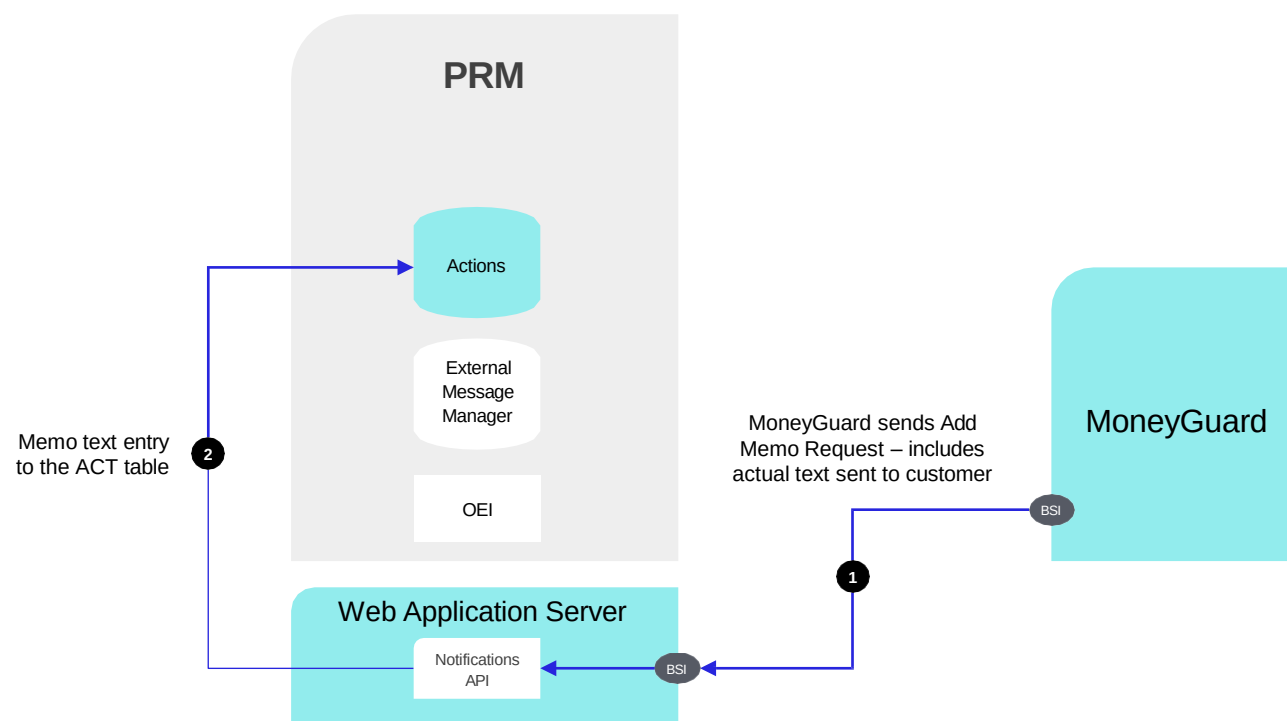
MoneyGuard & PRM

Detailed Workflow 1 of 4



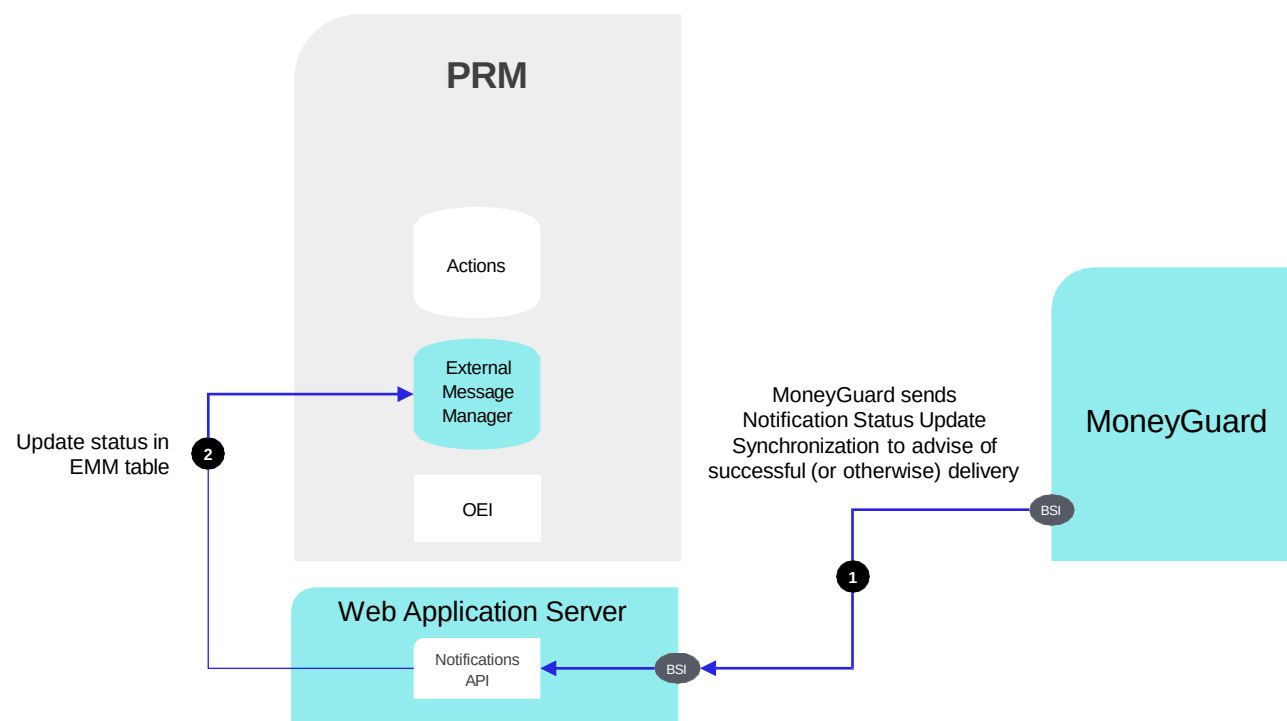
MoneyGuard & PRM

Detailed Workflow 2 of 4



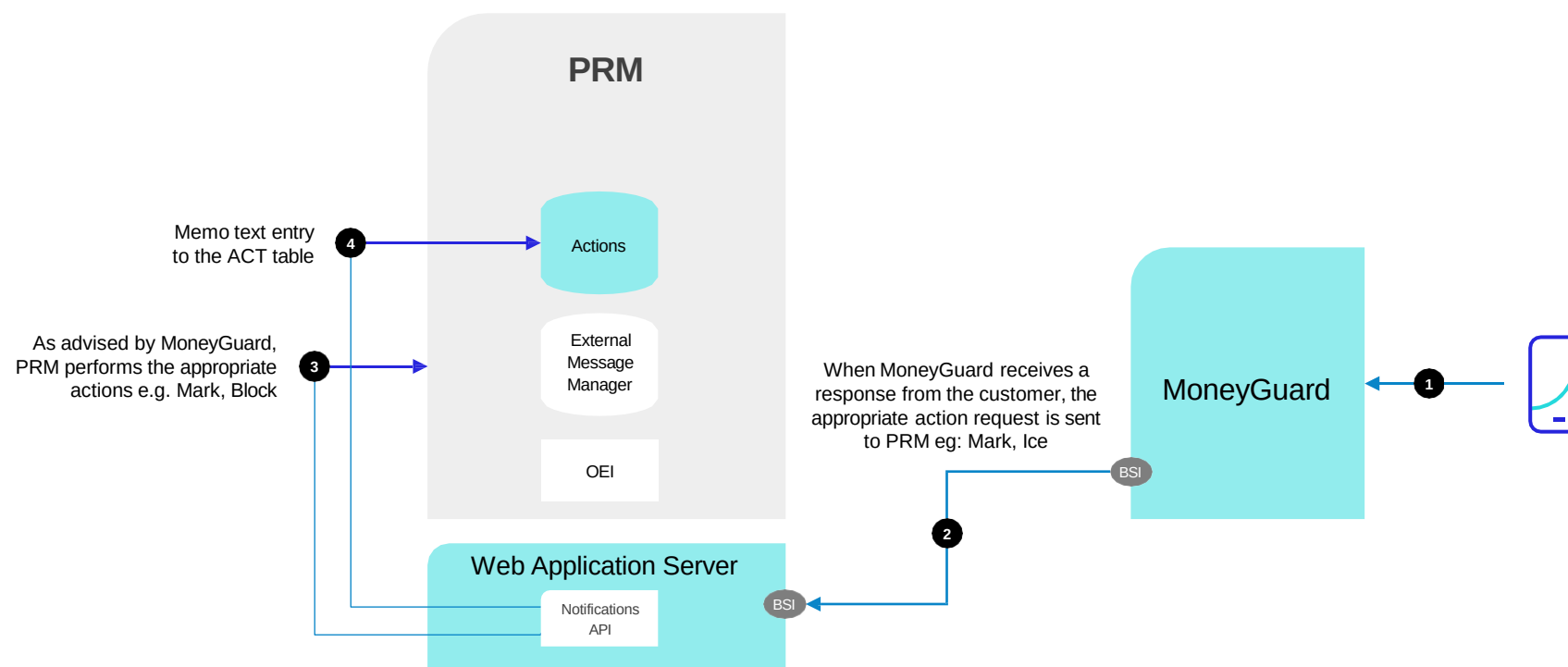
MoneyGuard & PRM

Detailed Workflow 3 of 4



MoneyGuard & PRM

Detailed Workflow 4 of 4



Access Management

Case Management

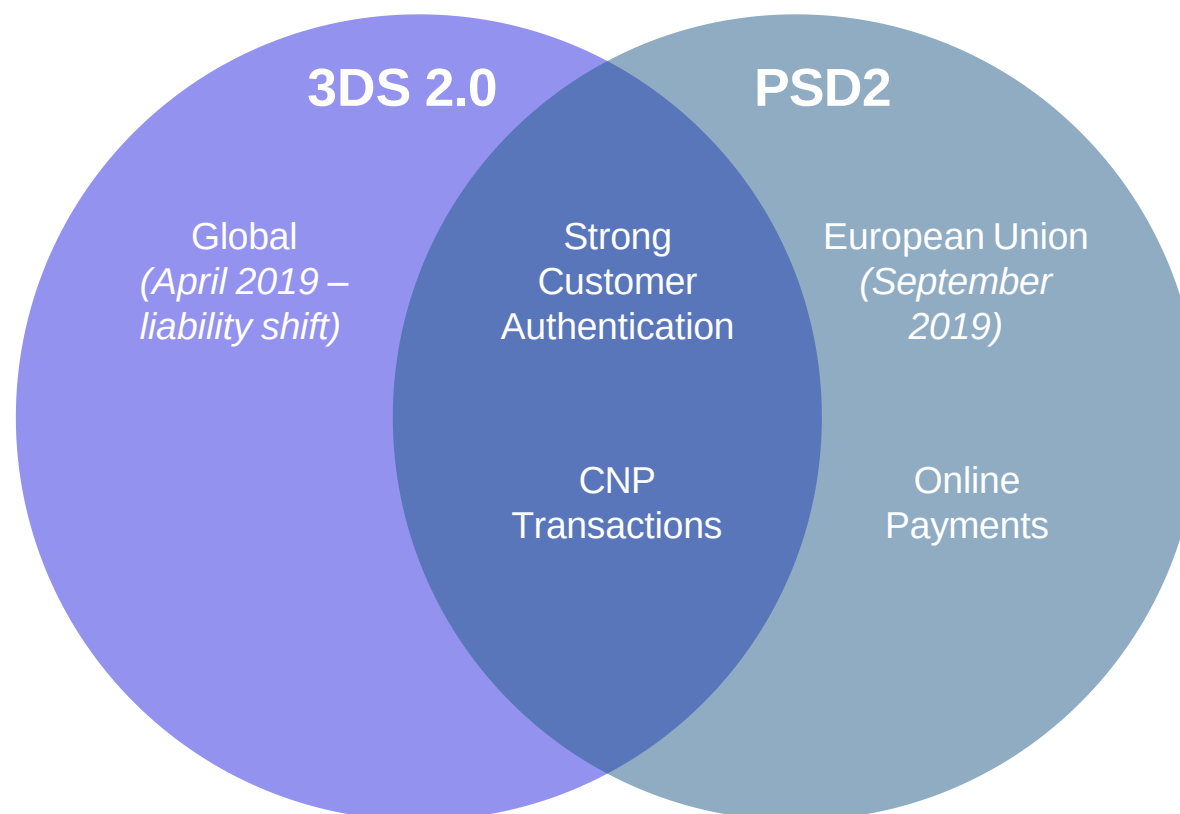
From our partner Polonious

3DS2 & STRONG CUSTOMER AUTHENTICATION

13 December 2018

3DS2 vs PSD2

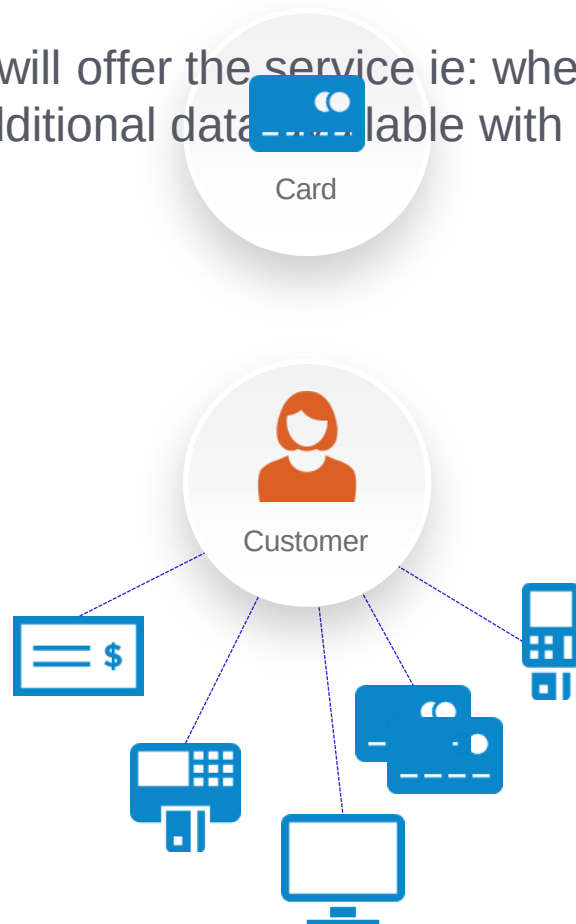
Same But Different



3DS2

Strong Customer Authentication

ACS providers will offer the service ie: whether to step-up authentication based on previous behaviour for the card and additional data available with 3DS2 (IP address, browser / device details)

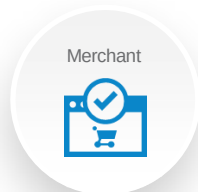


But nobody knows the cardholder like the issuer:

- Customer, not card, behaviour eg: same Device ID may be recorded on other cards, digital banking
- CNP purchase history across multiple cards
- POS / ATM history across multiple cards
- Suspicious activity on other accounts

3DS 2.0

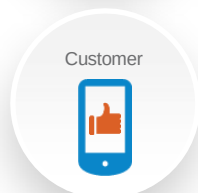
Customer Experience – Use Case 1



Customer checks out, merchant sends authentication request



ACS decides high risk – OOB authentication request to customer



Customer authenticates successfully eg: OTP, fingerprint



Customer authorisation request (financial message) deemed high risk



3DS 2.0

Customer Experience – Use Case 1

Decline in real-time



- Poor experience as customer will feel they have "authorised" but were then declined

Do not decline when SCA has been performed?



- Increase in fraud losses

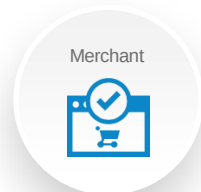
Second 2FA from issuer?



- Poor experience, customer will be asked to authenticate twice, likely lead to abandonment

3DS 2.0

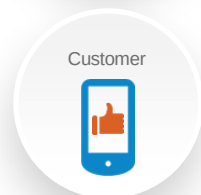
Customer Experience – Use Case 2



Customer checks out, merchant sends authentication request



ACS decides high risk – OOB authentication request to customer



Customer authenticates successfully eg: OTP, fingerprint



Customer authorisation request (financial message) deemed low risk



3DS 2.0

Customer Experience – Use Case 2

- With all the additional information available to PRM, transaction was deemed low risk and SCA assessment would have exempted
- Should have exempted, yet customer was required to authenticate
- Not the ultimate customer experience:



SCA

With PRM & MoneyGuard

- Authentication & authorisation in the one place
 - ACS just a "pass through"
 - PRM, with additional customer transaction and behaviour history, determines whether SCA is required or exempt with greater accuracy
 - If SCA is required, automated call to MoneyGuard
- MoneyGuard options:
 - OTP SMS
 - Push notification (can confirm the registered device was used)
 - Biometrics
 - Fingerprint
 - Facial recognition
 - Verifies both device and customer identity

Non-Payment Transactions

Visa Secure

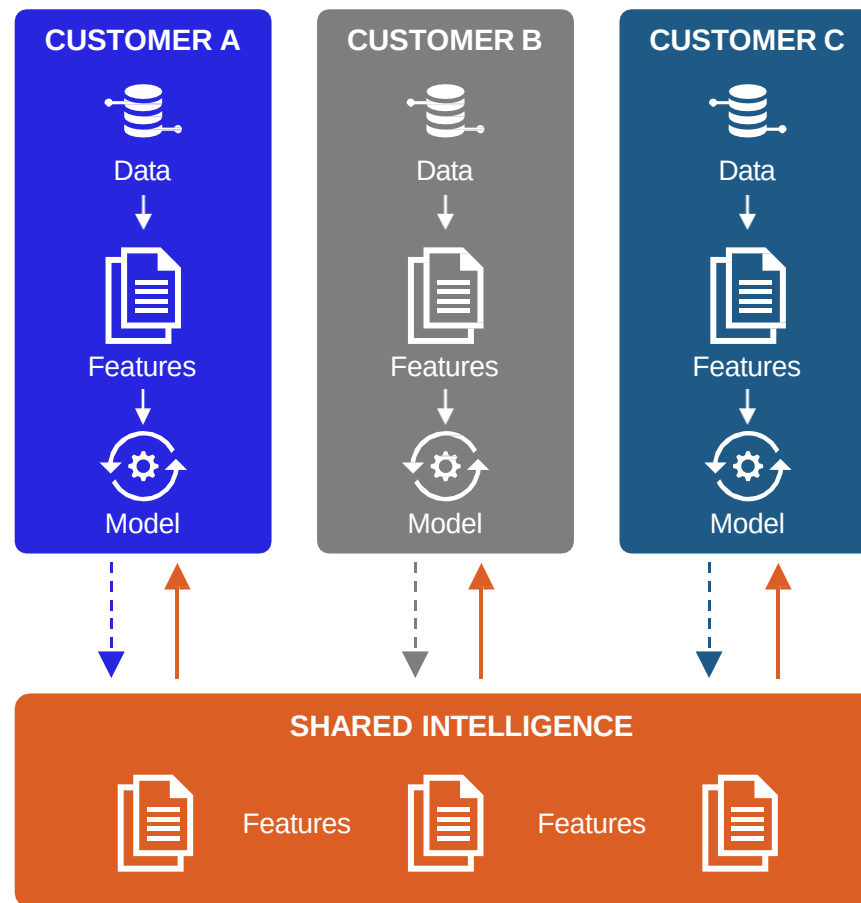
- NPA requests enable merchants to submit authentication requests for non-payment use cases e.g. adding a card, modifying existing cardholder information
- When a merchant sends an NPA AReq:
 - If authenticated with a challenge, the RReq must contain a) an ECI of 5 and b) a CAVV
 - If authenticated without a challenge (frictionless), ECI and CAVV not required
- **I don't understand the following:**
 - If a merchant sends an NPA as part of a 3DS Requestor Initiated (3RI) authentication request (e.g., a merchant-initiated authentication request when the cardholder is not available) and the transaction is successfully authenticated, the ECI and CAVV are not required.
 - a 3DS Server sends an NPA AReq with 3DS Requestor Authentication Indicator of “06 = Cardholder Verification as part of EMV token ID&V” an issuer must respond with a challenge request and the 3DS Server must proceed with initiating the challenge.
- Effective 14 April 2021, all issuers must support and respond to NPA transactions with the above requirements

PRM as CENTRAL INFRASTRUCTURE

Network Intelligence

Network Intelligence

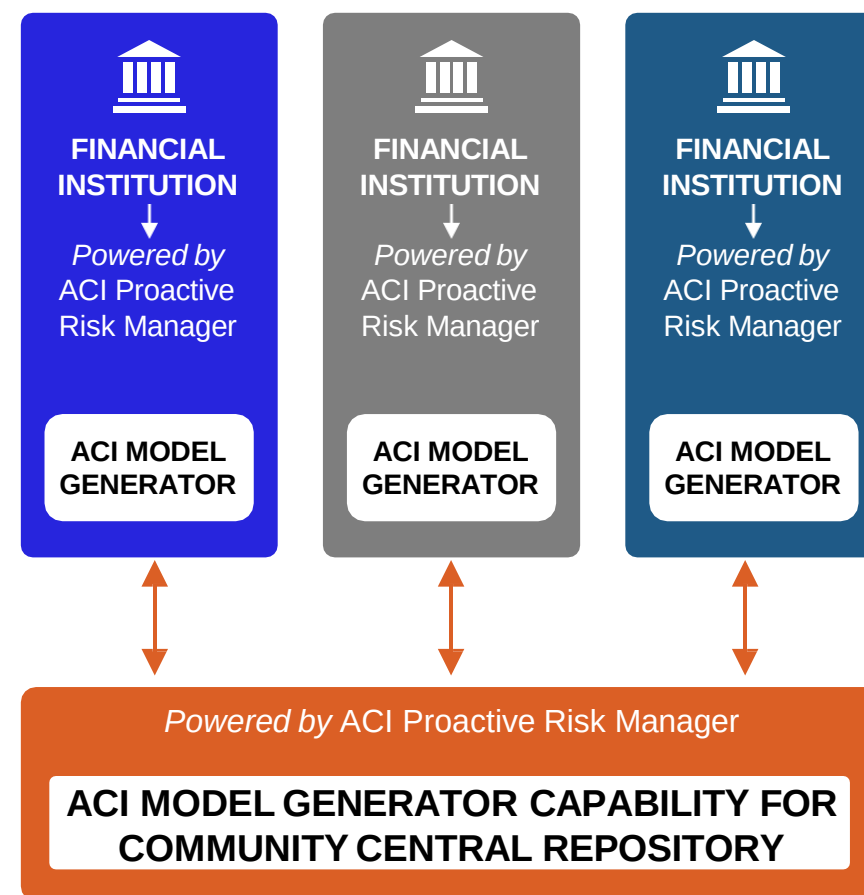
Empowers unprecedented collaboration among FIs



Network Intelligence

Empowers unprecedented collaboration among FIs

- Members receive feature values from the community central repository, feedback their **local** risk scores
- Central repository aggregates information, calculates new feature values, shares back through the community
- Members “pick and choose” decide whether to adopt (or adapt) new features for their **local** models



Reporting

Reports

User Activity Report

User Activity History Report

Filter Report

Users

Filter Users

☐ Guest1

☒ Guests

☐ fdfdfgd

☒ Guest2

☐ Guest3

☐ UL_SERVICE

Actions

Filter Actions

☒ All Actions

☒ Action Level Memo

☒ Alert

☒ Alert Delete

☒ Alert Email

☒ Alert Review Timeout

☒ Block

☒ Call

Statuses

Filter Statuses

☒ All Statuses

☒ Completed

☒ EM Statuses

☒ Requested

☒ Failed

☒ Submitted

Dimensions

Filter Dimensions

☒ All Dimensions

☒ Card

☒ Merchant

☒ Customer

☒ Terminal

☒ Employee

☒ User Account ID

☒ Account

From

2020-03-11

To

2020-03-11

Today

Past week

Past 2 weeks

Past 3 weeks

Past 4 weeks

Past 30 days

Past 60 days

Past 90 days

APPLY

RESET TO DEFAULT

Action Name	Dimension ID	Dimension	Action Date and Time	User Name	Action Details
Review Timed out	A03456***0ADE	Card	2020-03-11 10:49:20	Guest2	REVIEW LOCK REVOKED FOR USER - REVIEW OPEN BEYOND ALERTTIMEOUT INTERVAL
Opened Review	A03456***0ADE	Card	2020-03-11 05:03:07	Guest2	
Override/ice Ctr Req	A03456***0ADE	Card	2020-03-11 04:43:41	Guest2	Override/ice between <2020-03-11 04:43:07.00> and <2020-03-13 04:58:00.00>
Override/ice Req	A03456***0ADE	Card	2020-03-11 04:43:07	Guest2	Override/ice between <2020-03-11 04:43:07.272> and <2020-03-11 04:58:00.00>
Opened Review	A03456***0ADE	Card	2020-03-11 04:42:56	Guest2	
Review Timed out	EMPL 52044 04	Employee	2020-03-11 04:42:36	Guest2	REVIEW LOCK REVOKED FOR USER - REVIEW OPEN BEYOND ALERTTIMEOUT INTERVAL
Review Timed out	A03456***0ADE	Card	2020-03-11 04:42:36	Guest2	REVIEW LOCK REVOKED FOR USER - REVIEW OPEN BEYOND ALERTTIMEOUT INTERVAL
Alert Review Timed out	EMPL 2708 205	Employee	2020-03-11 04:42:36	Guest2	Account lock revoked for user - Account open beyond AlertTimeOut Interval

Reporting with MS Power BI

Data Sources

Any PRM Table

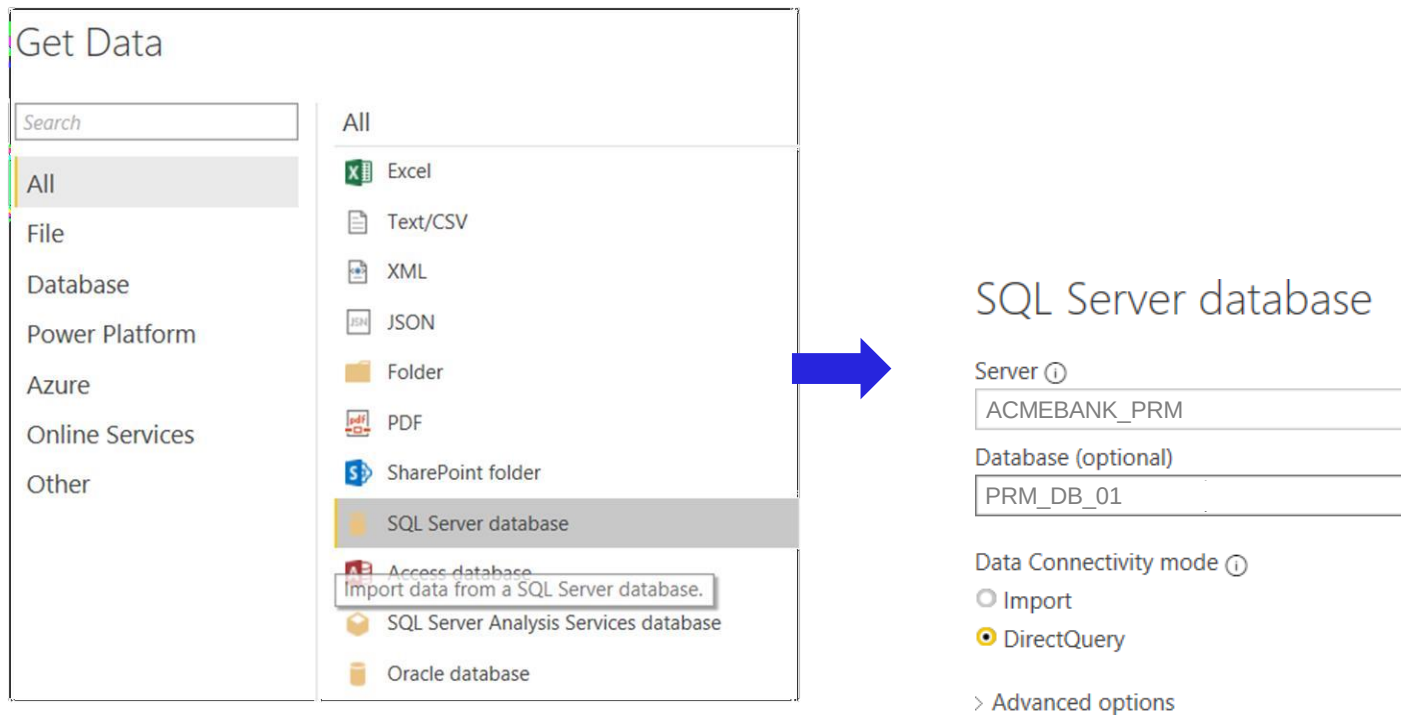
- Any PRM table within the database can be selected but the common starting points are:
 - DETAIL
 - DETAIL FLAG
 - FRAUD DETAIL
 - MASTER(S)
 - ACT
- Ideally, the required tables would be replicated into a new database

Data Sources

Connecting MS Power BI to PRM

- Click “Get Data”
- Select “SQL Server database”
- Click “DirectQuery”

NB: Oracle also supported

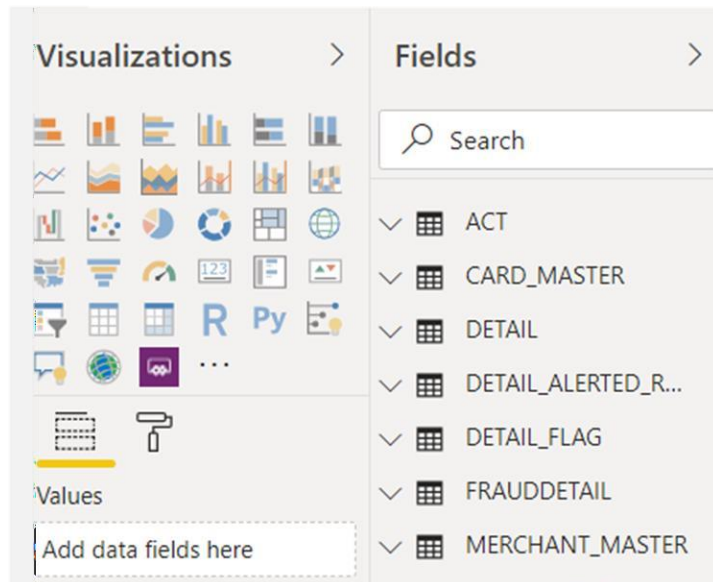


The image shows the 'Get Data' dialog box in Microsoft Power BI. On the left, a list of data sources is shown, with 'SQL Server database' selected. A blue arrow points from this selection to the right-hand configuration panel. The configuration panel is titled 'SQL Server database' and contains the following fields and options:

- Server ①**: A text box containing 'ACMEBANK_PRM'.
- Database (optional)**: A text box containing 'PRM_DB_01'.
- Data Connectivity mode ①**: Two radio buttons are present: 'Import' (unselected) and 'DirectQuery' (selected).
- > Advanced options**: A link to expand advanced settings.

Data Sources

Connecting MS Power BI to PRM



Navigator

Display Options

DE_UI_FIELD_SEARCH
DE_UI_GRIDBAG_CONFIG
DETAIL
DETAIL_ALERTED_RULE
DETAIL_AUX
DETAIL_FLAG
DETAIL_MARKSELECTION
DEVICE_3_DAYS_PROFILE
DEVICE_7_DAYS_PROFILE
DEVICE_DAILY_PROFILE
EMM_STATUS_TYPE
EMPLOYEE_30_DAYS_PEER_PROFILE
EMPLOYEE_30_DAYS_PROFILE
EMPLOYEE_7_DAYS_PEER_PROFILE
EMPLOYEE_7_DAYS_PROFILE
EMPLOYEE_90_DAYS_PEER_PROFILE
EMPLOYEE_90_DAYS_PROFILE
EMPLOYEE_DAILY_PEER_PROFILE
EMPLOYEE_DAILY_PROFILE
EMPLOYEE_MASTER

DETAIL_FLAG

DD_PARTITION_DATE	ND_KEY_HASH	RECORD_SOURCE_ID	SD_TIEBREAKER
18/07/2020	0	2	021059ab12266758202004
22/07/2020	0	32	m7olCPTQg0a/allowAr2/8ch
02/08/2020	0	10	0210224234744202004241
02/08/2020	0	10	0210224234749202004241
02/08/2020	0	10	0210224234794202004241
02/08/2020	0	10	0210224234796202004241
02/08/2020	0	10	0210224234810202004241
02/08/2020	0	10	0210224234812202004241
02/08/2020	0	10	0210224234858202004241
02/08/2020	0	10	0210224234869202004241
02/08/2020	0	10	0210224234872202004241
02/08/2020	0	10	0210224234881202004241
02/08/2020	0	10	0210224234890202004241
02/08/2020	0	10	0210224234903202004241
02/08/2020	0	10	0210224234908202004241
02/08/2020	0	10	0210224234918202004241
02/08/2020	0	10	0210224234931202004241
02/08/2020	0	10	0210224234977202004241
02/08/2020	0	10	0210224234983202004241
02/08/2020	0	10	0210224234984202004241
02/08/2020	0	10	0210224234991202004241
02/08/2020	0	10	0210224235010202004241
02/08/2020	0	10	0210224235023202004241

Select Related Tables Load Transform Data Cancel

Relationships

Create Relationships Between Tables

Create relationship

Select tables and columns that are related.

CARD_MASTER

SM_PAN	NM_DISPOSITION	DM_DISPDATE	NM_DISPREVID	NM_INUSEREVID	NM_DISPSTATUS
3743343505624129	0	null	0	0	0
3743343692893813	0	null	0	0	0
3743344859708308	0	null	0	0	0

DETAIL

	SD_PAN	SD_RETL_ID	SD_UAN	SD_TERM_ID	SD_EMPLOYEE_ID
9f	4111222233335564	RTL004560045412AA458955	34567890000000000478	TERM145788WER644	EMP0457963
9f	4111222233335564	RTL004560045412AA458955	34567890000000000478	TERM145788WER644	EMP0457963
9f	2020-06-17T15:18:20.388Z	RTL004560045412AA458955	34567890000000000478	TERM145788WER644	EMP0457963

Cardinality

One to many (1:*)

Cross filter direction

Single

☒ Make this relationship active

☐ Assume referential integrity [Learn more](#)

☐ Apply security filter in both directions

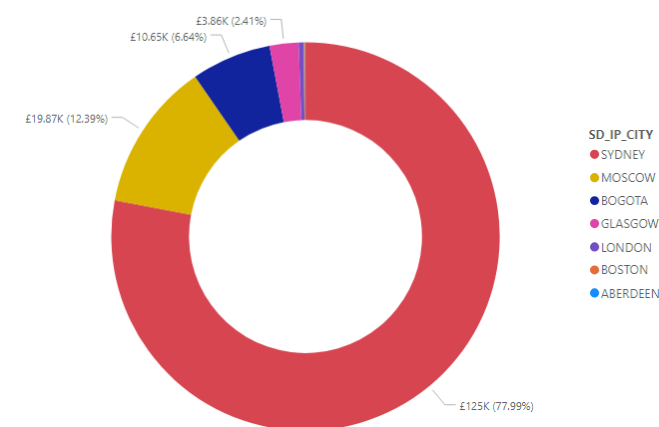
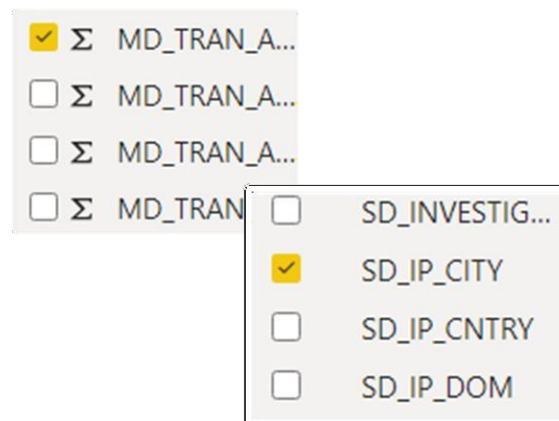
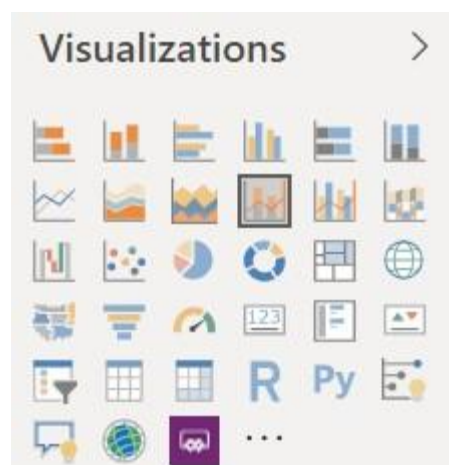
OK

Cancel

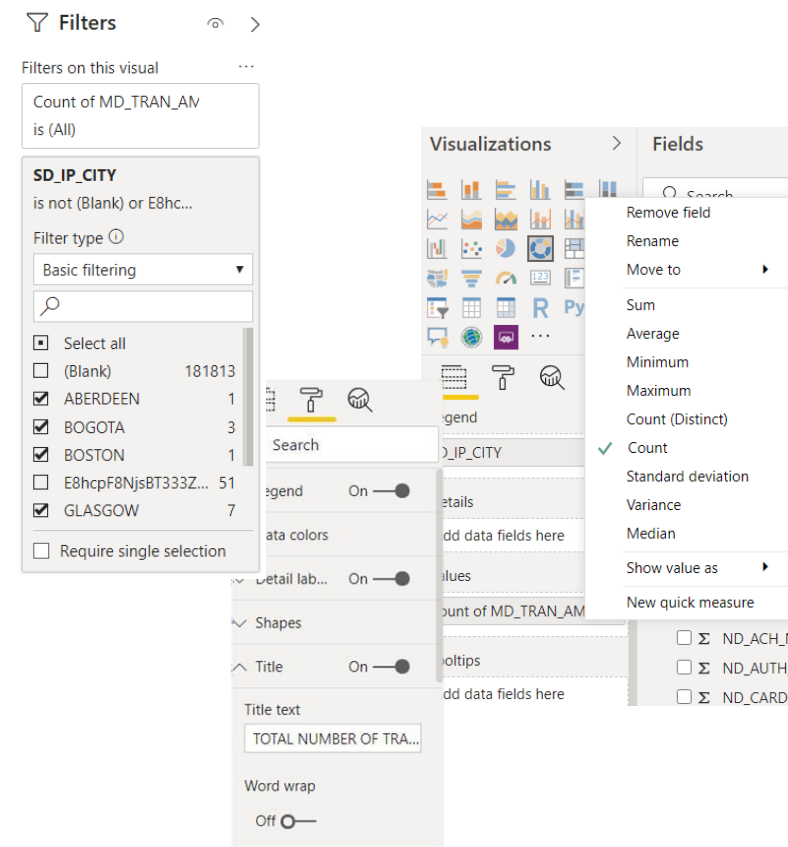
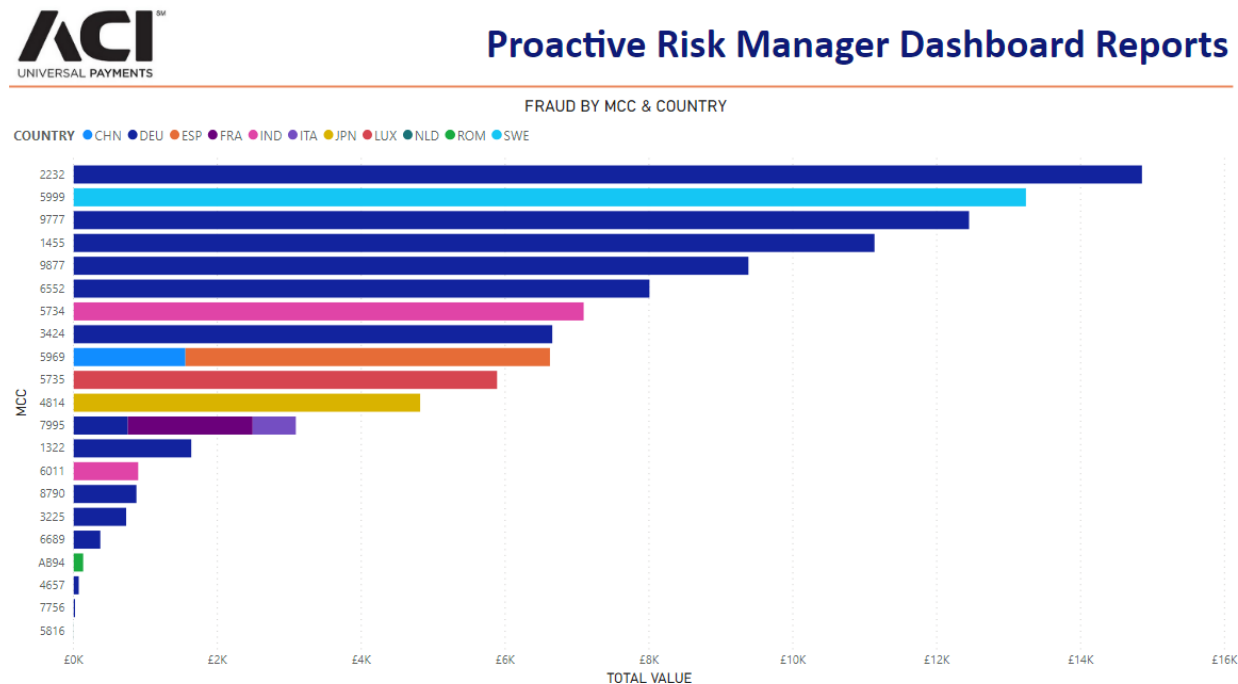
Reports

Creating Reports

- Click a visualization
- Select the required tables and fields
- Chart / graph will populate accordingly

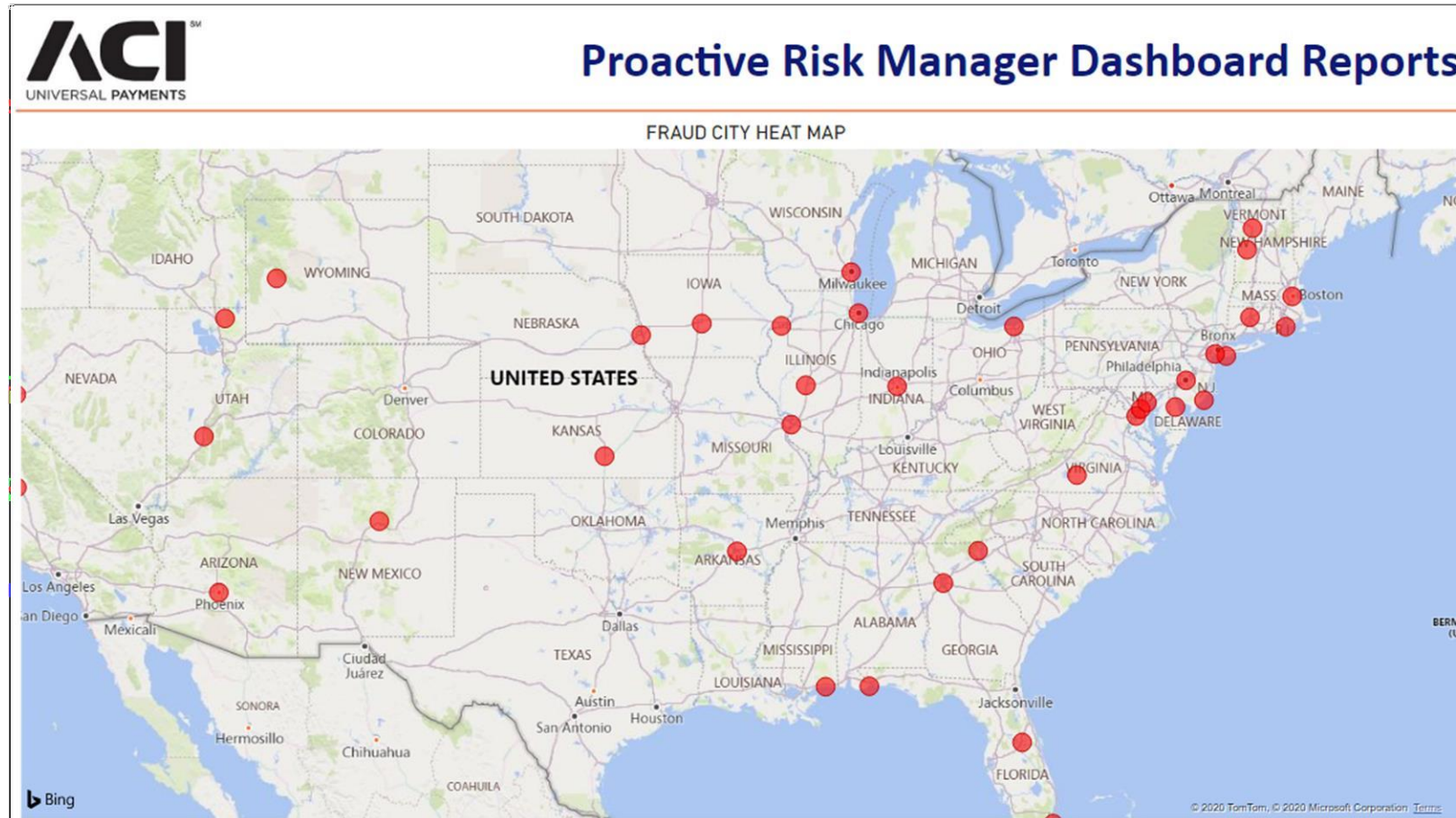


Customize as Required



Reports

Example

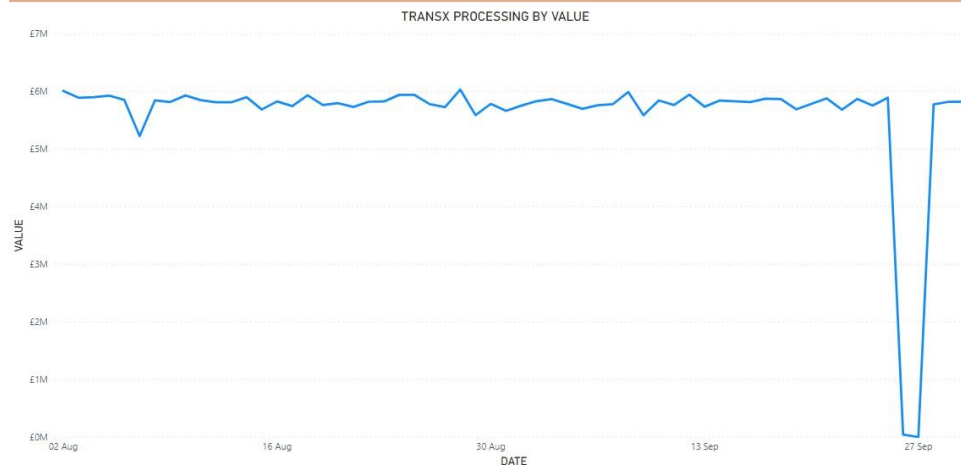


Reports

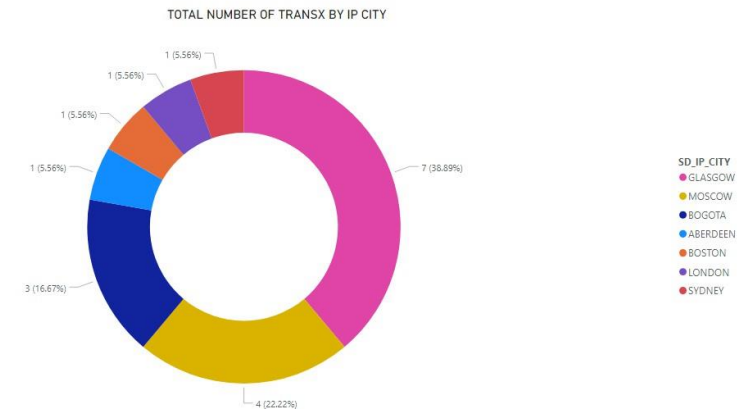
Examples



Proactive Risk Manager Dashboard Reports



Proactive Risk Manager Dashboard Reports



Reports

Ad-hoc Analysis



Proactive Risk Manager Dashboard Reports

SD_TERM_CNTRY	SD_TERM_CITY	SD_RETL_SIC_CDE	MD_TRAN_AMT1	Count of MD_TRAN_AMT1	Average of MD_TRAN_AMT1
GBR		6552	£57,677.11	118	£488.7891
USA	DES MOINES	6011	£37,177.67	60	£619.6278
USA	SPRINGFIELD	6011	£38,471.03	58	£663.2936
GBR		6689	£1,644.04	48	£34.2508
USA	LITTLE ROCK	6011	£27,903.37	41	£680.57
USA	WICHITA	6011	£23,682.57	41	£577.6237
USA	DOVER	6011	£27,524.23	39	£705.7495
USA	SACRAMENTO	6011	£28,523.2	39	£731.3641
USA	ATLANTA	6011	£21,416.92	36	£594.9144
USA	WASHINGTON	6011	£22,566.66	34	£663.7253
USA	BELLEVUE	6011	£22,307.21	32	£697.1003
USA	DAVENPORT	6011	£18,147.5	26	£697.9808
USA	DULUTH	6011	£18,343.53	26	£705.5204
USA	MONTPELIER	6011	£14,451.32	26	£555.82
USA	BOSTON	6011	£16,001.5	25	£640.06
USA	ST. LOUIS	6011	£14,026.72	25	£561.0688
USA	WALHALLA	6011	£19,145.39	25	£765.8156
USA	FREEPORT	6011	£14,292.02	24	£595.5008
USA	LAUREL	6011	£15,343.43	24	£639.3096
USA	PENSACOLA	6011	£14,511.78	24	£604.6575
USA	BEAVER	6011	£16,727.72	23	£727.2922
USA	OXFORD	6011	£15,621.07	21	£743.8605
USA	BILOXI	6011	£13,285.23	20	£664.2615
USA	KILLINGTON	6011	£15,803.75	19	£831.7763
USA	LOGAN	6011	£14,053.38	19	£739.6516
USA	PORTLAND	6011	£12,576.70	18	£698.7106
Total			£720,177.48	1148	£627.3323

SD_RETL_SIC_CDE ^ x

contains '7995' or sta...

Filter type ⓘ

Advanced filtering

Show items when the value:

contains

7995

☐ And
 ☒ Or

starts with

6

Apply filter

DD_DATE

is on or after 01/09/2...

Filter type ⓘ

Advanced filtering

Show items when the value:

is on or after

01/09/2020

12

:

00

AM

☒ And
 ☐ Or

Apply filter

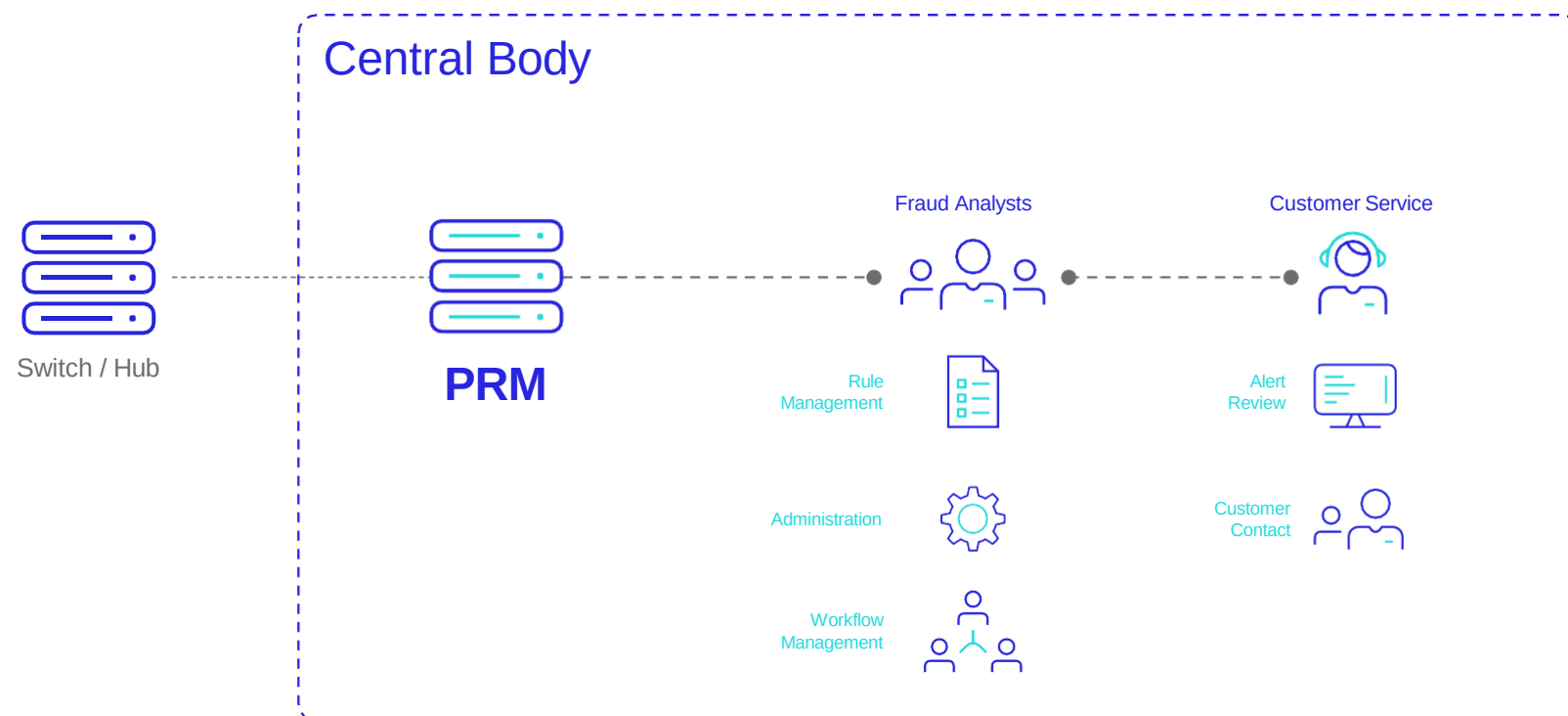
Service Options

For Central Infrastructure

Payments Intelligence Hub

Central body manages service for members

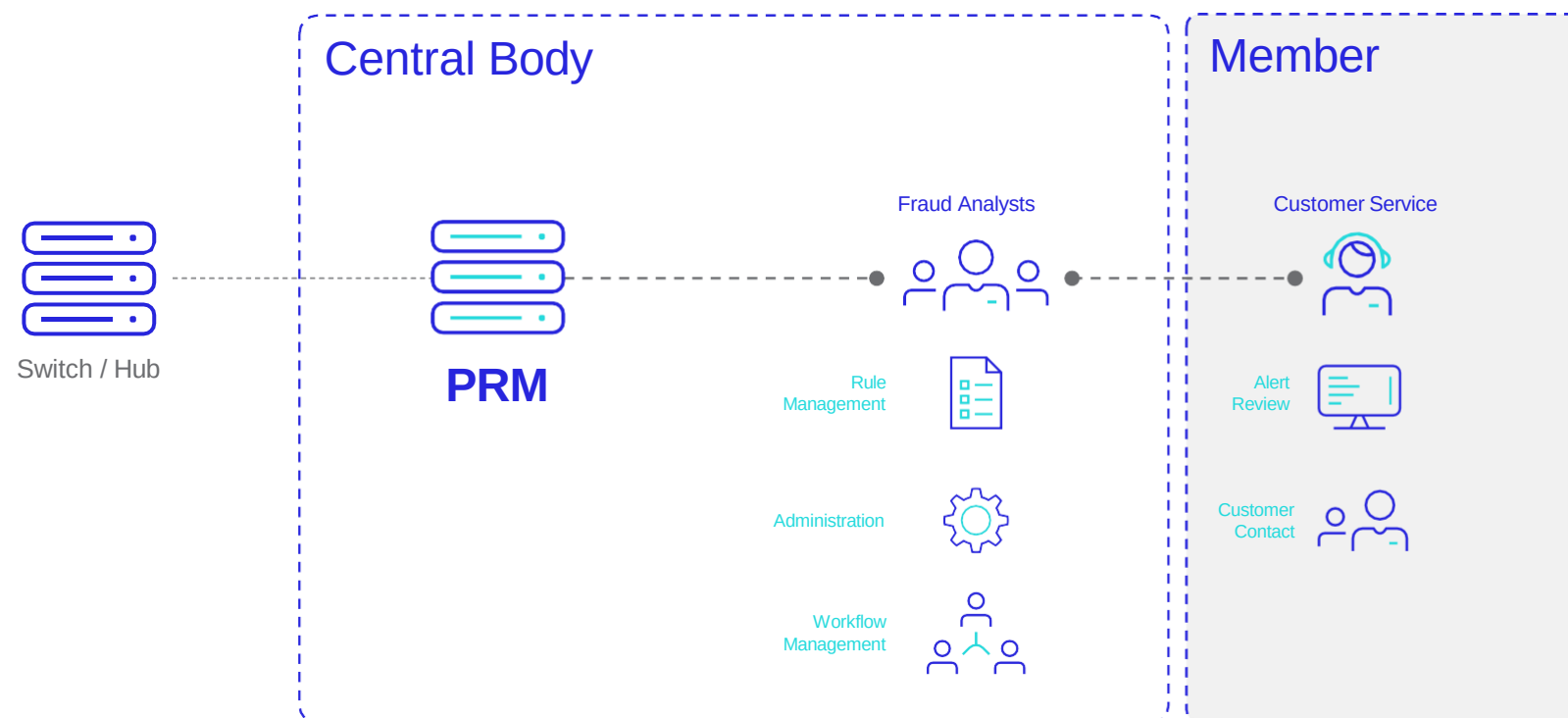
- Central body writes and deploys rules:
 - “Global” rules
 - Member specific rules
- Central body owns administration and workflow management (for their own staff)
- Central body works alerts and contacts the members’ customers



Payments Intelligence Hub

Central body administers service, members manage customers

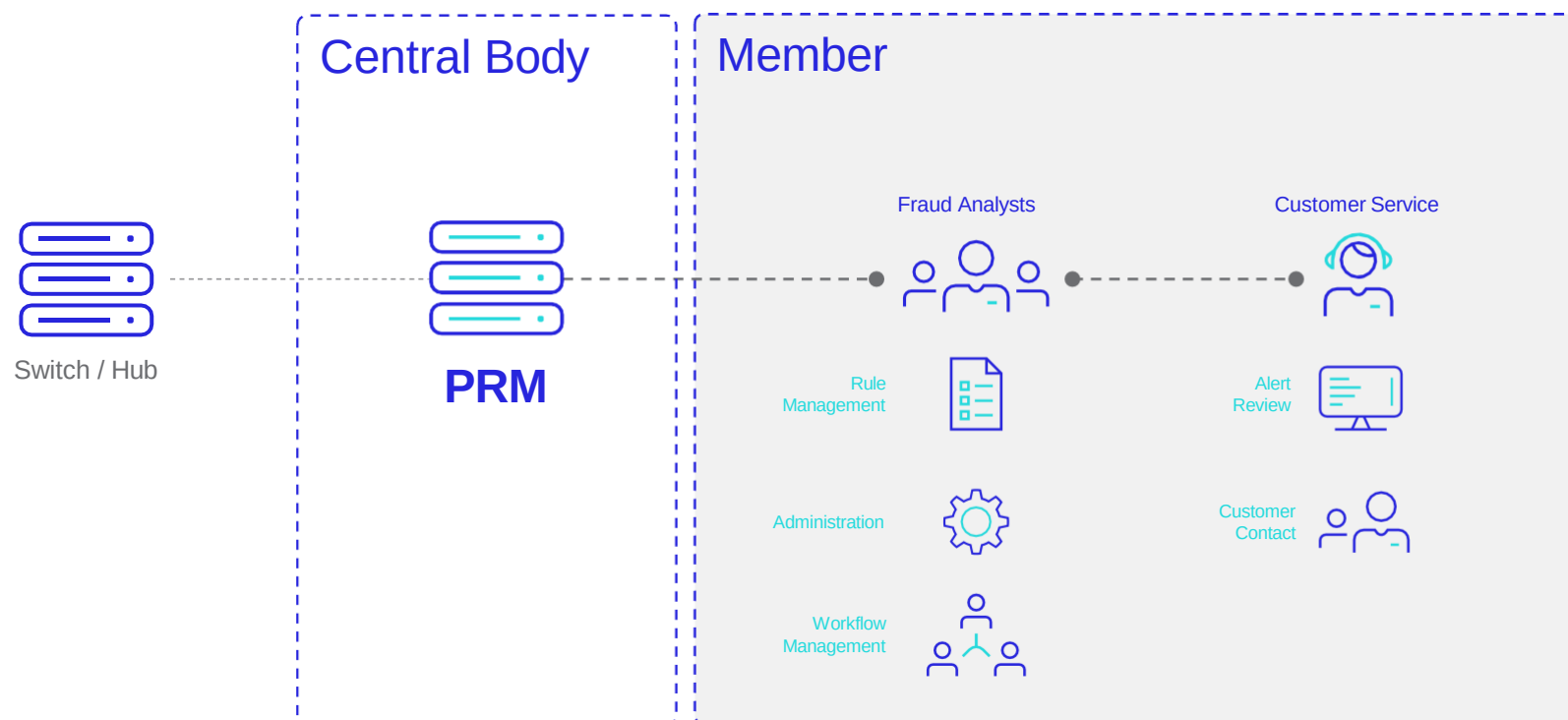
- Central body writes and deploys rules:
 - “Global” rules
 - Member specific rules
- Central body owns administration and workflow management (for their own staff)
- Members work alerts and contacts customers



Payments Intelligence Hub

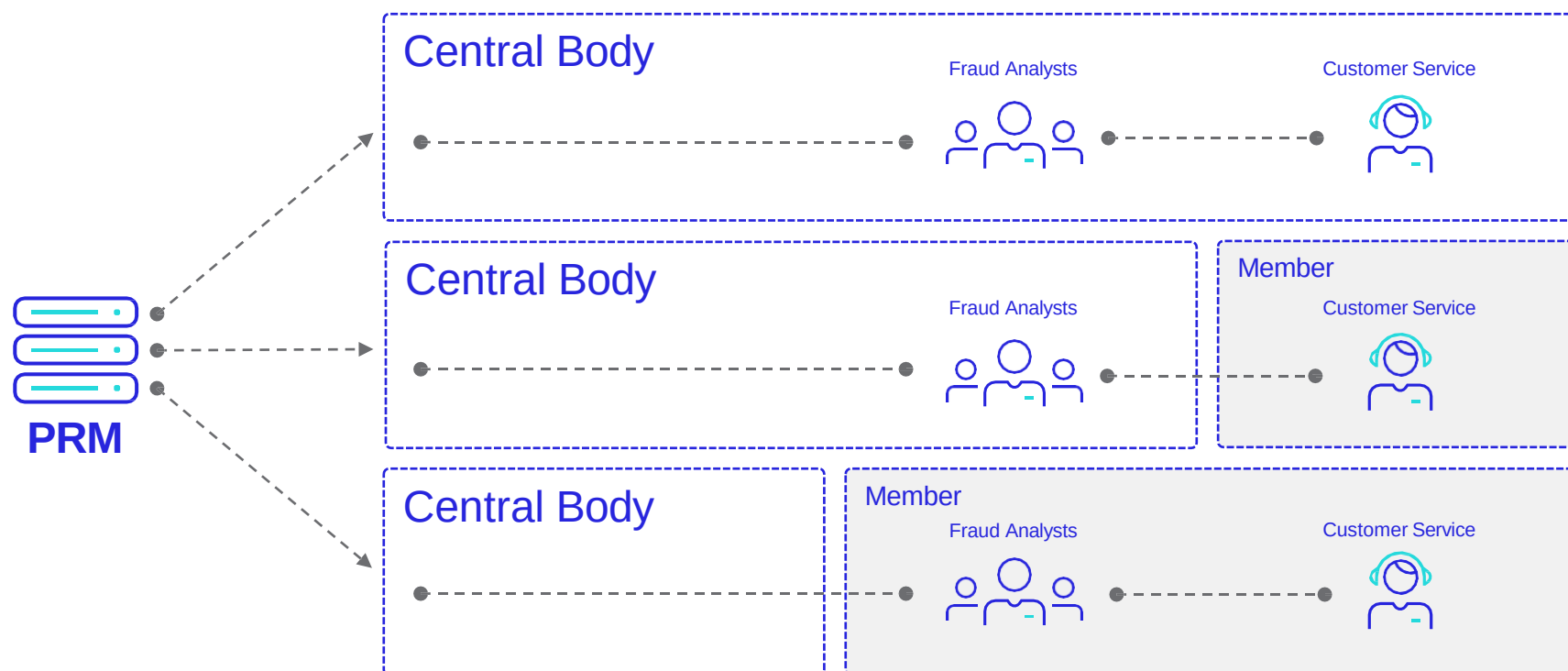
Member administers and manage customers

- Central body provides access
- Members write and deploys rules
- Members owns administration and workflow management
- Members work alerts and contacts customers



Payments Intelligence Hub

Mix and match



Software as a Service

Customer Needs

End-to-end requirements for a fraud management solution



Implement Infrastructure

Hardware & base software to run the application:

- Machine
- O/S
- Database
- WAS



Manage Infrastructure

Take care of the health of the infrastructure:

- Hardware
- CPUs
- Disks
- VMs



Manage Environment

Take care of the environment and everything around it:

- Maintenance jobs
- Interfaces
- Connectors



Implement Application

Implement ACI products:

- PRM
- AMG
- MoneyGuard



Manage Application

Ongoing maintenance:

- Configurations
- Patches
- Service packs
- Upgrades



Operate Application

Running the fraud shop:

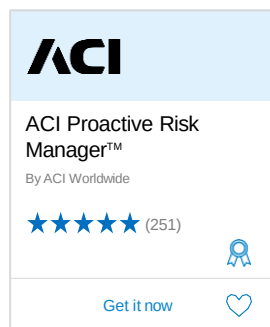
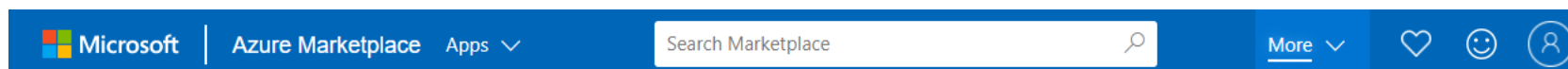
- Alert review and action
- Workflow
- Rule management
- Model development

PRM SaaS

Microsoft Azure



- With Microsoft Azure and the Microsoft Partner Network, ACI leverages its global professional services reach to serve customers in their native language and culture
- ACI Proactive Risk Manager is available from Microsoft's Azure Marketplace



PRM SaaS

Microsoft Azure



Microsoft Azure Search resources, services, and docs (G+/f)

Home > Marketplace > ACI Proactive Risk Manager (preview) >

Create ACI Proactive Risk Manager

1 Basics 2 Sql Managed Instance Settings 3 VMs Settings 4 Database Application properties 5 AG Application properties

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ Microsoft Azure Enterprise PRM Demo ▼

Resource group * ⓘ ▼
[Create new](#)

Instance details

Region * ⓘ East US ▼

Prefix for the environment (2-5 characters) * ⓘ cust1

RT Present * ⓘ Yes ▼

PRM License File * ⓘ Select a file 

PRM SaaS

Microsoft Azure



- One instance per customer i.e. not multi-tenancy
- Repeatable implementation, proven in different countries
- Complete within 6 hours
- PRM application is now available for configuration, with separate UAT and Production environments
- Licensing options:
 - Additional environments
 - Active/Passive or Active/Active
- SLAs:
 - Availability:
 - Real-Time 99.9%
 - Near Real-Time 99.9%
 - User Interface 99.0%
 - Response times:
 - Real-Time: 97% in less than 125ms
 - Near Real-Time: 97% at an average processing time of under 5 minutes

Cloud Deployments

Private Cloud, Public Cloud, Customer Cloud

SaaS



SaaS - Private Cloud (US)



The logo for getnet, featuring the word "getnet" in white lowercase letters on a red rectangular background, with a small grid icon to the right.

On-Premise

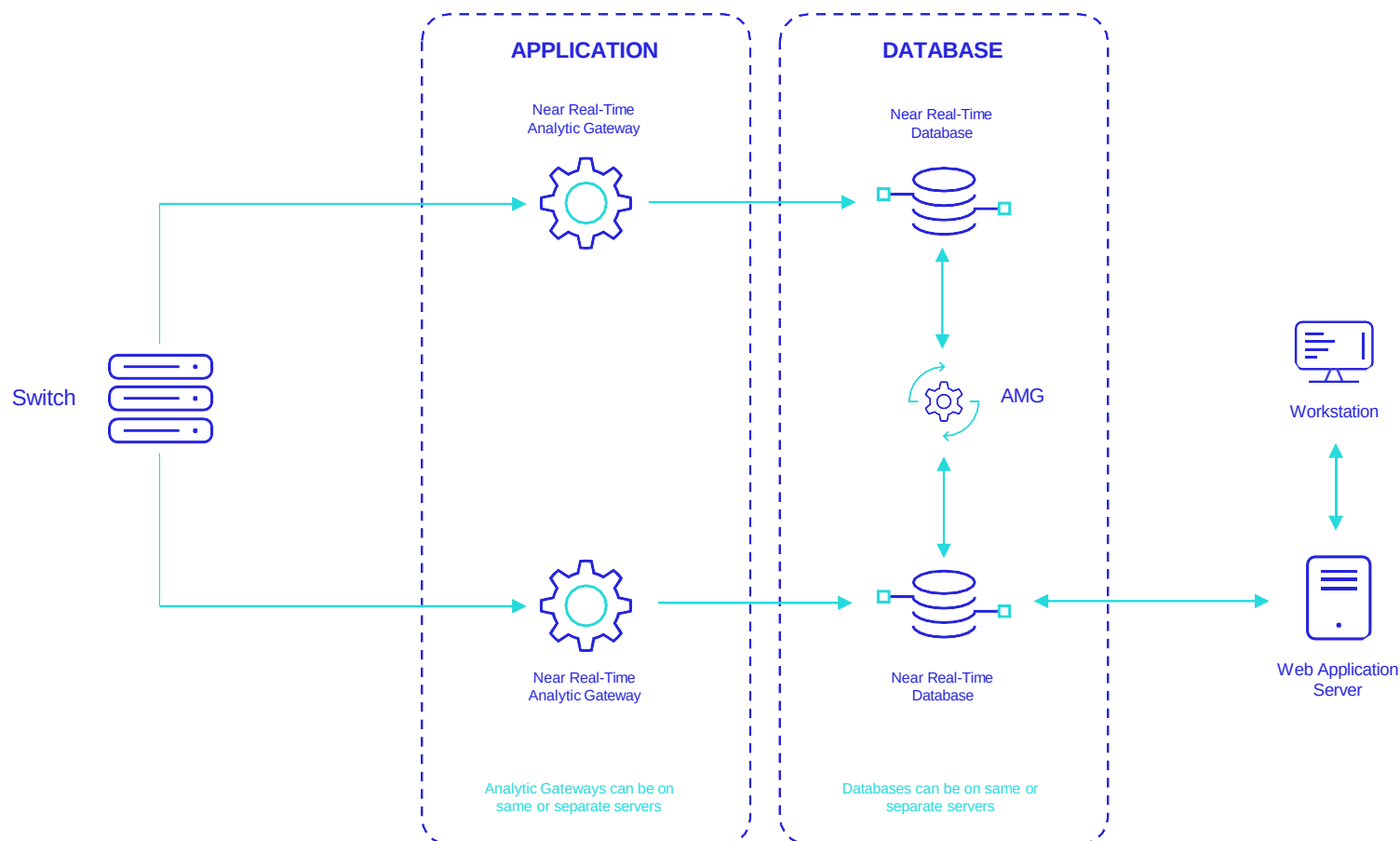


Architecture

ACI Proactive Risk Manager

Architecture

ACI Proactive Risk Manager

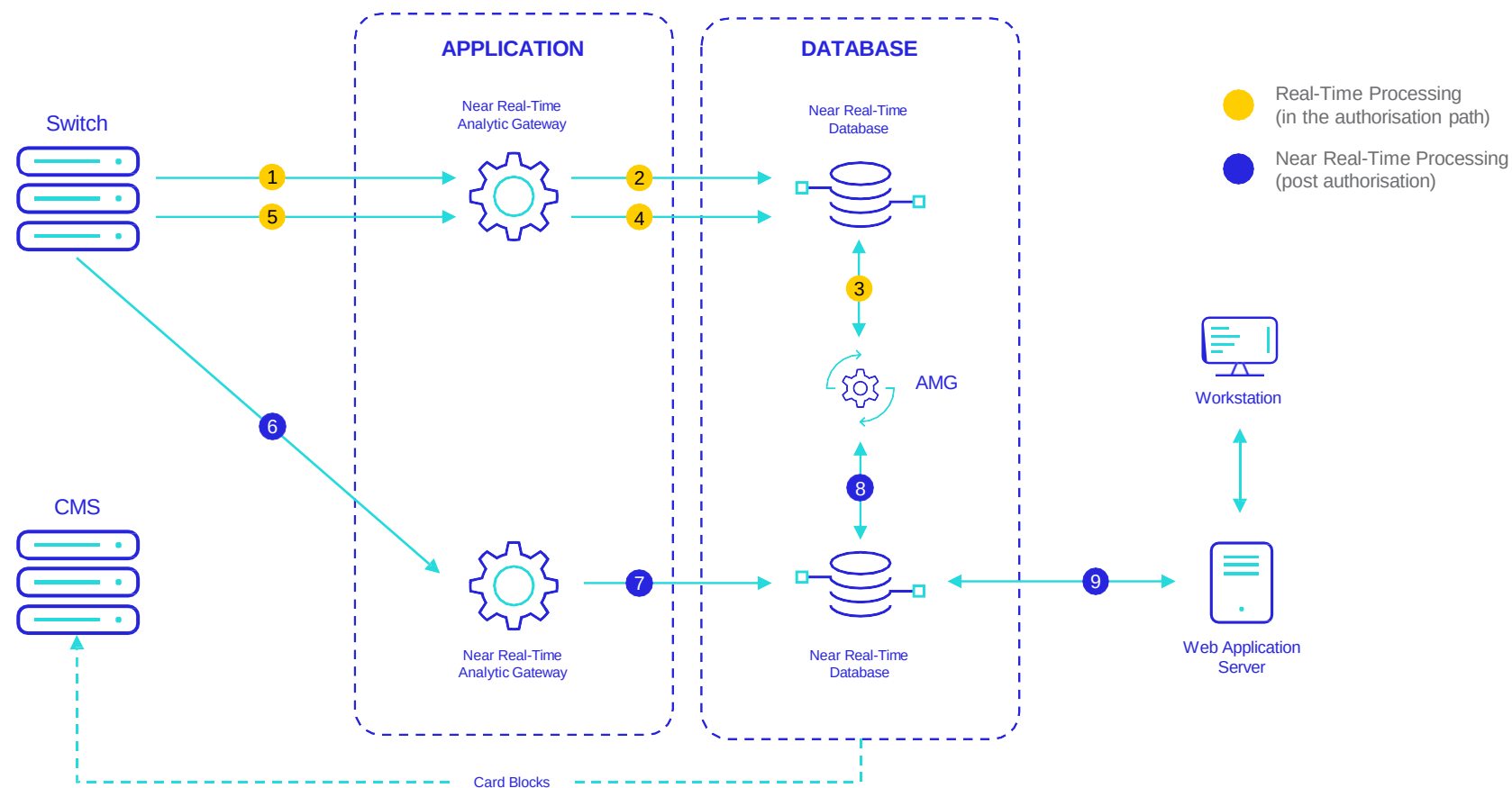


Data Flow

ACI Proactive Risk Manager

Data Flow

ACI Proactive Risk Manager

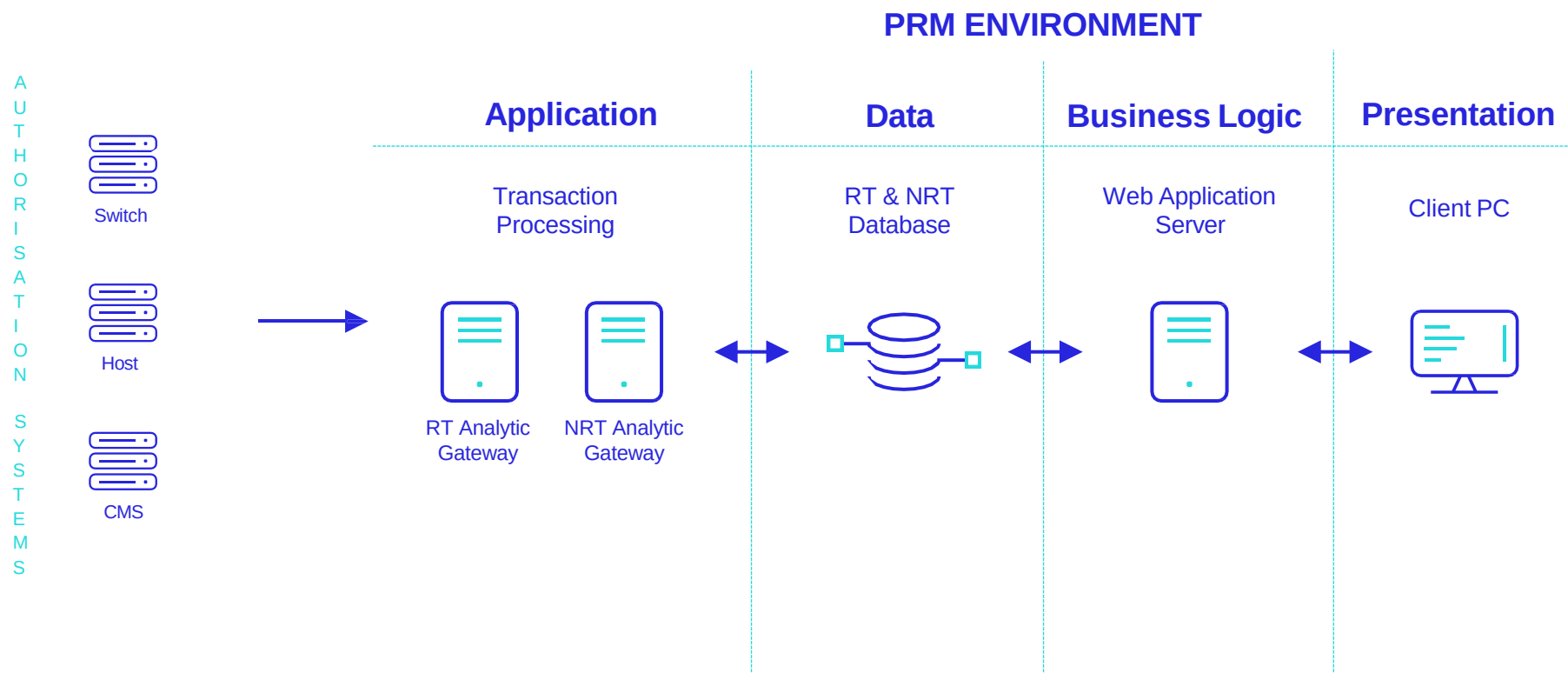


Logical Tier Topology

ACI Proactive Risk Manager

Topology

ACI Proactive Risk Manager



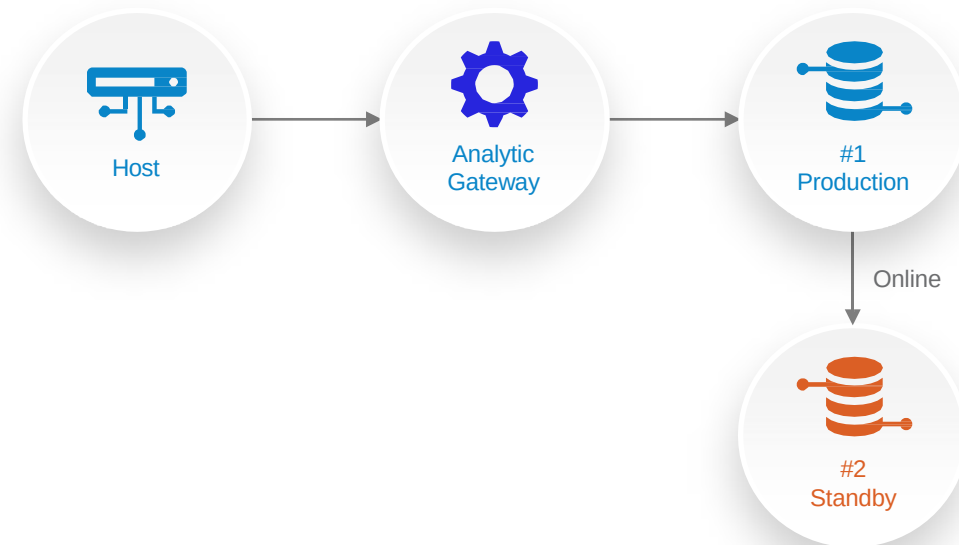
Standby Options

ACI Proactive Risk Manager

Standby

Hot Standby

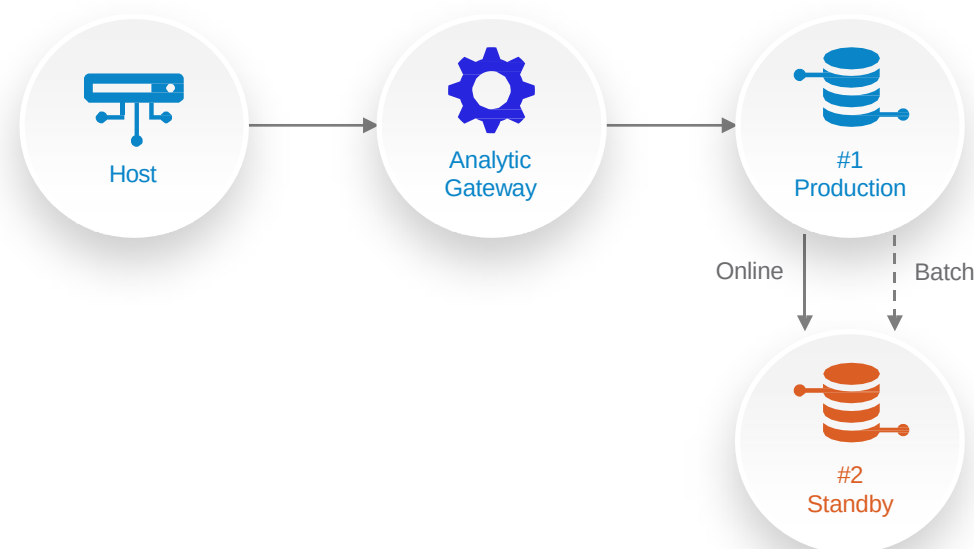
- Database #2 is up and running and is mirrored (100% replication)
- #2 will become production almost immediately, in a “flick of a switch”
- Database #2 will receive online transactions immediately after they occur in #1



Standby

Warm Standby

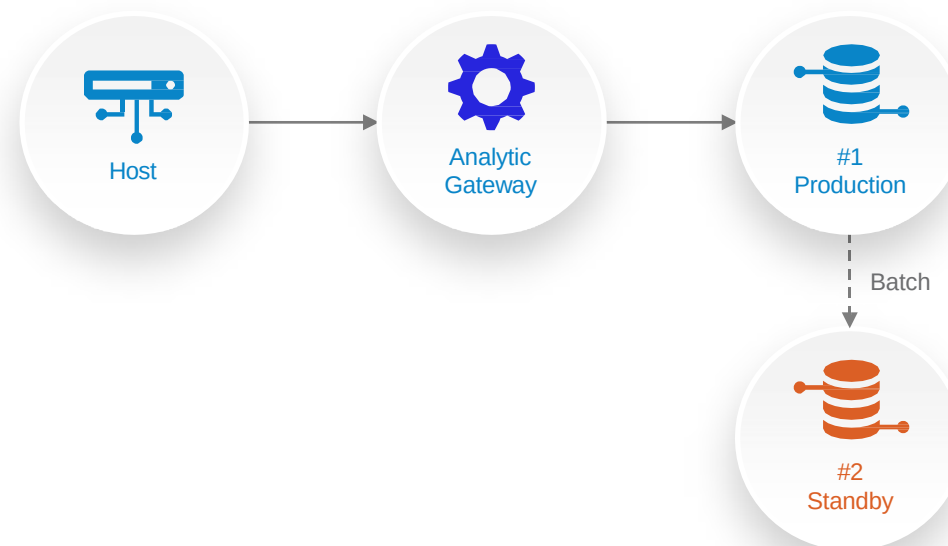
- Database #2 is not necessarily “up and running” but will be relatively in sync with production
- #2 can become production in a relatively short time
- Database #2 may receive transactions via message and/or batch



Standby

Cold Standby

- Database #2 is not “up and running”
- Standby may take hours to be in production
- Database #2 may receive transactions sent to #1 but will be at irregular intervals and typically batch



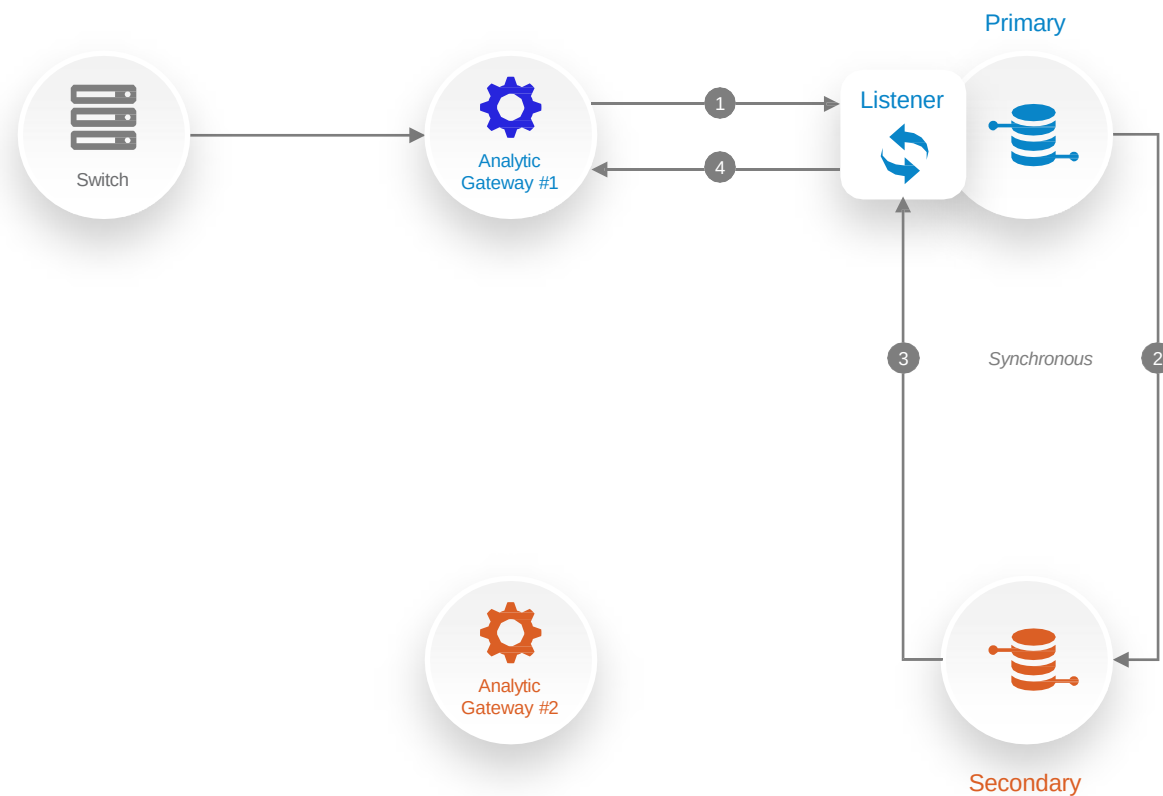
Disaster Recovery

ACI Proactive Risk Manager

DR

Synchronous DBs

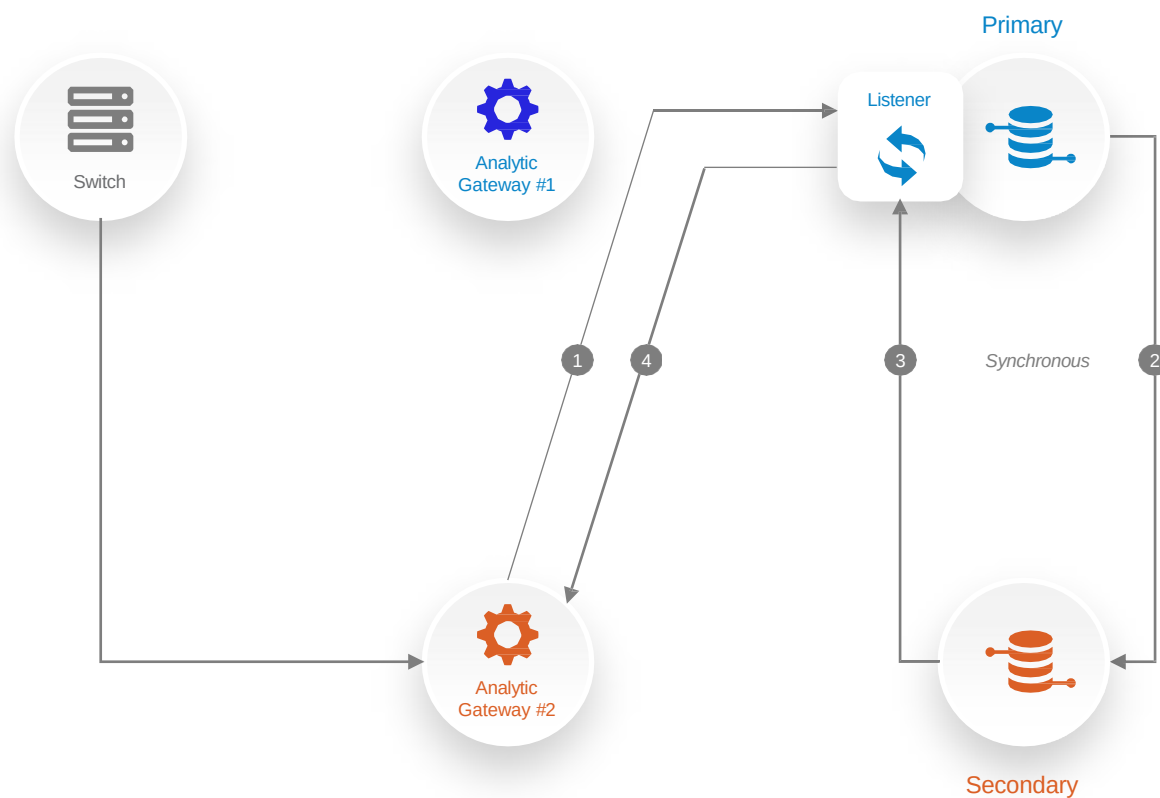
- Production using AG #1



DR

Synchronous DBs

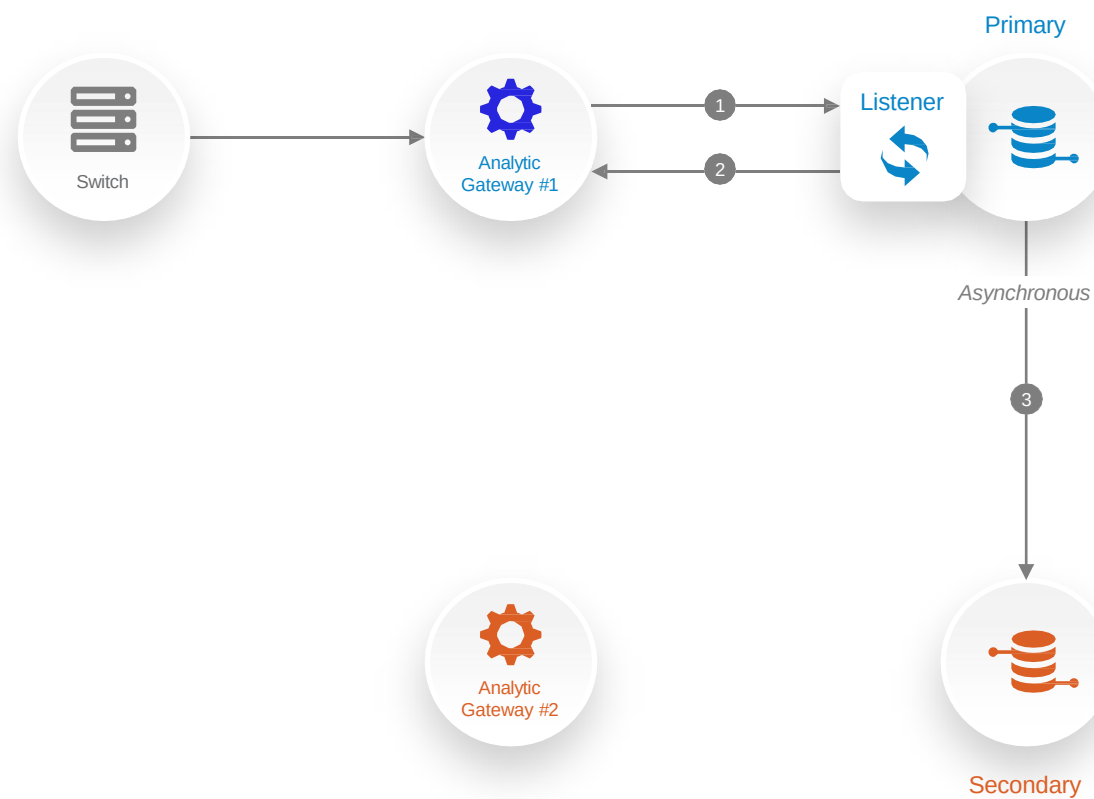
- AG #1 goes down
- Switch re-directed to AG #2



DR

Asynchronous DBs

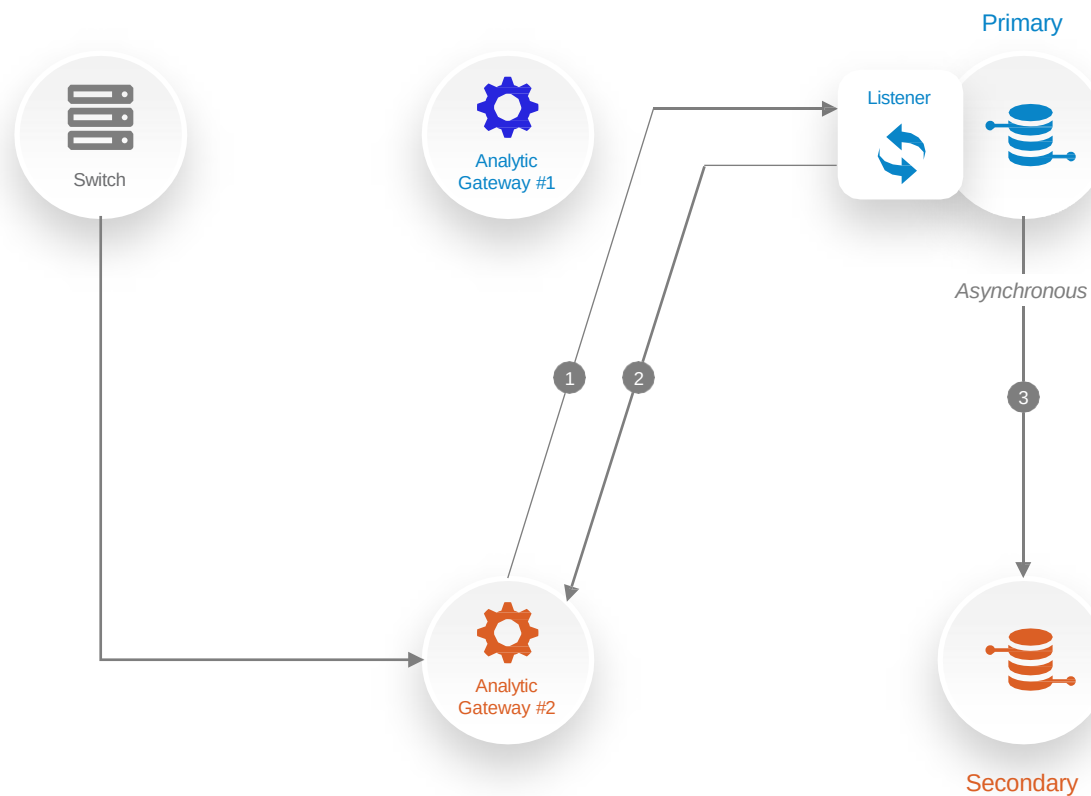
- Production uses AG #1



DR

Asynchronous DBs

- AG #1 goes down
- Switch re-directed to AG #2



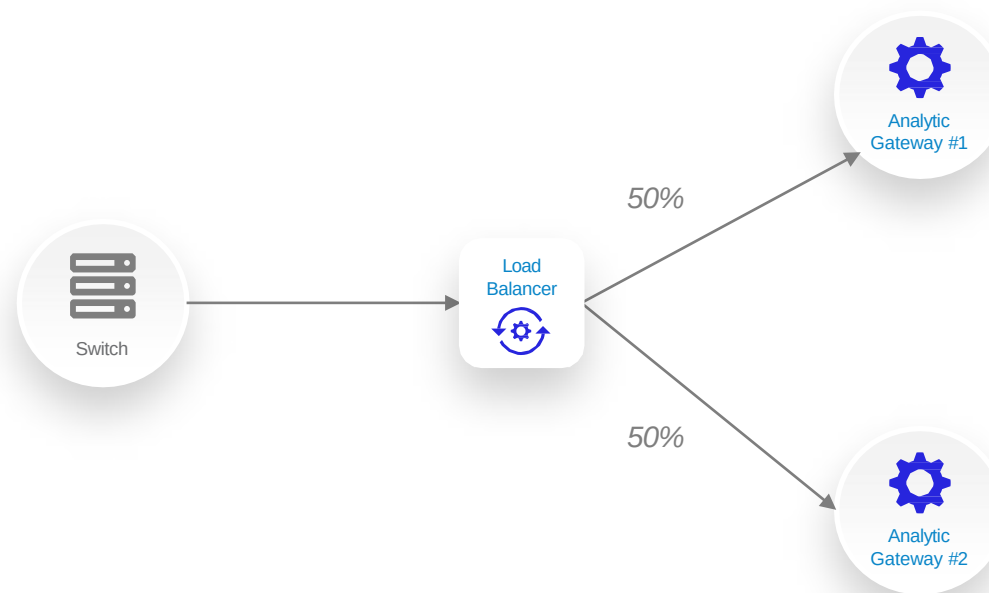
High Availability

ACI Proactive Risk Manager

High Availability

Second Site Licence

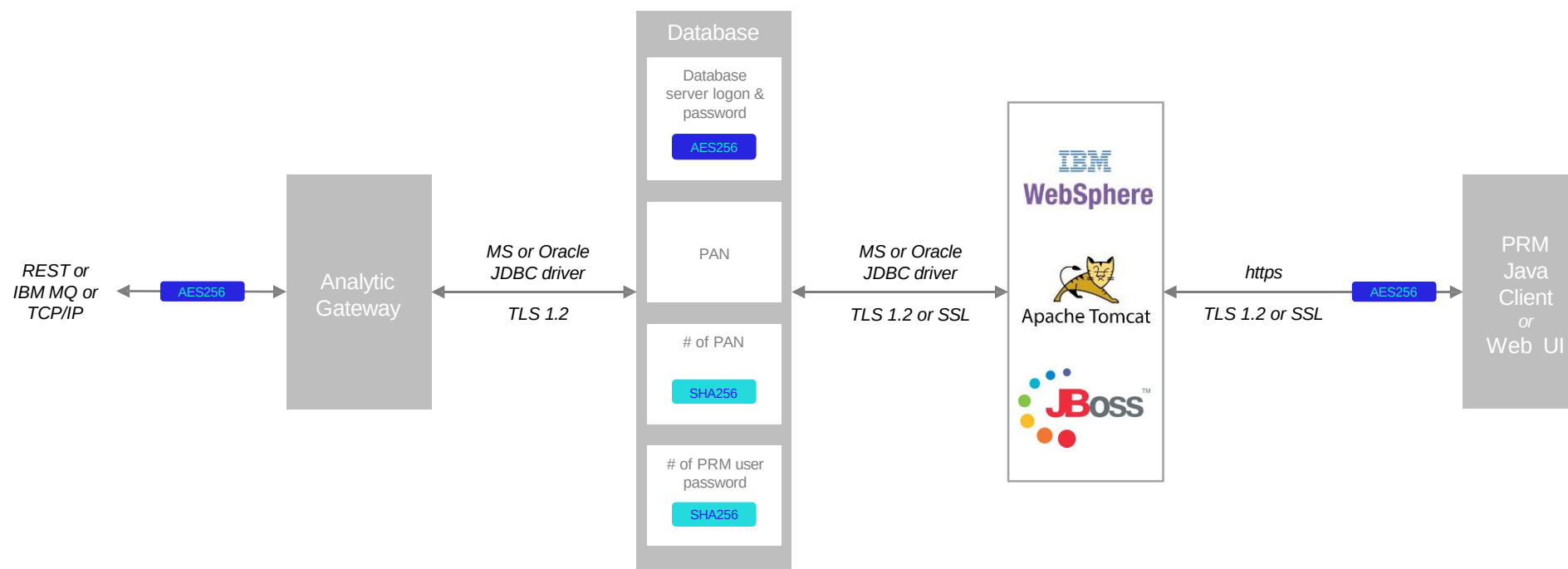
- Each AG processes a share of the production load in normal operations (Active/Active), but can process the full production load in the event of an outage
- AGs will be in different physical locations



Security

ACI Proactive Risk Manager

Encryption Points

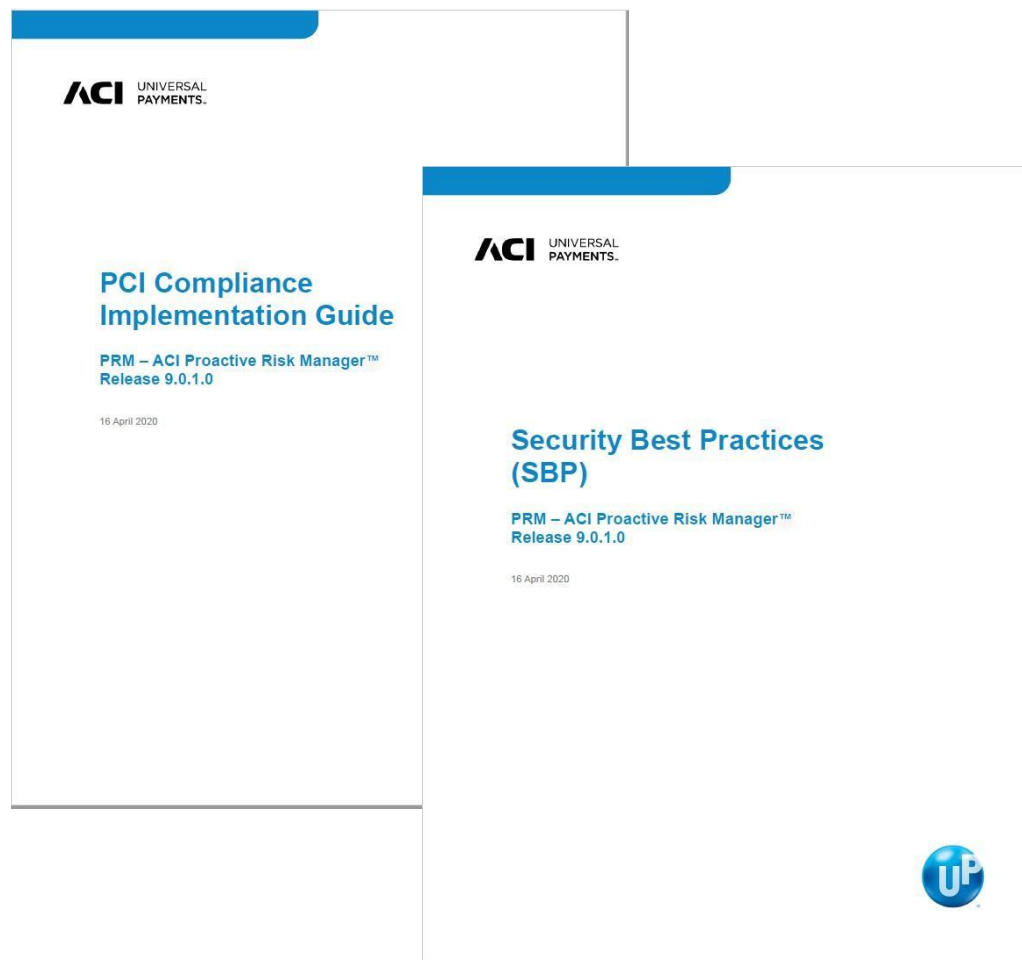


Data Security

- In motion
 - Communication channels between PRM application server's exposed services and external applications will be secured with 2-way TLS 1.2.
 - PRM UI Connections to Web Application Server will be HTTPS
- At rest
 - Passwords stored in the PRM database are encrypted using a one-way SHA-1 hash, SHA-256 hash, or salted SHA-256 hash (SSHA-256)
 - Standard database encryption tools are used to secure data at rest

Security Guides

- Comprehensive security documentation, including a *PCI Compliance Implementation Guide*, is available
- *Security Best Practices* provides recommendations for securing the application and surrounding logical environment



End of Deck

ACI Worldwide

Real-Time Payments